CVT System Model Derivation

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Abstract

This work develops a first-principles dynamic model for a mechanically actuated dry rubber V-belt continuously variable transmission (CVT) and its coupling to the vehicle drivetrain. The model is motivated by a persistent gap in the available literature: while quasi-static Baja-oriented models are common and detailed dynamic models exist for hydraulically actuated metal-belt CVTs, publicly available formulations for mechanically actuated rubber belt CVTs do not generally combine sheave inertia, flyweight actuation, torque-reactive secondary clamping, belt centrifugal effects, traction-limited torque transmission, and vehicle longitudinal dynamics within a single unified framework.

Starting from pulley geometry and belt-length closure, the derivation develops the kinematic ratio relation, rotational dynamics of the primary and secondary pulleys, and the axial equation of motion governing shift. The primary clutch model includes centrifugal flyweight actuation and spring preload, while the secondary model includes helix-generated torque feedback together with spring forces. A centrifugal belt correction is derived to account for the reduction in effective normal force at speed, and an analytical stick–slip switching law is introduced to distinguish no-slip operation from traction-limited torque transmission. These elements are assembled into a closed nonlinear state-space model suitable for direct numerical time integration.

The resulting formulation is intended both as a simulation model and as a derivation document that makes each modeling step physically interpretable. The simulation section is structured to verify recovery of expected quasi-static equilibria, illustrate the importance of the belt centrifugal correction, demonstrate transient slip behavior, and show coupled launch dynamics of the complete drivetrain. TODO: add the key quantitative result highlights once the final simulation figures and numerical comparisons are completed. TODO: add the final concluding implications and summary statement after the conclusion section is finalized.

Contents

1 Introduction

1.1 Background and Context

The continuously variable transmission (CVT) is a mechanical power transmission device capable of producing a continuously variable speed ratio between an input and output shaft, in contrast to the fixed discrete ratios of a conventional stepped gearbox. In the class of CVTs relevant to this work — dry rubber V-belt CVTs with mechanically actuated conical sheave pairs — the transmission ratio is not commanded by an external actuator or electronic controller. Instead, it is an emergent dynamic response: the equilibrium of axial forces acting on the primary and secondary sheave pairs, arising from centrifugal flyweight mechanisms, torque-reactive helix and ramp assemblies, and return springs, determines the radial position at which the belt seats within each sheave groove, and therefore the effective pitch radii and transmission ratio at every instant.

This self-regulating character is simultaneously the CVT's greatest practical asset and the primary source of difficulty in modeling it rigorously. Because no actuator explicitly commands the ratio, the ratio trajectory at any instant is the output of a coupled mechanical system whose state depends on engine speed, load torque, belt clamping geometry, and the prior history of the shift position — all evolving together in time.

This class of CVT is found throughout small-displacement powersport and utility vehicles. It is the mandated transmission in the SAE Baja competition series, where a 10 hp engine is paired with a CVT that must be designed, selected, and tuned entirely by student engineering teams from first principles. Beyond Baja, mechanically actuated rubber V-belt CVTs are standard equipment in snowmobiles, all-terrain vehicles (ATVs), utility task vehicles (UTVs), and small agricultural and industrial machinery. Despite their prevalence and the practical engineering importance of understanding their behavior, the quality of publicly available engineering models for these systems is strikingly uneven. The present work was motivated directly by this gap.

1.2 Motivation and Literature Review

1.2.1 Overview of the Modeling Landscape

A thorough review of CVT modeling literature reveals a consistent pattern: each body of work captures some aspects of the physics carefully while leaving equally important aspects absent, empiricized, or explicitly assumed away.

A comprehensive survey of CVT dynamics and control research through 2009 is provided by Srivastava and Haque in their review article *"A Review on Belt and Chain Continuously Variable Transmissions: Dynamics and Control"* Srivastava and Haque (2009).

Their review systematically covers the major CVT types — rubber V-belt, metal pushing V-belt, and chain CVT — across static, quasi-static, and fully dynamic formulations, and surveys both the physical models and the control strategies built upon them. Crucially, it emphasizes that rubber belt CVTs are mechanically and dynamically distinct from metal pushing belt CVTs in their actuation mechanisms, belt compliance, and contact mechanics, and that the two should not be treated interchangeably in modeling.

Their central conclusion — that there exists *"considerable disparity in the type of CVT models that have been used"* across the literature, and that fully dynamic, experimentally validated models remain scarce for several CVT classes — indicates that the field has developed in fragments rather than converging toward a unified framework.

The present section builds on that survey, focusing specifically on the subset of literature relevant to the mechanically actuated rubber V-belt CVT considered in this work.

1.2.2 Metal Pushing V-Belt CVT Models

The most analytically complete CVT dynamic models in the literature concern the Van Doorne-type metal pushing V-belt CVT used in automotive applications.

In this class of transmission, clamping force is imposed hydraulically through electronically controlled oil pressure rather than through centrifugal flyweights and spring mechanisms. The belt itself consists of a series of compressive steel elements rather than a flexible rubber belt operating primarily in tension.

Carbone, Mangialardi, and Mantriota Carbone et al. (2005) derive a dynamic model for the shifting behavior of the metal V-belt CVT and identify two distinct operating regimes governing ratio change. In the first regime, micro-slip redistribution within the contact arc governs the shift process, producing relatively slow ratio evolution. In the second regime, macro-slip occurs and rapid ratio change becomes possible.

Srivastava and Haque extend this class of modeling by incorporating dynamic effects associated with belt element inertia and transient force redistribution within the belt system Srivastava and Haque (2006). These studies represent some of the most detailed dynamic analyses of CVT behavior available in the literature.

Experimental characterization of the slip–torque relationship in metal belt CVTs has also been reported by Bonsen et al. Bonsen et al. (2004). Their results show that transmitted torque varies nonlinearly with slip ratio, increasing through a micro-slip region before reaching a maximum near the transition to macro-slip.

While these models achieve a high degree of dynamic fidelity, they describe a fundamentally different system class from the mechanically actuated rubber belt CVT considered here. The actuation mechanisms, belt mechanics, and contact physics differ substantially, limiting the direct applicability of these formulations.

1.2.3 Rubber Belt Mechanical CVT Models

For the mechanically actuated rubber V-belt CVT, the literature is considerably thinner.

The most detailed dynamic model in the open literature appears to be that of Julio and Plante Julio and Plante (2011). Their formulation discretizes the belt into nodal elements and applies Newton's second law to each node individually. Local friction forces are computed from the normal force at each contact point, allowing the torque transmission capacity of the belt to emerge from the cumulative tangential friction forces acting along the wrap arc.

This nodal formulation has several advantages. The belt tension distribution arises naturally from the equilibrium of individual elements rather than being imposed through the classical capstan relationship. The wedging behavior of the belt in the conical groove is captured directly through the element geometry, and centrifugal effects acting on belt elements are included through fictitious forces in the rotating reference frame.

However, two important limitations remain.

First, the inertia of the movable sheaves is neglected. The axial position of the sheaves is determined through instantaneous force balance rather than through an inertial equation of motion, meaning the model cannot capture the finite time required for sheave motion to respond to changing forces.

Second, the authors report difficulty obtaining stable predictions during the transition between clutching and shifting operation. Since mechanically actuated rubber belt CVTs use the belt clamping mechanism itself as the start-up clutch, the clutching regime forms an essential part of the system operating envelope.

Another related effort is presented in SAE technical paper *"Development of CVT Shift Dynamic Simulation Model with Elastic Rubber V-Belt"* Yi et al. (2011). This work develops a dynamic simulation model incorporating belt elasticity and derives a shift motion relationship based on axial force balance between pulleys. Because the full derivation is not publicly accessible, the detailed structure of the dynamic model cannot be evaluated fully from the available abstract and secondary descriptions.

1.2.4 Quasi-Static Baja-Specific Models

Within the SAE Baja community, most CVT modeling work follows a quasi-static force balance approach.

In these models the operating point of the transmission is obtained by solving the condition that the net axial force acting on the sheave assembly is zero. The shift position is therefore treated as an instantaneous function of operating conditions rather than as a dynamic state evolving over time.

A widely cited example is the thesis by Skinner Skinner (2020), which develops a quasi-static model of the flyweight-actuated primary and ramp-actuated secondary and investigates how parameters such as flyweight mass, spring preload, and ramp angle affect engagement speed and operating RPM.

While such models are useful for design tuning and steady-state analysis, they contain no explicit equation of motion for the shift coordinate, no sheave inertia, and no explicit treatment of belt slip. Consequently they cannot simulate transient behavior or predict the dynamic evolution of the transmission ratio.

1.2.5 The Persistent Gap

The literature survey above reveals a consistent pattern. For mechanically actuated dry rubber V-belt CVTs, no publicly available model simultaneously provides

1. an explicit inertial equation of motion for the axial shift coordinate,
2. first-principles modeling of flyweight actuation,
3. a torque-reactive secondary helix force derived from geometry,
4. centrifugal belt mass correction to effective clamping force,
5. an analytical traction-limit slip criterion, and
6. coupling to vehicle longitudinal dynamics.

Metal pushing-belt CVT models such as those of Carbone et al. (2005), Srivastava and Haque (2006), and Bonsen et al. (2004) achieve high levels of dynamic fidelity but describe a different transmission architecture under hydraulic actuation.

Rubber belt models such as Julio and Plante (2011) capture detailed belt mechanics but omit sheave inertia and encounter difficulties during clutching transitions.

Quasi-static Baja models such as Skinner (2020) provide useful design intuition but cannot capture transient behavior.

The absence of a unified dynamic treatment incorporating both mechanical actuation physics and traction-limited torque transmission motivates the development presented in this work.

Reference	Belt	Actuation	Shift ODE	Sheave Inertia	Clutching Mode	Slip Model	Vehicle Dyn.
Carbone et al. (2005)	Metal	Hydraulic	Yes	Yes	No	Yes	No
Srivastava and Haque (2006)	Metal	Hydraulic	Yes	Yes	No	Yes	No
Bonsen et al. (2004)	Metal	Hydraulic	Partial	Yes	No	Yes	No
Julio and Plante (2011)	Rubber	Mechanical	Partial	No	Unstable	Implicit	No
Yi et al. (2011)	Rubber	Mechanical	Partial	Unknown	Unknown	Not explicit	No
Skinner (2020)	Rubber	Mechanical	No	No	No	No	No
Present Work	Rubber	Mechanical	Yes	Yes	Yes	Analytical	Yes

Table 1: Comparison of representative CVT modeling approaches

1.3 Objectives and Scope

This work has two parallel objectives.

The first is technical: to derive a complete, coupled system of nonlinear ordinary differential equations governing the CVT from engine input shaft to vehicle wheel, suitable for direct numerical integration in time.

The second is pedagogical: to make each step of the derivation explicit and physically interpretable, allowing the reader to develop intuition for the mechanisms governing CVT behavior.

The model includes:

- Primary sheave axial shift dynamics
- Secondary helix and spring force model
- Belt geometry and kinematics
- Belt traction and slip-state model
- Rotational equations of motion for both pulleys
- Vehicle longitudinal dynamics

The model intentionally excludes thermal effects, belt wear, detailed tribological contact mechanics, and multi-body vehicle dynamics beyond one-dimensional longitudinal motion.

1.4 Contribution Statement

The primary technical contributions of this work are:

1. A unified nonlinear state-space model for the mechanically actuated rubber V-belt CVT.
2. First-principles derivation of the primary axial shift equation of motion including sheave inertia and centrifugal flyweight actuation.

3. First-principles derivation of the torque-reactive secondary helix force.
4. Derivation of the centrifugal belt mass correction to effective clamping force.
5. An analytical slip-state switching criterion separating no-slip and traction-limited regimes.
6. Full coupling of the CVT dynamics to vehicle longitudinal motion.

1.5 Document Organization

The remainder of this document is organized as follows.

Section 2 introduces the fundamental operating principles of a belt-sheave CVT and describes the specific drivetrain architecture considered in this work.

Section 3 defines notation, coordinate conventions, and symbol definitions used throughout.

Section 4 states the modeling assumptions that scope the analysis.

Section 5 presents the governing physical principles underlying the model.

Section 6 defines the dynamic state variables.

Section 7 through **??** contain the core derivations: pulley geometry (Section 7), axial force models (Section 8), torque transmission mechanics (**??**), traction limits (Section 11), and slip-state formulation (**??**).

Section 13 assembles the complete dynamic system.

Appendices provide supplementary material: belt length approximation validation (Appendix A), common ramp profile designs (Appendix B), numerical methods (Appendix C), and secondary capacity closure (Appendix D).

2 Conceptual Overview of a Continuously Variable Transmission

This section introduces the fundamental operating principles of a continuously variable transmission (CVT) and then presents the specific CVT and drivetrain architecture considered in this work.

The goal is to establish clear mechanical intuition before developing the mathematical model in later sections.

2.1 Geometric Principle of a Belt-Sheave CVT

A belt-type CVT consists of two pulleys connected by a flexible belt. Each pulley is formed by two opposing conical faces (sheaves). The axial separation between these sheaves determines the effective radius at which the belt rides.

When the sheaves move closer together, the belt is forced outward to a larger radius. When the sheaves move apart, the belt settles at a smaller radius. Because the belt length remains approximately constant, an increase in effective radius on one pulley necessarily causes a corresponding decrease on the other.

As a result, the transmission ratio changes continuously as the pulley geometries change. Unlike a traditional gearbox, there are no discrete gear steps; the ratio evolves smoothly between its minimum and maximum values.

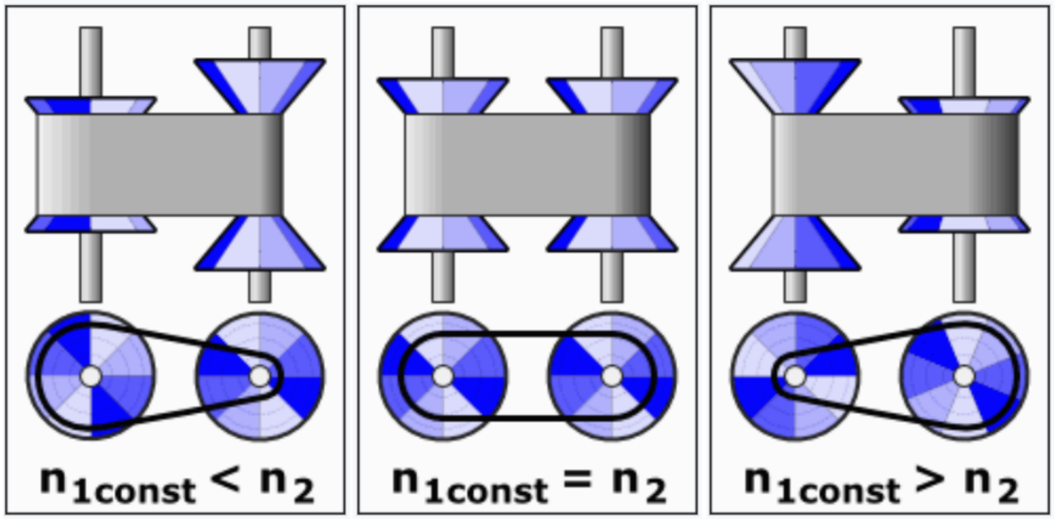

Figure 1: Conceptual illustration of a belt-sheave CVT at three different ratios. Changing the axial separation of the conical sheaves alters the effective belt radius, thereby modifying the transmission ratio continuously.

2.2 Continuous Ratio Versus Discrete Gearing

The defining feature of a CVT is that its transmission ratio can vary smoothly rather than in discrete steps. To understand why this matters, it is useful to first consider how a conventional transmission behaves.

In a stepped gearbox, each gear pair imposes a fixed relationship between engine speed and wheel speed. When vehicle speed increases beyond the useful range of one gear, the transmission must shift to another. This produces an abrupt change in engine speed for the same vehicle speed. As a result, the engine repeatedly accelerates and decelerates as the vehicle progresses through the gears.

A belt-sheave CVT operates differently. Because the effective pulley radii change continuously, the speed relationship between the engine and drivetrain can be adjusted smoothly at any instant. There are no fixed gear pairs that constrain the system to specific ratios. Instead, the ratio can evolve gradually as operating conditions change.

This continuous adjustability allows the engine to operate near a preferred speed while the vehicle accelerates. For example, if an engine produces its peak power within a narrow speed band, a CVT can maintain the engine near that region while vehicle speed increases steadily. Rather than stepping through gears, the transmission adapts its ratio to match the engine's most effective operating condition.

The result is smoother acceleration, improved power utilization, and greater flexibility in matching engine behavior to load demands.

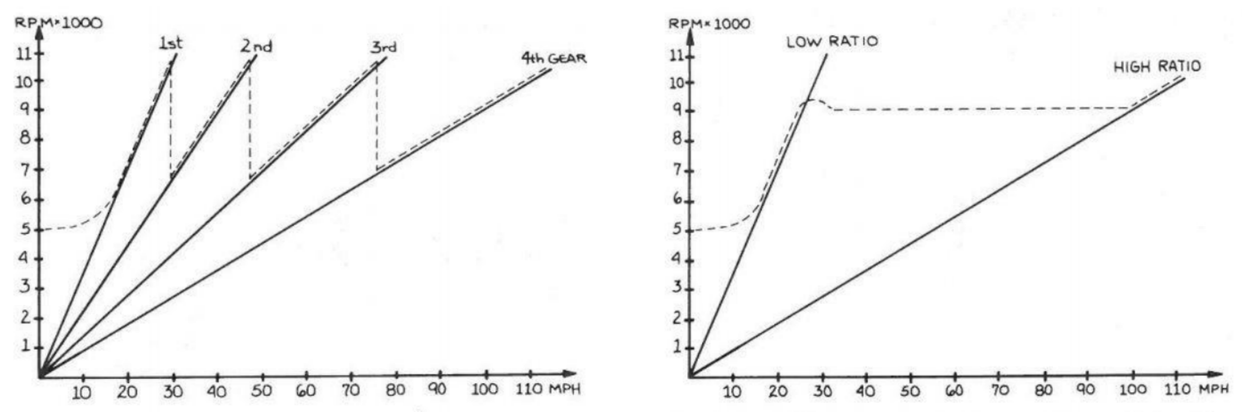

Figure 2: Illustrative comparison of engine speed versus vehicle speed. Left: stepped transmission showing discrete shifts. Right: continuously variable transmission showing smooth ratio evolution.

2.3 Traction-Based Torque Transmission

While the geometric adjustment of pulley radii determines the speed ratio, torque is transmitted through frictional interaction between the belt and the sheave faces.

When a pulley attempts to rotate, it applies a tangential force to the belt. For that force to be transmitted without relative motion, sufficient normal clamping force must press the belt against the conical surfaces. The greater the normal force, the greater the tangential force that can be supported before slip occurs.

This requirement arises directly from the nature of friction. The belt does not contain rigid gear teeth that mechanically lock the pulleys together. Instead, torque transfer depends on traction generated by contact pressure and surface friction. If the transmitted torque exceeds the available traction capacity, the belt will begin to slip relative to the sheave faces.

Slip reduces efficiency, generates heat, and can accelerate component wear. For this reason, CVT designs incorporate mechanisms that actively regulate clamping force in response to operating conditions.

It is therefore useful to distinguish between two interacting aspects of CVT behavior:

- The **geometric ratio**, which determines relative speeds,
- The **traction capacity**, which determines how much torque can be transmitted without slip.

Both must be considered to fully understand performance.

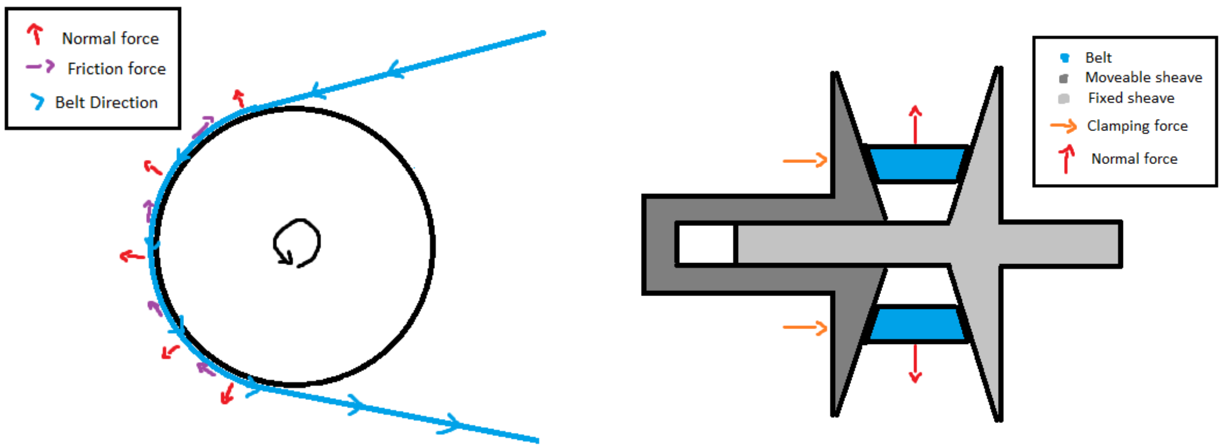

Figure 3: Belt-pulley contact mechanics and force transmission. Left: belt wrapped around a pulley showing normal forces (pressing belt against sheave surface) and tangential friction forces (transmitting torque through traction). Right: cross-sectional view of a CVT pulley assembly showing how axial clamping force between opposing sheaves generates the normal force required for friction and torque transmission.

2.4 Actuation Mechanisms in the Present CVT

While the geometric principles described above apply to all belt-type CVTs, the manner in which axial motion and clamping forces are generated depends on the specific hardware implementation.

The CVT considered in this work employs:

- A **centrifugal flyweight and ramp mechanism** on the primary pulley,
- A **torque-feedback helix mechanism** on the secondary pulley,
- Compression and torsional springs that regulate axial clamping forces.

On the primary side, rotating flyweights move outward under centrifugal effects. Through a ramped interface, this outward motion generates axial force, which adjusts the sheave separation and alters the effective radius.

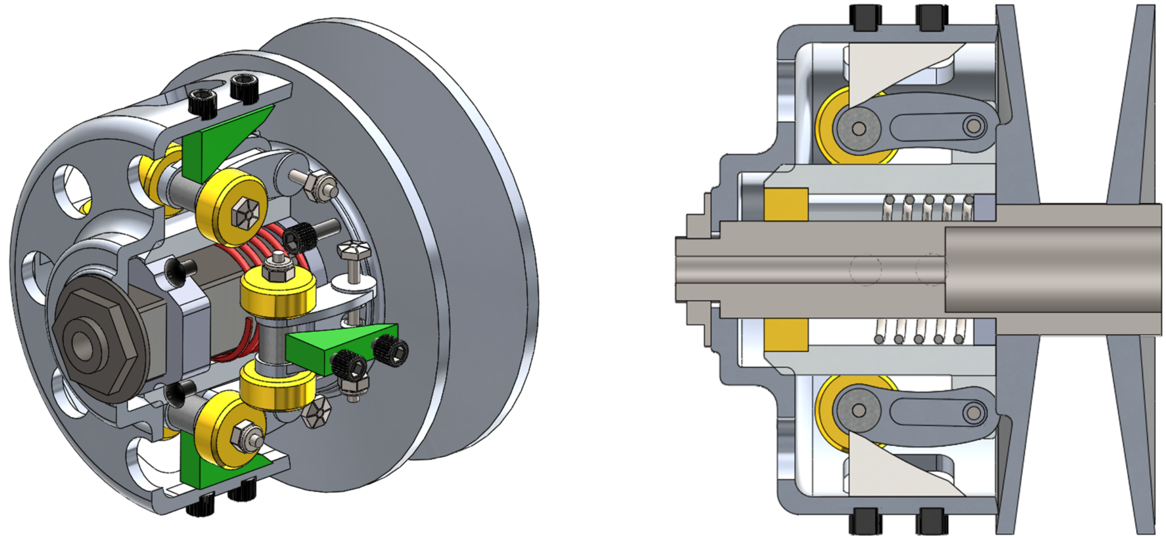

Figure 4: Primary CVT pulley assembly. Left: exterior view. Right: cross-sectional view showing centrifugal flyweights (yellow), ramps (green), and compression spring (red). As engine speed increases, centrifugal force drives the flyweights outward along the ramps, generating axial clamping force.

On the secondary side, transmitted torque produces a rotational reaction within a helical interface. This interaction generates axial force that increases clamping in response to torque demand. Springs supplement this mechanism to regulate baseline clamping and system stability.

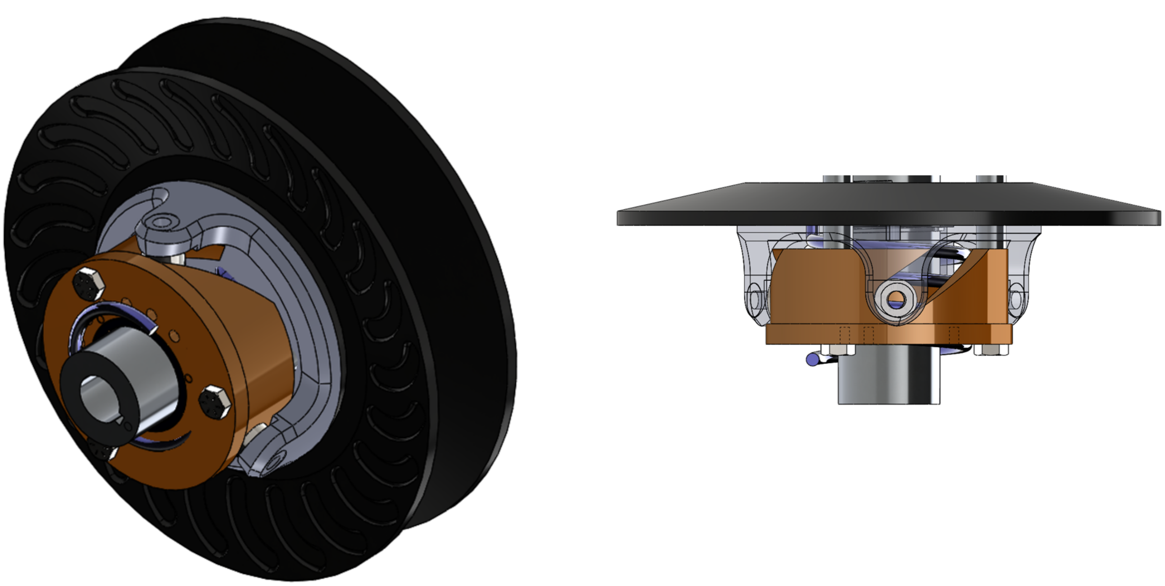

Figure 5: Secondary CVT pulley assembly. Left: exterior view. Right: cross-sectional view showing torque-feedback helix mechanism (bronze) and torsional spring (black). Transmitted torque generates a rotational reaction in the helix, which produces axial clamping force proportional to load.

Although these mechanisms are specific to the hardware used in this system, the underlying principles of variable effective radius and traction-based torque transmission remain common to belt-driven CVTs.

2.5 Full Drivetrain Architecture

The CVT does not operate in isolation. In the system considered here, the drivetrain consists of:

- An internal combustion engine,
- The primary CVT pulley,
- A belt connecting to the secondary CVT pulley,
- A fixed-ratio gearbox,
- Final drive components and driven wheels,
- Tire-ground interaction generating tractive forces.

Engine torque enters the primary pulley, is transmitted through the belt to the secondary pulley, passes through a constant gear reduction, and ultimately produces tractive force at the ground.

The interaction between engine dynamics, CVT ratio evolution, drivetrain inertia, and external load forces governs overall vehicle performance.

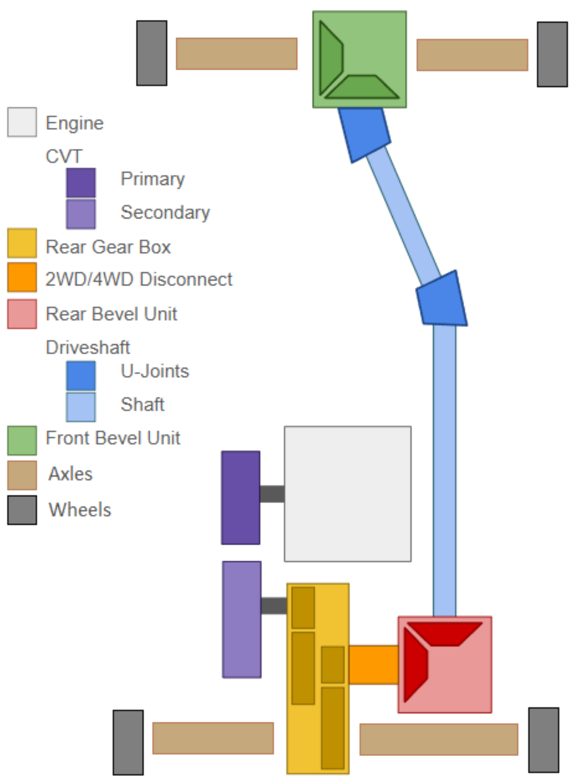

Figure 6: Overview of the complete drivetrain architecture considered in this work.

3 Reference Material

This section records any information for easy reference.

3.1 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions.

3.1.1 Vehicle Longitudinal Coordinate and Incline Angle

Vehicle motion is modeled along a single longitudinal coordinate that follows the surface of the road.

The coordinate x is defined locally along the direction of travel, tangent to the road surface. The positive direction, $+x$, corresponds to the forward direction of vehicle motion.

Because the road may be inclined relative to horizontal, the local direction of the x axis depends on the road slope. The incline angle is denoted by α and is measured between the road surface and a horizontal reference line.

$$\alpha > 0 \quad : \text{uphill in the } +x \text{ direction}$$

$$\alpha < 0 \quad : \text{downhill in the } +x \text{ direction}$$

As illustrated in Figure 7, the x direction always lies tangent to the road surface. Consequently, the orientation of the longitudinal axis varies with the local road incline α.

Vehicle velocity is defined along this coordinate. A velocity $v > 0$ corresponds to motion in the forward $+x$ direction along the road, while $v < 0$ corresponds to motion in the opposite direction.

These definitions establish the geometric orientation of vehicle motion and the sign convention for road incline used throughout the model.

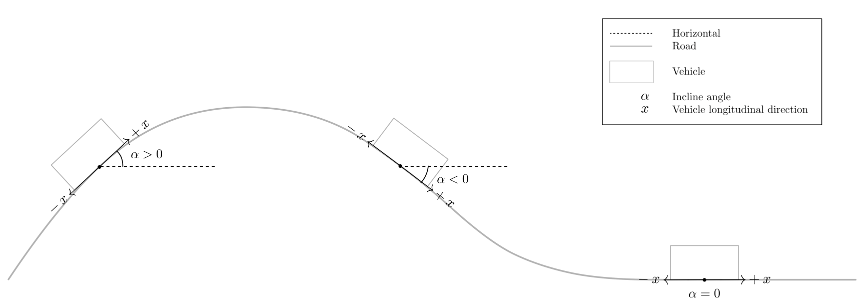

Figure 7: Definition of the longitudinal vehicle coordinate and incline angle. The coordinate x is defined tangent to the road surface and points in the forward direction of travel. The road incline α is measured relative to the horizontal reference line. Positive α corresponds to uphill motion in the $+x$ direction, while negative α corresponds to downhill slope.

3.1.2 Rotational Sign Convention

All rotating components in the drivetrain follow a consistent angular sign convention defined relative to forward vehicle motion.

The positive longitudinal direction $+x$ was defined previously as the forward direction of travel along the road surface. The rotational directions of all drivetrain components are chosen such that positive rotation produces vehicle motion in this $+x$ direction.

As illustrated in Figure 8, positive wheel angular velocity $\omega_w > 0$ corresponds to wheel rotation that drives the vehicle forward along $+x$.

The same convention is propagated upstream through the drivetrain. Positive rotation of the secondary pulley, $\omega_s > 0$, corresponds to the direction of rotation that produces this forward wheel motion through the final reduction gearbox and axle.

Likewise, positive rotation of the primary pulley, $\omega_p > 0$, corresponds to the direction that drives the secondary pulley in this same sense through the CVT belt.

Engine angular velocity follows the same convention. The engine rotates positively when it drives the primary pulley in the defined positive direction.

Torque follows the same sign convention. A torque is considered positive when it acts in the defined positive rotational direction and therefore tends to increase the corresponding angular velocity.

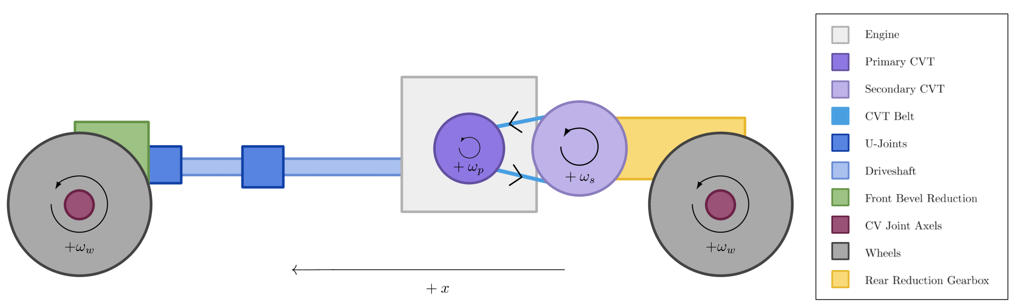

Figure 8: Rotational sign convention for the drivetrain. Positive angular velocities for the wheels (ω_w), secondary pulley (ω_s), and primary pulley (ω_p) are defined such that drivetrain rotation produces forward vehicle motion along the $+x$ direction.

3.1.3 CVT Axial Direction

The axial coordinate of the primary pulley is defined along the axis of the pulley shaft, parallel to the pulley centerline and perpendicular to the plane of belt rotation.

In the reference configuration shown in Figure 9 (left), the primary pulley is fully open. The movable sheave is separated from the fixed sheave by the initial gap, and this configuration defines the origin of the axial coordinate.

The axial variable s measures the translation of the movable sheave relative to this open configuration.

$$s = 0 \quad : \text{fully open configuration}$$

A positive value of s corresponds to the movable sheave translating toward the fixed sheave along the pulley axis.

$$s > 0 \quad : \text{sheave closing motion}$$

As the movable sheave closes, the belt is squeezed between the conical faces and is forced outward along the pulley ramps. This increases the effective belt radius on the primary pulley and corresponds to an upshift of the transmission.

The maximum displacement $s = S_{\max}$ corresponds to the fully closed configuration shown in Figure 9 (right).

Although s is defined using the primary pulley, the motion cannot occur independently. Because the belt length remains constrained, closure of the primary pulley forces the secondary pulley to open. The precise geometric relationship between these motions will be derived later.

Axial velocity follows the same sign convention: $\dot{s} > 0$ corresponds to the movable sheave translating toward the fixed sheave.

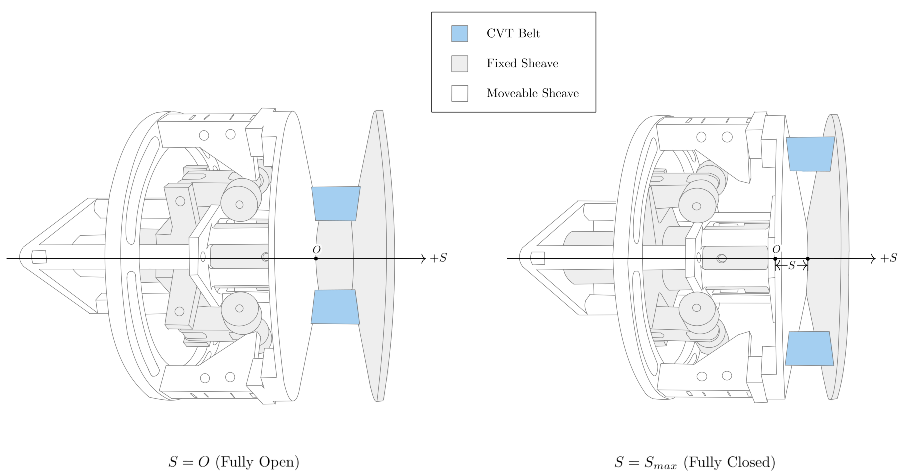

Figure 9: Definition of the primary axial coordinate. The system begins in an open configuration. The axial displacement s measures how much the movable sheave has closed from this initial position. Increasing s increases primary effective radius and is coupled to secondary axial motion.

3.1.4 Radial and Tangential Directions

Within the plane of pulley rotation, directions are defined using the standard polar coordinate convention commonly used in rotational mechanics.

At any point in the plane, two orthogonal directions are defined: the radial direction $\hat{e}_r$ and the tangential direction $\hat{e}_\theta$. These directions form a local coordinate basis centered on the pulley axis.

The radial direction lies in the plane of rotation and points directly outward from the center of the pulley. A radial force is defined as positive when it acts away from the center of rotation. Thus, $F_r > 0$ corresponds to a force directed outward along $\hat{e}_r$.

The tangential direction lies in the plane of rotation and is perpendicular to the radial direction. The positive tangential direction $\hat{e}_\theta$ is defined to be consistent with the positive rotational convention introduced in Section 3.1.2. Tangential motion in this direction corresponds to motion produced by positive angular velocity.

Figure 10 illustrates these directions using physical examples from the primary pulley. The centrifugal flyweights generate outward radial forces F_r, while the belt moves along the tangential direction with velocity v_t. These quantities serve as representative examples of forces and motion that act along the radial and tangential directions.

The previously defined axial direction is perpendicular to this plane of rotation and points along the pulley shaft, into the page.

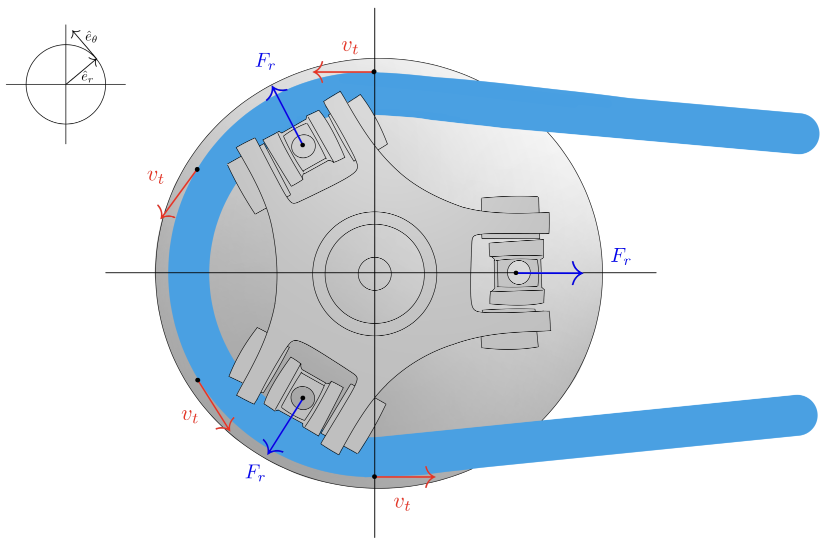

Figure 10: Radial and tangential directions in the plane of pulley rotation. The radial direction $\hat{e}_r$ points outward from the center of the pulley, while the tangential direction $\hat{e}_\theta$ is perpendicular to $\hat{e}_r$ and aligned with positive rotation. Example quantities illustrated include outward radial forces F_r generated by flyweights and tangential belt velocity v_t. For reference, the previously defined axial coordinate s is perpendicular to the page in this view and points along the pulley shaft, into the page.

3.2 Symbol Summary

Symbol	Meaning
State and kinematics	
$\boldsymbol{x}(t)$	state vector
t	time
$\omega_p,\ \omega_s$	primary and secondary angular speed
$\alpha_p,\ \alpha_s$	primary and secondary angular acceleration

Symbol	Meaning
$s,\ \dot{s},\ \ddot{s}$	shift position, velocity, and acceleration
$R,\ \dot{R}$	CVT ratio and ratio rate
v_{rel}	relative slip speed at belt–pulley interface
Geometry	
$r_{p,\text{outer}}(s),\ r_{s,\text{outer}}(s)$	primary and secondary outer radius
$r_{p,\text{eff}}(s),\ r_{s,\text{eff}}(s)$	primary and secondary effective torque radius
$r_{p,\text{outer,min}}$	minimum primary outer radius
$r_{s,\text{outer,max}}$	maximum secondary outer radius
$s_{\text{dz}},\ s_{\max}$	deadzone threshold and maximum shift
β	sheave face half-angle (radial reference)
C	pulley center distance
L_b	total belt length
ϕ	belt wrap angle used in distributed contact model
$r,\ \dot{r}$	local bConceptual Overview of Variable Geoelt path radius and its time derivative
r_{cm}	belt element center-of-mass radius
Constraint and mapping functions	
$f(r_p, r_s, C)$	belt-length compatibility root function ($f = 0$)
$\text{root}_{r_s}(\cdot)$	numerical root solve for secondary radius
$r_f(s),\ \dfrac{dr_f}{ds}$	primary flyweight radius map and gradient
$\theta_s(s),\ \dfrac{d\theta_s}{ds}$	secondary helix angle map and gradient
Forces and torques	
$F_{p,\text{ax}},\ F_{s,\text{ax}}$	primary and secondary axial mechanism force
$F_{c,p},\ F_{c,s}$	primary and secondary belt centrifugal axial contribution
F_{ax}	generic axial clamp force at one pulley contact
$\tau_{\text{eng}},\ \tau_{\text{load}}$	engine input torque and reflected load torque
$\tau_p,\ \tau_s$	coupling torque seen at primary and secondary
$\tau,\ \tau_c$	generic interface torque and applied coupling/friction torque
τ_{ns}	no-slip requested coupling torque
$\tau_{\max}$	traction-limited torque capacity
Mass/inertia and material parameters	
$I_p,\ I_s$	primary and secondary rotational inertia
m_f	effective flyweight mass
$m_{p,m},\ m_{s,m}$	equivalent primary and secondary translating mechanism mass
ρ_b	belt material density
A_b	belt cross-sectional area
μ	belt–pulley traction coefficient
$k_p,\ x_{p,0}$	primary spring stiffness and preload displacement
$k_{s,\theta},\ \theta_{s,0}$	secondary torsional stiffness and preload angle
$k_{s,x},\ x_{s,0}$	secondary axial stiffness and preload displacement

Table 2: Core symbols used throughout the derivation.

3.3 Base Units

All quantities in this document are expressed in SI units unless otherwise stated.

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Plane angle	radian	rad

Table 3: Base SI units used by the model variables and parameters.

3.4 Common Derived Units

Quantity	Unit	Symbol
Velocity	meter per second	m/s
Acceleration	meter per second squared	$\mathrm{m/s^2}$
Angular velocity	radian per second	rad/s
Angular acceleration	radian per second squared	$\mathrm{rad/s^2}$
Area	square meter	$\mathrm{m^2}$
Volume	cubic meter	$\mathrm{m^3}$
Density	kilogram per cubic meter	$\mathrm{kg/m^3}$
Linear spring stiffness	newton per meter	N/m
Torsional spring stiffness	newton meter per radian	N m/rad
Force	newton ($\mathrm{kg\,m/s^2}$)	N
Torque	newton meter ($\mathrm{kg\,m^2/s^2}$)	N m
Moment of inertia	kilogram meter squared	$\mathrm{kg\,m^2}$
Power	watt ($\mathrm{kg\,m^2/s^3}$)	W
Dimensionless ratios/coefs.	unitless	1

Table 4: Derived units and unitless groups used throughout the CVT model.

3.5 Constants and Parameters

All non-varying model parameters are centralized below as the canonical reference. Unless explicitly treated as inputs or state-dependent maps, symbols listed here are held constant during simulation.

Symbol	Name	Nominal value	Unit	Description
Geometry and belt				
L_b	Belt length	–	m	Fixed total belt length
C	Center distance	–	m	Primary–secondary shaft center distance (fixed after initialization)
β	Sheave half-angle	–	rad	Cone half-angle used for radial/axial conversion
h_b	Belt thickness	–	m	Radial belt thickness
b_{out}	Belt outer width	–	m	Belt width at outer face
b_{in}	Belt inner width	–	m	Belt width at inner face
A_b	Belt cross-sectional area	–	$\mathrm{m^2}$	Trapezoid area from h_b, b_{out}, b_{in}
$r_{p,\mathrm{outer,min}}$	Primary minimum outer radius	–	m	Radius at fully open primary

Symbol	Name	Nominal value	Unit	Description
$r_{s,\text{outer,max}}$	Secondary maximum outer radius	–	m	Radius at fully closed secondary
s_{dz}	Shift deadzone threshold	–	m	Onset of effective radial motion
$s_{\max}$	Maximum shift	–	m	Physical upper travel bound
Primary/secondary actuation				
m_f	Flyweight mass	–	kg	Effective flyweight mass
$r_{f,0}$	Initial flyweight radius	–	m	Flyweight radius offset at $s = 0$
k_p	Primary spring stiffness	–	N/m	Primary compression spring rate
$x_{p,0}$	Primary spring preload	–	m	Initial compression of primary spring
$k_{s,\theta}$	Secondary torsional stiffness	–	N m/rad	Torsional spring rate in helix assembly
$\theta_{s,0}$	Secondary torsional preload	–	rad	Initial torsional deflection preload
r_h	Helix reference radius	–	m	Effective cylindrical radius in helix kinematics
$k_{s,x}$	Secondary axial stiffness	–	N/m	Secondary axial compression spring rate
$x_{s,0}$	Secondary axial preload	–	m	Initial compression of secondary axial spring
Inertia and drivetrain				
I_{eng}	Engine inertia	–	kg m^2	Engine rotational inertia
I_{prim}	Primary pulley inertia	–	kg m^2	Primary CVT pulley inertia
I_p	Effective primary inertia	–	kg m^2	$I_p = I_{\text{eng}} + I_{\text{prim}}$
I_{sec}	Secondary pulley inertia	–	kg m^2	Secondary CVT pulley inertia
I_{gb}	Gearbox inertia	–	kg m^2	Gearbox/intermediate shaft inertia
I_{wheel}	Wheel inertia	–	kg m^2	Wheel/hub inertia about wheel axis
I_s	Effective secondary inertia	–	kg m^2	Lumped secondary-side inertia about secondary axis
G	Final drive ratio	–	1	Fixed ratio defined by $\omega_s = G\,\omega_w$
m	Vehicle mass	–	kg	Total vehicle mass
r_w	Wheel rolling radius	–	m	Effective tire rolling radius
Contact and environment				
ρ_b	Belt density	–	kg/m^3	Belt material density
μ	Belt–pulley friction coefficient	–	1	Effective Coulomb traction coefficient
C_{rr}	Rolling resistance coefficient	–	1	Tire-road rolling resistance coefficient
C_d	Drag coefficient	–	1	Aerodynamic drag coefficient
A_f	Frontal area	–	m^2	Vehicle aerodynamic reference area
ρ	Air density	–	kg/m^3	Ambient air density used in drag model

g	Gravitational acceleration	–	m/s^2	Gravity constant

Table 5: Canonical non-varying parameters used by the CVT model equations.

4 Modeling Assumptions

The following assumptions define the modeling scope and physical idealizations used in the CVT dynamic formulation. These assumptions establish the limits within which the derived equations remain valid.

4.1 Material and Structural Idealizations

Assumption 1 (Constant Material Properties)

Material properties of structural components are treated as constant. No temperature-dependent material variation is modeled. Thermal effects are not solved as part of this system.

Assumption 2 (Rigid Structural Components)

All structural components (pulleys, shafts, housing) are modeled as rigid bodies. Elastic deformation under operational loading is neglected. The belt is assumed axially rigid (no longitudinal stretch), while still conforming geometrically around pulleys.

Assumption 3 (Negligible Wear and Surface Evolution)

Component wear, belt degradation, and pulley surface evolution are neglected. The model represents short-term operational behavior.

4.2 Contact and Friction Modeling

Assumption 4 (Idealized Belt–Pulley Friction Law)

Belt traction is modeled using a simplified friction law (e.g., Coulomb friction with a capstan formulation and an explicit stick–slip transition rule). Microscopic contact mechanics, rubber hysteresis, lubrication films, and detailed Stribeck effects are not explicitly modeled.

Assumption 5 (Constant Effective Friction Coefficient)

The belt–pulley coefficient of friction is treated as constant. It does not evolve with temperature, wear, or surface conditioning.

Assumption 6 (Uniform Contact Distribution)

Contact pressure and traction over the pulley wrap angle are treated as effectively uniform or averaged. Localized edge loading and non-uniform stress distributions are neglected.

4.3 Geometric and Kinematic Idealizations

Assumption 7 (Perfect Axis Alignment)

Primary and secondary pulley axes are assumed perfectly aligned. Misalignment, shaft eccentricity, and geometric runout are neglected.

Assumption 8 (Constant Belt Length)

The belt length is treated as constant. Longitudinal elasticity, transient stretch, permanent creep, and progressive elongation are not modeled. Effective pulley radii are determined solely from geometry and belt-length constraints.

Assumption 9 (Uniform Contact Radius Around Wrap)

At any given instant, the belt contacts each pulley at a single, uniform effective radius throughout the entire wrap angle. The contact radius is constant around the circumference of contact, with no spiral or progressive radial variation along the wrap. While the effective radius may evolve with axial shift position, it remains spatially uniform at each time instant.

4.4 Dynamic and Inertial Modeling

Assumption 10 (Lumped Inertia Representation)

Rotational inertias of the engine, pulleys, drivetrain, and wheels are represented as lumped equivalent inertias. Distributed shaft flex and torsional compliance are neglected.

Assumption 11 (Negligible Vibrational Effects)

High-frequency vibrational effects, structural resonances, and drivetrain oscillations are neglected. The model captures dominant rigid-body dynamics only.

Assumption 12 (Negligible Parasitic Friction)

Frictional losses in non-critical components (e.g., bearings, seals, auxiliary shafts) are neglected. Only friction directly relevant to belt traction and CVT torque transfer is modeled.

Assumption 13 (Ideal CVT Power Transmission)

Power transmission through the CVT is modeled as ideal (i.e., no internal dissipation). Although belt slip is explicitly modeled and affects torque capacity and shift dynamics, the associated power loss P_{loss} is neglected.

Assumption 14 (Single Active Slip Interface)

At any instant, the belt is assumed to either adhere to both pulleys or slip at exactly one pulley contact. Simultaneous slip at both pulley contacts is not modeled.
If operating conditions would otherwise cause both contacts to exceed their traction limits, the model assumes an instantaneous transition between single-slip states.

4.5 Environmental and Operational Scope

Assumption 15 (Negligible Gravitational Influence on CVT Internals)

Gravitational forces acting on CVT internal components are negligible relative to centrifugal and contact forces at operating speeds.

Assumption 16 (Full-Throttle Engine Model)

The engine is modeled using a prescribed full-throttle torque curve. Load feedback modifies engine speed dynamically, but part-throttle dynamics and throttle transients are not explicitly modeled.

5 Governing Principles

Theory 1: Newton's Second Law (Translation)

Equation

$$\sum F = ma$$

Description

- $\sum F$ is the net force (N)
- m is mass (kg)
- a is acceleration (m/s^2)

Source

Classical mechanics (textbook / lecture notes).

Derivation

From the definition of acceleration and empirical laws of motion; used here as a governing relationship.

Theory 2: Newton's Second Law (Rotation)

Equation

$$\sum \tau = I\alpha$$

Description

- $\sum \tau$ is net torque (N m)
- I is moment of inertia (kg m^2)
- α is angular acceleration (rad/s^2)

Source

Classical rigid body dynamics (textbook / lecture notes).

Derivation

Standard rigid-body rotational dynamics relationship.

Theory 3: Torque Definition

Equation

$$\tau = Fr$$

Description

- τ is torque (N m)
- F is tangential force (N)
- r is moment arm radius (m)

Source

Classical mechanics (textbook / lecture notes).

Derivation

Defines torque as the product of tangential force and moment arm. Equivalently, $F = \tau/r$ gives the tangential force from a known torque.

Theory 4: Kinetic Energy (Translational)

Equation

$$T = \frac{1}{2}mv^2$$

Description

- T is kinetic energy (J)
- m is mass (kg)
- v is velocity (m/s)

Source

Classical mechanics (textbook / lecture notes).

Derivation

Translational kinetic energy of a point mass or rigid body moving with velocity v.

Theory 5: Centrifugal Force

Equation

$$F_c = m\omega^2 r$$

Description

- F_c is centrifugal force (N)
- m is mass (kg)
- ω is angular velocity (rad/s)
- r is radius from the axis of rotation (m)

Source

Bogna Szyk and Steven Wooding (2024).

Derivation

Derived from circular motion kinematics where $a_c = \omega^2 r$, then applying $F = ma$.

Theory 6: Hooke's Law (Compressional)

Equation

$$F = k\Delta x$$

Description

- F is force (N)
- k is spring constant (N/m)
- Δx is displacement (m)

Source

Encyclopaedia Britannica (2024).

Derivation

Linear spring constitutive relationship in the elastic regime.

Theory 7: Hooke's Law (Torsional)

Equation

$$F = \frac{k\Delta\theta}{r}$$

Description

- F is force (N)
- k is torsional spring constant (N m/rad)
- $\Delta\theta$ is angular deflection (rad)
- r is torque arm radius (m)

Source

Dr. Templeman (2024).

Derivation

Starting from $\tau = k\Delta\theta$ and $\tau = Fr$, then $F = \frac{k\Delta\theta}{r}$.

Theory 8: Engine Torque from Power

Equation

$$\tau = \frac{P}{\omega}$$

Description

- τ is torque (N m)
- P is power (W)
- ω is angular velocity (rad/s)

Source

2024 Power Test, LLC (2024).

Derivation

From the rotational power relationship $P = \tau\omega$.

Theory 9: Gearing (Torque Scaling)

Equation

$$\tau_{\text{out}} = \tau_{\text{in}}\, R_{\text{gear}}$$

Description

- τ_{out} is output torque (N m)
- τ_{in} is input torque (N m)
- R_{gear} is gear reduction ratio (unitless)

Source

Brain (2023).

Derivation

Ideal gear relationship assuming no losses; power balance implies inverse scaling of speed.

Theory 10: Coulomb Friction

Equation

$$F_f = \mu_s F_n$$

Description

- F_f is friction force (N)
- μ_s is coefficient of static friction (unitless)
- F_n is normal force (N)

Source

Wikipedia contributors (2024).

Derivation

Coulomb friction model for impending slip.

Theory 11: Capstan Equation

Equation

$$T_1 = T_2 e^{\mu\theta}$$

Description

- T_1 is tight-side belt tension (N)
- T_2 is slack-side belt tension (N)
- μ is belt–pulley friction coefficient (unitless)
- θ is wrap angle (rad)

Source

Hackaday (2021).

Derivation

Differential force balance on an infinitesimal belt element on a pulley yields $\frac{\mathrm{d}T}{T} = \mu \, \mathrm{d}\theta$, then integrate over wrap.

Theory 12: Density

Equation

$$\rho = \frac{m}{V}$$

Description

- ρ is density ($\mathrm{kg/m^3}$)
- m is mass (kg)
- V is volume ($\mathrm{m^3}$)

Source

The Editors of Encyclopaedia Britannica (2024).

Derivation

Definition of density.

Theory 13: Air Resistance (Quadratic Drag)

Equation

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

Description

- F_d is drag force (N)
- ρ is fluid density (kg/m^3)
- v is object speed relative to fluid (m/s)
- C_d is drag coefficient (unitless)
- A is reference area (m^2)

Source

SoftSchools (2020).

Derivation

Standard empirical/derived quadratic drag model; applicable at moderate to high Reynolds number.

Theory 14: Rolling Resistance

Equation

$$F_r = C_r N$$

Description

- F_r is rolling resistance force (N)
- C_r is coefficient of rolling resistance (unitless)
- N is normal force (N)

Source

?.

Derivation

Rolling resistance arises from deformation of contact surfaces (tire and road). For a vehicle on level ground, $N = mg$ where m is vehicle mass and g is gravitational acceleration.

Theory 15: Gravitational Force

Equation

$$F_g = mg$$

Description

- F_g is gravitational force (N)
- m is mass (kg)
- g is gravitational acceleration (m/s^2)

Source

?.

Derivation

Near Earth's surface, $g \approx 9.81$ m/s^2. On an incline with angle θ, the component parallel to the surface is $F_g \sin\theta$.

6 System State Definition

The dynamic behavior of the CVT system is described using a set of *state variables*. A state variable is a quantity whose future evolution is determined by its current value and the governing differential equations of the system.

In numerical simulation, state variables are the quantities that are explicitly integrated forward in time. All other model quantities — such as torque transfer, transmission ratio, belt forces, and contact conditions — are computed algebraically from these states.

6.1 State Vector

State Vector **(6.1)**

$$\boldsymbol{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ \hat{v}_b \end{bmatrix}$$

where:

- ω_p is the angular speed of the primary pulley (rad/s),
- ω_s is the angular speed of the secondary pulley (rad/s),
- s is the axial shift position of the primary sheave (m),
- $\dot{s}$ is the axial shift velocity (m/s).
- $\hat{v}_b$ is the modeled belt transport speed (m/s).

6.2 Independent Dynamic Degrees of Freedom

These five quantities represent the independent dynamic motions of the CVT system.

The primary pulley can rotate independently under engine input and belt interaction. Its angular speed ω_p therefore represents one rotational degree of freedom.

The secondary pulley also rotates and interacts with both the belt and the external drivetrain load. Its angular speed ω_s represents a second independent rotational degree of freedom.

In addition to rotation, the CVT includes axial motion of the primary sheave. This axial displacement s determines the effective pulley geometry and therefore the transmission ratio. The axial motion is mechanically independent of pure rotation and must be modeled explicitly.

Because axial motion follows second-order dynamics, both the axial position s and its rate $\dot{s}$ are required to fully describe its evolution.

Finally, the belt itself moves through the pulley system and carries power between the two pulleys. The quantity $\hat{v}_b$ describes the transport speed of the belt and is included as a state variable so that this belt motion can be represented directly within the dynamic model.

Together, these variables capture the complete rotational and axial motion of the system. Removing any one of them would eliminate an independent dynamic degree of freedom from the model.

6.3 Time Integration Perspective

During numerical simulation, the state vector is advanced forward in time.

At each timestep Δt:

- The angular speeds ω_p and ω_s are updated according to their angular accelerations.
- The axial position s is updated using its current velocity $\dot{s}$.
- The axial velocity $\dot{s}$ is updated according to the axial acceleration.
- The modeled belt transport speed $\hat{v}_b$ is updated according to its governing evolution law.

The angular and axial accelerations are determined from the governing equations derived in subsequent sections. Once the state vector has been updated, all other quantities in the model are computed directly from the current values of ω_p, ω_s, s, $\dot{s}$, and $\hat{v}_b$.

7 Pulley Geometry

This section establishes the geometric relationships that govern how axial shift modifies the effective pulley radii and therefore the transmission ratio.

The analysis presented here is purely kinematic. No force balances or dynamic effects are considered. The goal is to describe how pulley geometry alone determines belt position and ratio evolution.

The objective is to derive explicit expressions for:

- Primary outer radius as a function of shift,
- Secondary outer radius as a function of shift,
- Center-to-center distance,
- CVT ratio and its rate of change.

Radius Definitions Used in This Section Because the belt has finite radial height, two geometric radii are used here: the *outer radius* (belt outer surface/contact geometry) and the *inner radius* (belt inner surface). Let h_b denote the belt height (belt radial thickness). Then,

$$r_{\text{inner}} = r_{\text{outer}} - h_b,$$

The effective torque radius used in torque and ratio relations is introduced later in Section 7.8.

7.1 Conceptual Overview of Variable Geometry

As introduced in Section 2, each pulley consists of two opposing conical sheaves. One sheave is fixed axially, while the other translates along the shaft.

The key controllable motion in the system is this *axial separation* between the sheaves. When the movable sheave translates toward the fixed sheave, the gap narrows and the belt is forced outward along the conical faces. When the sheaves separate, the belt settles deeper between them.

This axial motion does not directly change the pulley radius. Instead, the effective radius arises geometrically from where the belt sits within the sheave gap. In other words, axial displacement determines the belt's radial position, and therefore the effective pulley radius. The precise relationship between these quantities will be derived in the following subsection.

Because the belt length remains constrained, the effective radii of the primary and secondary pulleys cannot vary independently. An increase in the primary effective radius must therefore be accompanied by a corresponding decrease in the secondary effective radius. This geometric coupling is fundamental to CVT operation and forms the basis of the ratio model derived below.

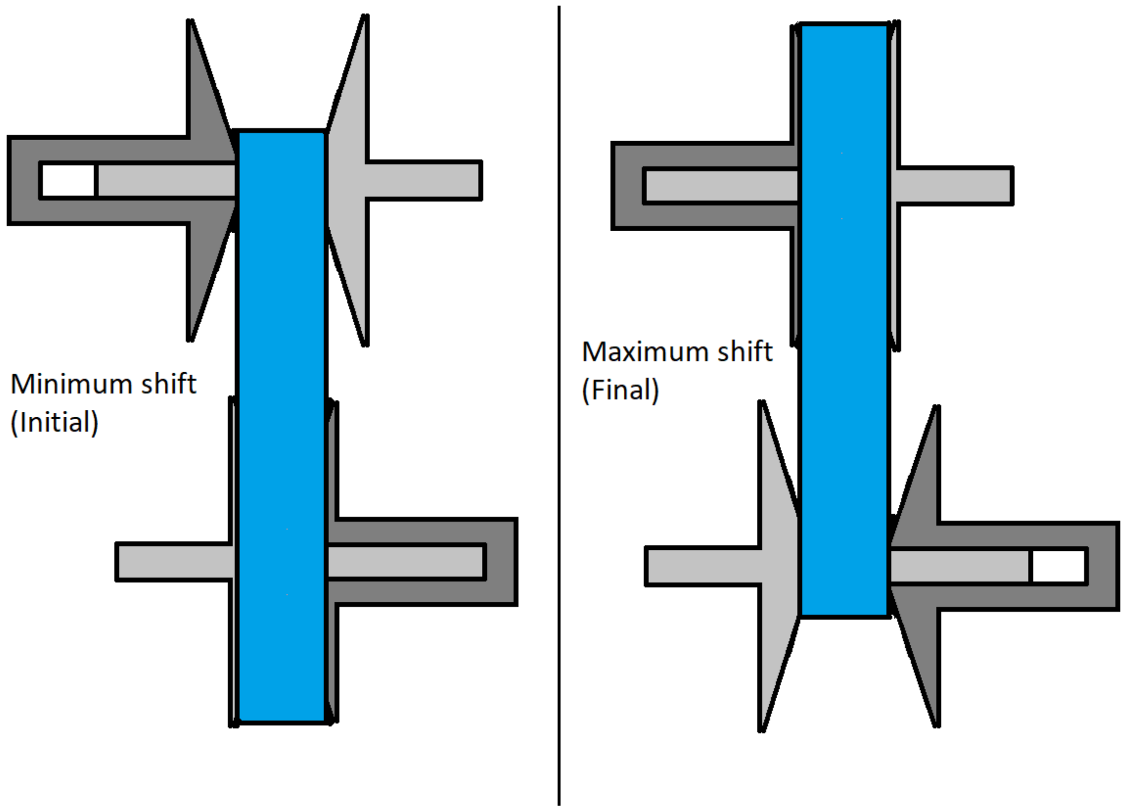

Figure 11: Conceptual illustration of variable pulley geometry. Changes in axial separation alter effective radii. Because belt length is constrained, an increase in primary effective radius corresponds to a decrease in secondary effective radius.

7.2 Definition of Axial Shift Variable

Let s denote the axial displacement of the movable primary sheave, defined according to the sign convention in Section 3.1.3.

At $s = 0$, the primary pulley is in its fully open configuration. The sheaves are at their maximum axial separation, and the belt rests at its minimum effective radius.

The shift coordinate s measures the amount of closure relative to this initial configuration. Increasing s corresponds to the movable sheave translating toward the fixed sheave, reducing the axial gap.

Deadzone and Travel Limits In practice, the initial sheave gap at $s = 0$ is larger than the belt width. As a result, small axial motion does not immediately alter the belt's contact position on the conical faces.

The interval

$$0 \leq s < s_{\mathrm{dz}}$$

defines this *deadzone*. Within this region, the movable sheave translates axially but does not yet sufficiently compress the belt to change its effective contact radius. The belt remains at the same radial position despite the reduction in gap.

Once the sheave gap becomes small enough to fully engage the belt, further axial motion alters the belt position. For

$$s_{\mathrm{dz}} \leq s \leq s_{\max},$$

additional closure forces the belt outward along the conical faces, producing a proportional change in primary effective radius.

The shift coordinate is therefore bounded as

$$0 \leq s \leq s_{\max}.$$

The upper limit $s_{\max}$ corresponds to the physical travel limit of the sheave assembly. At this position, the sheaves are fully closed and mechanically contact one another, forming a hard geometric boundary beyond which further axial motion is not possible.

7.3 Primary Outer Radius as a Function of Shift

When the primary sheaves close beyond the deadzone, the belt is forced outward along the conical faces. The relationship between axial closure and radial expansion follows directly from the cone geometry.

As shown in Figure 12, the sheave face forms a straight line in cross-section. The cone half-angle β is defined as the angle between the sheave face and the *radial direction* (As defined in Section 3.1.4).

For a small axial displacement Δs beyond the deadzone, the total closure is shared symmetrically between the two opposing sheaves, so each sheave contributes $\Delta s/2$.

From the right triangle formed by the sheave face,

$$\tan\beta = \frac{\Delta s/2}{\Delta r},$$

which gives

$$\Delta r = \frac{\Delta s}{2\tan\beta}. \tag{7.1}$$

Thus, the cone angle determines how axial motion is converted into radial growth: smaller β produces greater radial change for the same axial displacement.

Taking the infinitesimal limit of Equation (7.1) yields the local geometric relationship between axial shift and effective radius,

$$\frac{dr}{ds} = \frac{1}{2\tan\beta}. \tag{7.2}$$

Because the sheave face is straight, this derivative is constant, meaning that radial position varies linearly with axial displacement.

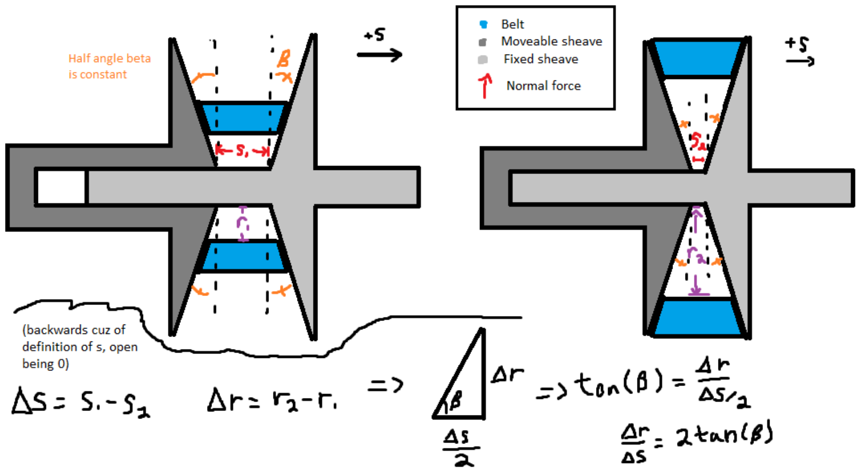

Figure 12: Axial–radial relationship for opposing conical sheaves. The angle β is defined between the sheave face and the radial direction.

Piecewise Radius Model Let $r_{p,\text{outer,min}}$ denote the minimum primary outer radius, corresponding to the fully open configuration ($s = 0$).

Within the deadzone, axial motion does not alter belt position. Once $s \geq s_{\text{dz}}$, only the displacement *beyond the deadzone* contributes to radial growth.

Primary Outer Radius **(7.3)**

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\text{outer,min}}, & 0 \leq s < s_{\text{dz}}, \\ r_{p,\text{outer,min}} + \dfrac{s - s_{\text{dz}}}{2\tan\beta}, & s_{\text{dz}} \leq s \leq s_{\max}. \end{cases}$$

7.4 Belt Length Constraint

In the previous subsection, the primary effective radius was expressed as a function of axial shift. The secondary radius, however, cannot be chosen independently. The belt has a fixed total length, and therefore any change in primary radius must be accompanied by a corresponding geometric adjustment of the secondary radius.

This section derives the geometric compatibility constraint imposed by constant belt length. By expressing the total belt length in terms of the primary radius r_p, secondary radius r_s, and center-to-center distance C, we obtain an implicit relationship that couples the two pulley radii.

The derivation follows directly from the belt path geometry shown in Figure 13. The geometric quantities used in the derivation are

- r_p — radius of the primary pulley
- r_s — radius of the secondary pulley
- C — center-to-center distance between pulley shafts
- ϕ_p — belt wrap angle on the primary pulley
- ϕ_s — belt wrap angle on the secondary pulley

- ℓ — length of a single straight tangent span

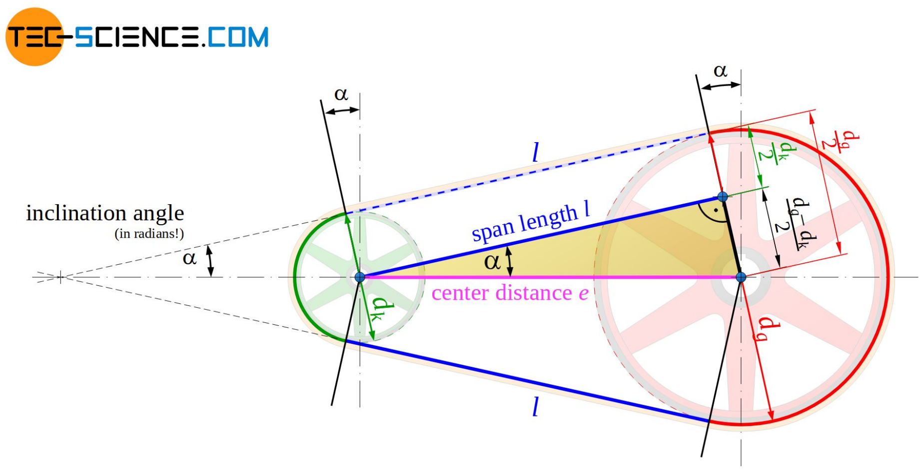

Figure 13: Belt geometry for two pulleys of radii r_p and r_s separated by center distance C. The belt wraps the pulleys with angles ϕ_p and ϕ_s and connects via two straight tangent spans ℓ.

Belt Length Decomposition The total belt length L_b is the sum of: (i) wrapped arc length on the primary pulley, (ii) wrapped arc length on the secondary pulley, and (iii) two straight tangent spans. Therefore,

$$L_b = r_p\phi_p + r_s\phi_s + 2\ell, \tag{7.4}$$

where ℓ is the length of a single straight tangent span segment.

Straight Span Length From the right-triangle geometry formed by the pulley centers and tangent points,

$$\ell = \sqrt{C^2 - (r_s - r_p)^2}. \tag{7.5}$$

Wrap Angles The wrap angles on the primary and secondary pulleys are

$$\phi_p = \pi - 2\lambda, \tag{7.6}$$
$$\phi_s = \pi + 2\lambda. \tag{7.7}$$

If $\lambda < 0$, these expressions automatically redistribute wrap between pulleys while preserving total belt length.

Tangency Angle (Signed Convention) Let λ denote the signed angle between the line of centers and the straight tangent belt segment (as defined in Figure 13). We define

$$\sin\lambda = \frac{r_s - r_p}{C}, \tag{7.8}$$

and equivalently,

$$\lambda = \sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.9}$$

Under this convention, λ may be negative when $r_p > r_s$. This signed definition ensures the wrap distribution automatically adjusts when the pulley ordering reverses.

Closed Form Belt-Length Expression Substituting Equation (7.5) and Equation (7.6)–Equation (7.7) into Equation (7.4) yields

$$\begin{aligned} L_b &= r_p(\pi - 2\lambda) + r_s(\pi + 2\lambda) + 2\sqrt{C^2 - (r_s - r_p)^2} \\ &= \pi(r_p + r_s) + 2\lambda(r_s - r_p) + 2\sqrt{C^2 - (r_s - r_p)^2}. \end{aligned} \tag{7.10}$$

Finally, substituting Equation (7.9) gives

$$L_b = \pi(r_p + r_s) + 2(r_s - r_p)\,\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{7.11}$$

Root Form (Geometric Compatibility Constraint) Rearranging Equation (7.11) yields

Belt Length Constraint Root Function **(7.12)**

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\,\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b.$$

The belt-length constraint is therefore

$$f(r_p, r_s, C) = 0. \tag{7.13}$$

The form of Equation (7.13) is unchanged if radii are defined at the belt outer surface, inner surface, or effective mid-height radius, provided all terms are referenced consistently. In particular, L_b must correspond to the same reference as r_p and r_s (e.g., outer-length with outer radii, inner-length with inner radii, or effective-length with effective radii).

Remark (Transcendental Nature and Approximation)

Equation (7.12) is transcendental due to the inverse-sine and square-root terms and therefore does not admit a closed-form solution.
In this work, the constraint is solved numerically using a bracketed root-finding method (e.g., bisection or Brent's method), which is deterministic and robust over the physically admissible interval.
It is common in simplified treatments to assume

$$\frac{r_s - r_p}{C} \approx 0,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this small-angle assumption, the belt-length equation reduces to a much simpler approximate form.
However, this approximation can introduce substantial geometric error when $\frac{r_s - r_p}{C}$ is not small. Appendix A quantifies this deviation and demonstrates that the approximation is not acceptable under realistic operating conditions.

7.5 Center-to-Center Distance Determination

In-order to make use of the belt-length constraint derived in Section 7.4, the center-to-center distance C must be determined.

The center-to-center distance C is a fixed geometric parameter of the assembled CVT system. Once the transmission is constructed, C does not vary during operation.

To determine C, we evaluate the belt-length constraint Equation (7.13) at the *known* initial configuration of the system.

Initial Geometric Configuration At startup, the CVT is assumed to be in its lowest ratio configuration:

- The primary pulley is at its minimum outer radius,

$$r_p = r_{p,\text{outer,min}}.$$

- The secondary pulley is at its maximum outer radius,

$$r_s = r_{s,\text{outer,max}}.$$

These values are determined by the physical geometry of the sheaves and represent the fully separated primary and fully closed secondary configuration.

Center Distance from Belt-Length Compatibility Substituting these known radii into Equation (7.12) yields

$$f(r_{p,\text{outer,min}}, r_{s,\text{outer,max}}, C) = 0. \tag{7.14}$$

This equation contains a single unknown, C, and is solved numerically to determine the fixed center-to-center distance consistent with the specified belt length L_b.

$$C = \underset{C}{\mathrm{root}}\; f(r_{p,\text{outer,min}}, r_{s,\text{outer,max}}, C)\,. \tag{7.15}$$

Remark (One-Time Geometric Solve)

The center distance C is determined once during model initialization. Thereafter, C is treated as a constant in all subsequent geometric and dynamic calculations.

7.6 Secondary Outer Radius as a Function of Shift

With the center-to-center distance C determined from Section 7.5, and the primary radius $r_{p,\text{outer}}(s)$ defined by Equation (7.3), the secondary outer radius is obtained by enforcing the belt-length compatibility constraint.

Geometric Coupling At any shift position s, the primary radius $r_{p,\text{outer}}(s)$ is known. The secondary radius r_s must therefore satisfy

$$f(r_{p,\text{outer}}(s), r_s, C) = 0, \tag{7.16}$$

where $f(\cdot)$ is defined in Equation (7.12).

Numerical Determination of $r_s(s)$ Because Equation (7.16) is transcendental in r_s, the secondary radius is obtained numerically for each value of s:

Secondary Outer Radius **(7.17)**

$$r_{s,\text{outer}}(s) = \underset{r_s}{\mathrm{root}}\; f(r_{p,\text{outer}}(s), r_s, C)\,.$$

Remark (Dynamic Root Solve)

Unlike the center-distance calculation in Section 7.5, which is performed once during initialization, this root solve is performed repeatedly as the shift coordinate s evolves during simulation. Because the physically admissible interval for r_s is bounded, a bracketed solver (e.g., Brent's method) converges rapidly.

With both $r_{p,\text{outer}}(s)$ and $r_{s,\text{outer}}(s)$ determined, all geometric quantities describing the belt path can now be expressed explicitly as functions of the shift coordinate s.

7.7 Wrap Angles as Functions of Shift

The wrap angles on the primary and secondary pulleys vary as the pulley radii change. Using the geometric relations from Section 7.4, the tangency angle becomes

$$\lambda(s) = \sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right). \tag{7.18}$$

Substitution into the wrap-angle definitions yields

$$\phi_p(s) = \pi - 2\sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right), \tag{7.19}$$

$$\phi_s(s) = \pi + 2\sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right). \tag{7.20}$$

7.8 Effective Torque Radius

All geometric relations derived thus far use the outer surface of the belt as the reference radius. This simplifies the belt-length formulation and pulley geometry.

Torque transmission, however, occurs through the belt cross-section. The appropriate moment arm for torque is therefore located approximately at the mid-height of the belt, as illustrated in Figure 14.

The effective torque radius for each pulley is defined as

$$r_{p,\text{eff}} = r_{p,\text{outer}} - \frac{h_b}{2}, \tag{7.21}$$

$$r_{s,\text{eff}} = r_{s,\text{outer}} - \frac{h_b}{2}. \tag{7.22}$$

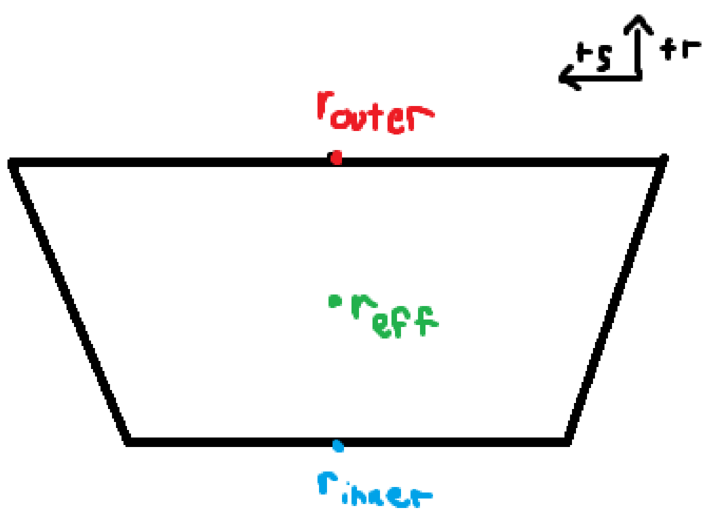

Figure 14: Belt cross-section showing the outer radius r_{outer}, inner radius r_{inner}, and effective torque radius r_{eff} located at the belt mid-height. The effective radius is used for all torque and power transmission calculations.

7.9 CVT Ratio Definition

The instantaneous transmission ratio is determined purely by pulley geometry. It is defined as the ratio of secondary to primary effective radii:

CVT Ratio **(7.23)**

$$R(s) = \frac{r_{s,\mathrm{eff}}(s)}{r_{p,\mathrm{eff}}(s)}.$$

Because both effective radii are functions of the shift coordinate s, the transmission ratio is likewise a function of s alone.

As the primary sheaves close, $r_{p,\mathrm{eff}}(s)$ increases. By belt-length compatibility, the secondary radius decreases. Consequently, the ratio $R(s)$ decreases with increasing s.

Thus:

- Small s (minimum primary radius) corresponds to high ratio.
- Large s (maximum primary radius) corresponds to low ratio.

The CVT ratio is therefore a purely geometric quantity determined by the axial shift position.

7.10 Ratio Rate of Change

The CVT ratio is not only a function of shift position, but also changes in time whenever the sheaves are moving. This time variation matters because a changing ratio introduces additional kinematic coupling between the primary and secondary rotations. In other words, it is not sufficient to know the instantaneous ratio $R(s)$; the model also requires how quickly the ratio is changing.

This section derives an explicit expression for $\dot{R}$ in terms of the shift velocity $\dot{s}$ and geometric derivatives.

7.10.1 Start from the Ratio Definition

The ratio is defined by the effective radii as

$$R(s) = \frac{r_{s,\mathrm{eff}}(s)}{r_{p,\mathrm{eff}}(s)}. \tag{7.24}$$

Because R depends on time only through $s(t)$, the chain rule gives

$$\dot{R} = \frac{dR}{ds}\dot{s}. \tag{7.25}$$

Thus, the objective is to compute dR/ds.

7.10.2 Differentiate R with Respect to s

Differentiating Equation (7.24) with respect to s (quotient rule) gives

$$\frac{dR}{ds} = \frac{d}{ds}\left(\frac{r_{s,\text{eff}}}{r_{p,\text{eff}}}\right) = \frac{\left(\frac{dr_{s,\text{eff}}}{ds}\right) r_{p,\text{eff}} - r_{s,\text{eff}}\left(\frac{dr_{p,\text{eff}}}{ds}\right)}{r_{p,\text{eff}}^2}. \tag{7.26}$$

Rearranging into a form that exposes the required components:

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}}\frac{dr_{s,\text{eff}}}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\frac{dr_{p,\text{eff}}}{ds}. \tag{7.27}$$

At this point, the remaining task is to express the two derivatives $dr_{p,\text{eff}}/ds$ and $dr_{s,\text{eff}}/ds$ in computable form.

7.10.3 Primary Radius Derivative

The primary radius derivative follows directly from the sheave geometry, since the primary radius is prescribed explicitly as a function of shift. From the piecewise primary radius model Equation (7.3), the primary outer radius satisfies

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\text{outer,min}}, & 0 \le s < s_{\text{dz}}, \\ r_{p,\text{outer,min}} + \dfrac{s - s_{\text{dz}}}{2\tan\beta}, & s_{\text{dz}} \le s \le s_{\text{max}}. \end{cases}$$

Differentiating with respect to s gives

Primary Radius Derivative **(7.28)**

$$\frac{dr_p}{ds} = \frac{dr_{p,\text{outer}}}{ds} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\ \dfrac{1}{2\tan\beta}, & s_{\text{dz}} \le s \le s_{\text{max}}. \end{cases}$$

Because $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ and h_b is constant, the same derivative applies:

$$\frac{dr_{p,\text{eff}}}{ds} = \frac{dr_p}{ds}. \tag{7.29}$$

Remark (At the Travel Limit)

If the model clamps s at the physical limit s_{max}, then $r_{p,\text{outer}}$ (and thus $r_{p,\text{eff}}$) is constant beyond this point and the derivative is zero for $s > s_{\text{max}}$. In this document we restrict to the admissible domain $0 \le s \le s_{\text{max}}$.

7.10.4 Secondary Radius Derivative

Unlike the primary radius, the secondary radius is not prescribed explicitly as a function of shift. Instead, it is determined implicitly by the belt-length compatibility condition derived in Section 7.4. Thus, the secondary radius derivative must be obtained by differentiating the constraint itself.

The belt-length constraint is written as

$$f(r_p, r_s, C) = 0, \tag{7.30}$$

where $f(\cdot)$ is defined in Equation (7.12) and C is constant.

Since both $r_p = r_p(s)$ and $r_s = r_s(s)$ depend on the shift coordinate s, differentiating Equation (7.30) with respect to s gives, by the multivariable chain rule,

$$\frac{\partial f}{\partial r_p}\frac{dr_p}{ds} + \frac{\partial f}{\partial r_s}\frac{dr_s}{ds} = 0. \tag{7.31}$$

Solving Equation (7.31) for the secondary radius derivative yields

$$\frac{dr_s}{ds} = -\frac{\dfrac{\partial f}{\partial r_p}}{\dfrac{\partial f}{\partial r_s}}\frac{dr_p}{ds}. \tag{7.32}$$

Because the effective secondary radius differs from the outer radius only by the constant belt offset $h_b/2$, the same derivative applies to the effective radius:

$$\frac{dr_{s,\text{eff}}}{ds} = \frac{dr_s}{ds}. \tag{7.33}$$

Equation Equation (7.32) shows that the secondary radial motion is not independent, but is instead determined by the belt-length constraint and the primary radial motion.

At this point, the remaining task is to evaluate the two belt-constraint sensitivities $\partial f/\partial r_p$ and $\partial f/\partial r_s$, which appear in Equation (7.32). The following subsections derive these quantities explicitly for the belt-length function defined in Equation (7.12).

7.10.5 Partial Derivatives of the Belt Constraint

Recall the belt-length constraint written in root form as

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b = 0. \tag{7.34}$$

We now compute the two partial derivatives required in Equation (7.32), differentiating term-by-term and applying the product rule directly to the inverse-sine term.

Product rule for the inverse-sine factor Consider the factor

$$(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right).$$

Differentiating with respect to r_p (product rule) gives

$$\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \underbrace{\frac{\partial}{\partial r_p}(r_s - r_p)}_{(1)}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\underbrace{\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right)}_{(2)}. \tag{7.35}$$

We therefore compute the two sub-derivatives.

Derivative (1)

$$\frac{\partial}{\partial r_p}(r_s - r_p) = -1. \tag{7.36}$$

Derivative (2) Using the chain rule,

$$\frac{d}{dx}\sin^{-1}(x) = \frac{1}{\sqrt{1 - x^2}},$$

let

$$u = \frac{r_s - r_p}{C}.$$

Then

$$\frac{\partial u}{\partial r_p} = -\frac{1}{C}.$$

Therefore,

$$\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) = \frac{1}{\sqrt{1 - u^2}}\left(-\frac{1}{C}\right). \tag{7.37}$$

Now simplify the denominator:

$$\sqrt{1 - u^2} = \sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2} = \frac{\sqrt{C^2 - (r_s - r_p)^2}}{C}.$$

Substituting into Equation (7.37) yields

$$\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) = -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{7.38}$$

Substitute into the product rule Substituting Equation (7.36) and Equation (7.38) into Equation (7.35) gives

$$\begin{aligned}\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] &= (-1)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\left(-\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}\right) \\ &= -\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}. \end{aligned} \tag{7.39}$$

Derivative of the square-root term For the straight-span contribution,

$$2\sqrt{C^2 - (r_s - r_p)^2},$$

we apply the chain rule:

$$\frac{\partial}{\partial r_p}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{7.40}$$

Assemble $\partial f/\partial r_p$ Differentiating Equation (7.34) term-by-term gives

$$\begin{aligned}\frac{\partial f}{\partial r_p} &= \pi + 2\left(-\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) + \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \end{aligned} \tag{7.41}$$

The square-root contributions cancel exactly.

Derivative with respect to r_s The same procedure yields

$$\frac{\partial}{\partial r_s}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}, \tag{7.42}$$

and

$$\frac{\partial}{\partial r_s}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = -\frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{7.43}$$

Assembling terms gives

$$\begin{aligned}\frac{\partial f}{\partial r_s} &= \pi + 2\left(\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) - \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \end{aligned} \tag{7.44}$$

Summary of Belt Constraint Partial Derivatives From the derivations above, the partial derivatives of the belt-length constraint Equation (7.34) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{7.45}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.46}$$

These expressions are globally valid under the geometric definition of λ given in Equation (7.9) and require no piecewise case distinction.

7.10.6 Final Expression for the Secondary Radius Derivative

Equation Equation (7.32) expresses the secondary radius derivative in terms of the belt-constraint partial derivatives and the primary radius derivative:

$$\frac{dr_s}{ds} = -\frac{\dfrac{\partial f}{\partial r_p}}{\dfrac{\partial f}{\partial r_s}} \frac{dr_p}{ds}. \tag{7.47}$$

From Section 7.10.5, the partial derivatives of the belt-length constraint Equation (7.12) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{7.48}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.49}$$

Substituting Equation (7.48) and Equation (7.49) into Equation (7.47) gives

$$\frac{dr_s}{ds} = -\frac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} \frac{dr_p}{ds}. \tag{7.50}$$

Finally, substituting the piecewise primary radius derivative Equation (7.28) yields

Secondary Radius Derivative **(7.51)**

$$\frac{dr_s}{ds} = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\[2ex] -\dfrac{1}{2\tan\beta} \dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}, & s_{\mathrm{dz}} \le s \le s_{\max}. \end{cases}$$

Because $r_{s,\mathrm{eff}} = r_s - h_b/2$ with h_b constant, the same expression applies to the effective secondary radius derivative:

$$\frac{dr_{s,\mathrm{eff}}}{ds} = \frac{dr_s}{ds}. \tag{7.52}$$

7.10.7 Final Expression for dR/ds and $\dot{R}$

From Equation (7.27), the ratio derivative is

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}}\frac{dr_{s,\text{eff}}}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\frac{dr_{p,\text{eff}}}{ds}. \tag{7.53}$$

Using Equation (7.29) and Equation (7.52), this becomes

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}}\frac{dr_s}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\frac{dr_p}{ds}. \tag{7.54}$$

Substituting Equation (7.50) into Equation (7.54) gives

$$\frac{dR}{ds} = \left[-\frac{1}{r_{p,\text{eff}}}\frac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\right]\frac{dr_p}{ds}. \tag{7.55}$$

Finally, substituting the piecewise primary derivative Equation (7.28) yields the fully expanded result

$$\frac{dR}{ds} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\[2ex] \dfrac{1}{2\tan\beta}\left[-\dfrac{1}{r_{p,\text{eff}}}\dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} - \dfrac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\right], & s_{\text{dz}} \le s \le s_{\max}. \end{cases} \tag{7.56}$$

Using the chain rule Equation (7.25), $\dot{R} = (dR/ds)\dot{s}$. Substituting Equation (7.56) and rearranging gives

Rate of Change of Transmission Ratio (7.57)

$$\dot{R} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\[2ex] -\dfrac{\dot{s}}{2\tan\beta\, r_{p,\text{eff}}}\left[\dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} + \dfrac{r_{s,\text{eff}}}{r_{p,\text{eff}}}\right], & s_{\text{dz}} \le s \le s_{\max}. \end{cases}$$

7.11 Section Summary and Forward Connection

This section established the complete *geometric foundation* of the CVT model. The development was intentionally kinematic: the objective was to derive a closed, computable mapping from axial shift to pulley geometry and transmission ratio, without introducing force balance, slip behavior, or torque relations.

At a high level, the geometry defines the chain

$$s(t) \longrightarrow r_{p,\text{eff}}(s),\ r_{s,\text{eff}}(s) \longrightarrow R(s) \longrightarrow \dot{R}(s, \dot{s}),$$

with each intermediate relationship derived explicitly from pulley and belt geometry.

Key Results Established The following geometric components are now defined and ready for use in subsequent modeling:

1. **Primary radius law.** Equation (7.3) provides the piecewise relationship between axial shift s and primary outer radius, including deadzone behavior and cone-angle dependence.
2. **Belt-length compatibility constraint.** Equation (7.13) enforces constant belt length and defines the implicit, nonlinear coupling between r_p, r_s, and the center distance C.
3. **Center distance determination.** Section 7.5 fixes the assembled center-to-center distance C by evaluating the belt constraint at a known geometric configuration. This solve is performed once during initialization.
4. **Secondary radius as a function of shift.** Section 7.6 defines $r_s(s)$ implicitly by solving the belt constraint at each shift position, producing the required redistribution of belt radius between pulleys.
5. **Wrap angles as functions of shift.** Section 7.7 defines the wrap angles $\phi_p(s)$ and $\phi_s(s)$ follow directly from the tangency geometry and are available for later contact, work, and belt-distribution calculations.
6. **Effective torque radii.** Section 7.8 corrects outer radii to effective radii using belt thickness, ensuring subsequent torque relations use an appropriate moment arm.
7. **Ratio and ratio rate.** Equation (7.23) defines the geometric ratio $R(s)$, and Section 7.10 provides $\dfrac{dR}{ds}$ and $\dot{R}$ via Equation (7.56) and Equation (7.57), including the deadzone condition where R is constant.

Structural Role in the Full Model These results form the geometric backbone of the CVT simulation. All subsequent sections treat the relationships derived here as fixed mappings. In particular, later torque and force models will use $r_{p,\text{eff}}(s)$ and $r_{s,\text{eff}}(s)$ as inputs, and shifting kinematics will use $R(s)$ and $\dot{R}(s, \dot{s})$ to relate primary and secondary motion during ratio change.

Several key properties are now clear: (i) the geometric coupling is inherently nonlinear (through the belt-length constraint), (ii) the deadzone is explicitly represented, and (iii) within the admissible travel range the mappings are differentiable and suitable for continuous-time simulation.

Transition to Dynamics With the geometry fully specified, the next objective is to determine how the shift coordinate evolves in time. Subsequent sections introduce axial force generation (from pulley mechanisms and belt interaction) to determine $\dot{s}$ and $\ddot{s}$. Once $\dot{s}$ is known, the ratio-rate expression Equation (7.57) provides the corresponding ratio evolution and the required kinematic coupling between rotating components during shifting.

8 Pulley Axial Force Models

The geometric relations developed in the previous section define how the transmission ratio varies with axial shift. What remains is to determine *why* the shift coordinate $s(t)$ changes in time.

Axial motion of the sheaves is driven by clamping forces generated within the primary and secondary assemblies. On the primary side, centrifugal flyweights and springs produce an inward axial force that tends to close the sheaves and increase radius. On the secondary side, torque reaction and spring preload generate an opposing axial force.

The evolution of the shift coordinate is governed by the net axial force acting on the movable sheave assembly. Applying Newton's second law in the axial direction (Theory 1 (Newton's Second Law (Translation))) gives

$$m\,\ddot{s} = F_{p,\mathrm{ax}} - F_{s,\mathrm{ax}}, \tag{8.1}$$

where $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ denote the axial forces generated by the primary and secondary mechanisms, respectively, and m is the effective axial inertia of the moving components.

The purpose of this section is to derive expressions for $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ as functions of the current system state. Once these force models are established, the shift acceleration $\ddot{s}$ follows directly, completing the dynamic description of axial motion.

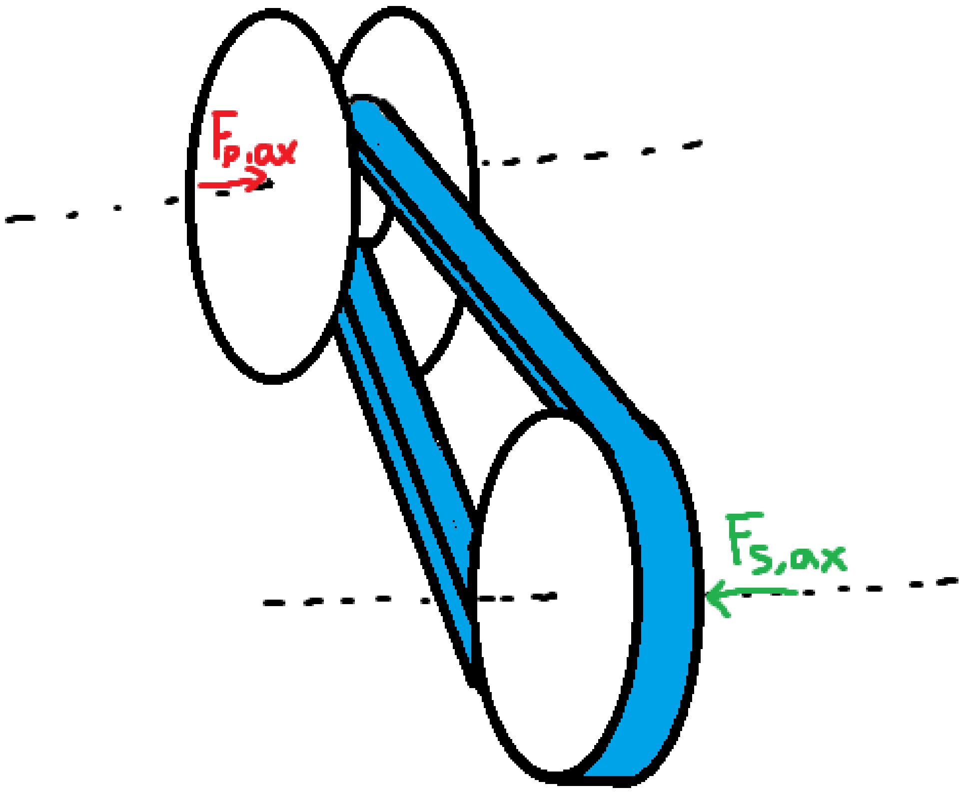

Figure 15: Axial force balance governing sheave motion. Primary and secondary mechanisms generate opposing axial forces. The net force determines the shift acceleration $\ddot{s}$.

Although the specific actuator mechanisms modeled here correspond to a flyweight-driven primary and a torque-reactive helix secondary, the formulation is modular: any actuator model may be substituted provided it returns the net axial force as a function of state.

8.1 Ramp Geometry Parameterization

Both pulley actuators redirect forces through inclined contact surfaces. As the shift coordinate s evolves, mechanical elements slide along these surfaces, converting radial or circumferential forces into axial clamping force.

To keep the formulation general and modular, the ramp geometries are parameterized directly as functions of the axial shift coordinate s. Any actuator profile may therefore be substituted provided it supplies the appropriate geometric mapping and derivative.

Primary Ramp Geometry (Axial-to-Radial Mapping) On the primary pulley, centrifugal force acting on the flyweights is redirected into axial closing force through a ramp surface (see Figure 16). As the movable sheave translates axially, the flyweight slides along this ramp.

We represent the ramp by an axial-to-radial profile

$$r_f = r_f(s), \tag{8.2}$$

where $r_f(s)$ is the effective flyweight radius.

An incremental axial displacement ds produces a radial displacement dr_f. The local slope of the ramp is therefore dr_f/ds. The instantaneous primary ramp angle is denoted by $\alpha_p(s)$ and is defined with respect to the axial direction as

$$\tan \alpha_p(s) = \frac{dr_f}{ds}, \qquad \alpha_p(s) = \tan^{-1}\left(\frac{dr_f}{ds}\right). \tag{8.3}$$

This definition follows directly from the geometry of the sliding motion: a steeper ramp (larger dr_f/ds) produces greater radial change per unit axial travel and therefore greater axial gain when radial force is redirected through the surface.

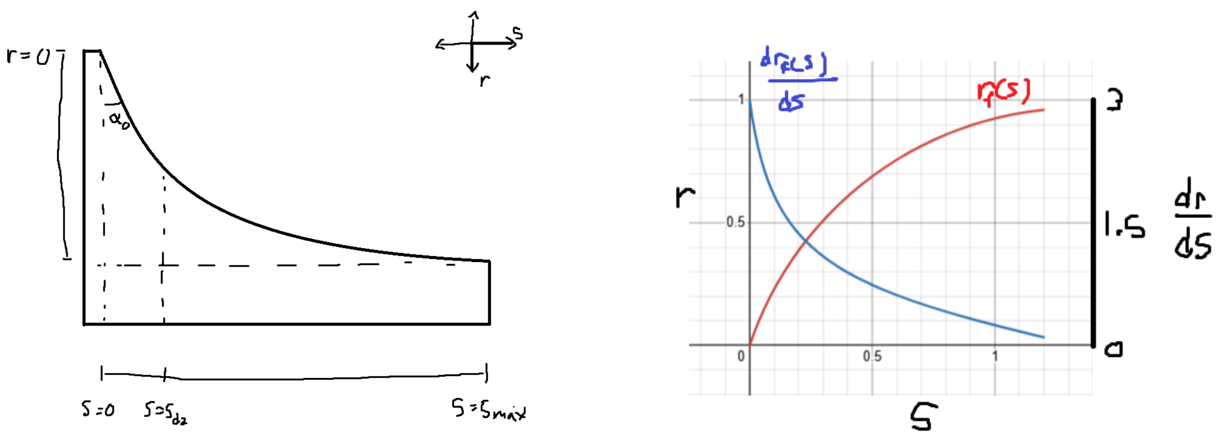

Figure 16: Primary ramp geometry. **Left:** As axial displacement s increases, the flyweight slides along the ramp, producing radial change dr_f. The instantaneous ramp angle satisfies $\tan \alpha_p = dr_f/ds$. **Right:** Example profiles of $r_f(s)$ and its slope dr_f/ds.

Primary Admissibility Conditions In order to be a valid primary ramp profile $r_f(s)$, it must satisfy:

1. $r_f(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{dr_f}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{dr_f}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the ramp angle $\alpha_p(s)$ is well-defined and continuous. Condition (2) guarantees that increasing axial displacement does not reduce flyweight radius. Condition (3) ensures the ramp angle remains strictly below $\pi/2$ and the axial gain $\tan \alpha_p$ remains finite.

Secondary Helix Geometry (Axial-to-Rotation Mapping) On the secondary pulley, axial travel produces relative rotation of a helical ramp mechanism, loading a torsional spring and generating torque-reactive clamping (see Figure 17).

We parameterize this kinematically as

$$\theta_s = \theta_s(s), \tag{8.4}$$

where $\theta_s(s)$ is the helix-induced rotation.

An incremental rotation $d\theta_s$ corresponds to a circumferential displacement $r_h d\theta_s$ along a cylindrical surface of effective radius r_h. The instantaneous secondary helix angle is denoted by $\alpha_s(s)$ and is defined from the local slope of the ramp surface:

$$\tan\alpha_s = \frac{\text{axial rise}}{\text{circumferential run}} = \frac{ds}{r_h d\theta_s}.$$

Rewriting in differential form gives

$$\tan\alpha_s(s) = \frac{1}{r_h \dfrac{d\theta_s}{ds}}. \tag{8.5}$$

Thus, a shallower helix (smaller α_s) produces greater rotation per unit axial travel, while a steeper helix produces less rotation. This inverse relationship arises naturally from the geometry of motion on a cylindrical surface.

Remark (Contrast with Primary Ramp)

The primary ramp behaves like a simple wedge: radial displacement is directly proportional to axial travel, so the ramp angle is naturally defined by $\tan\alpha_p = dr_f/ds$.
The secondary mechanism instead behaves like a power screw. Axial motion is produced through rotation around a cylindrical surface, so the relevant geometric ratio is how much rotation occurs per unit axial travel. This leads to the slope definition $ds/(r_h d\theta_s)$ and the inverse relationship in Equation (8.5).
In short, the primary redirects force through a wedge, while the secondary converts torque through a screw-like helix. The difference is purely geometric and reflects the distinct mechanisms by which axial motion is generated.

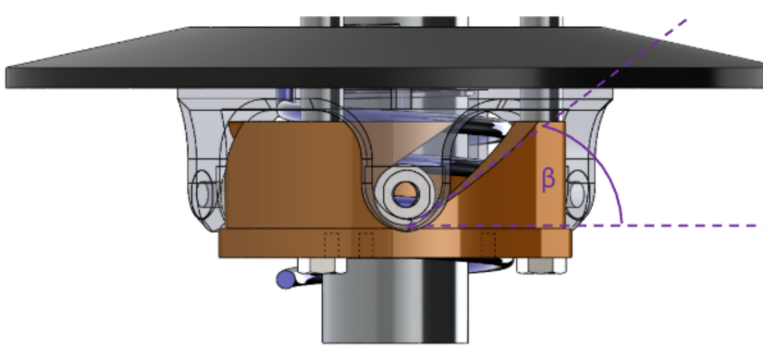

Figure 17: Secondary helix geometry. Axial travel s induces rotation $\theta_s(s)$ around a cylindrical surface of radius r_h. The instantaneous helix angle satisfies $\tan\alpha_s = ds/(r_h d\theta_s)$.

Secondary Admissibility Conditions The helix rotation mapping $\theta_s(s)$ must satisfy:

1. $\theta_s(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{d\theta_s}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{d\theta_s}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the torque-to-axial conversion remains continuous. Condition (2) guarantees that increasing axial displacement increases rotation. Condition (3) ensures the effective helix angle remains strictly below $\pi/2$ and the geometric gain remains finite.

Remark (Piecewise Profiles in Practice)

The C^1 requirement above is sufficient for all derivations in this document. In practice, piecewise C^1 profiles (continuous everywhere, continuously differentiable except at finitely many junction points where left and right derivatives exist) are also admissible for numerical simulation. At junction points, one-sided derivatives provide the required slope values, and the resulting force discontinuities pose no difficulty for standard ODE solvers. However, large slope discontinuities can introduce local stiffness into the system: the abrupt change in force gradients may require adaptive solvers to reduce the timestep significantly near the junction, degrading computational efficiency. In extreme cases, explicit integrators may struggle to maintain stability without impractically small step sizes, and implicit or stiff-aware solvers may be preferable.

Derivations for common ramp and helix profiles are provided in Appendix B.

8.2 Primary Axial Force Model

The primary axial clamping force is generated by the interaction between centrifugal loading of the flyweights and the restoring force of the primary compression spring.

As shown in Figure 18, each flyweight of mass m_f rotates with angular speed ω_p and experiences an outward radial centrifugal force. This radial force acts along the ramp profile $r_f(s)$ and is redirected into an axial closing force through the ramp geometry described in Section 8.1.

Opposing this motion is the primary compression spring, characterized by stiffness k_p and preload $x_{p,0}$. As the sheaves close (s increases), the spring compresses and produces a restoring axial force resisting further closure.

The net axial force on the movable primary sheave is therefore the axial component of the ramp-redirected centrifugal force minus the spring force. This net force determines the axial acceleration $\ddot{s}$ of the primary assembly.

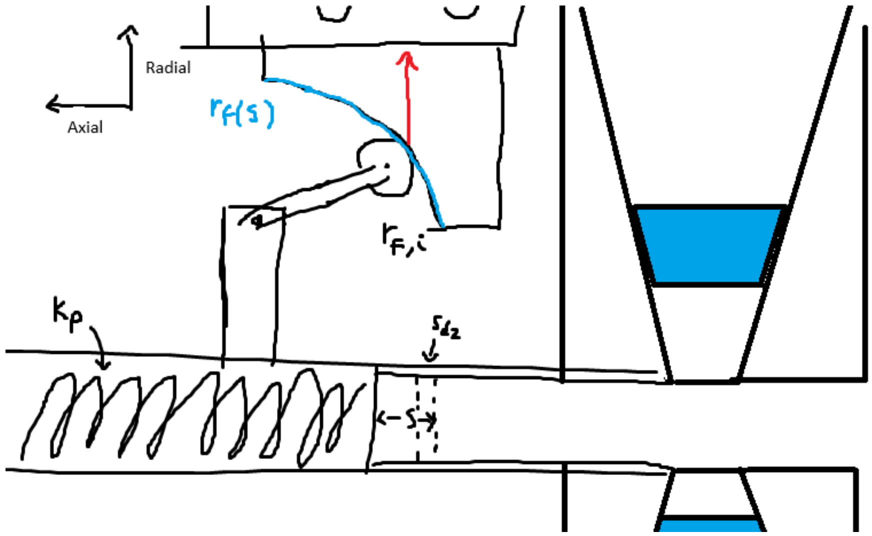

Figure 18: Primary CVT assembly showing flyweight mass m_f, ramp profile $r_f(s)$, compression spring with stiffness k_p and preload $x_{p,0}$, and axial shift coordinate s.

The state variable governing primary rotation is ω_p.

Centrifugal Loading of Flyweights From Theory 5 (Centrifugal Force),

$$F_c = m\omega^2 r.$$

Let

- m_f denote the flyweight mass,
- $r_{f,0}$ denote the initial flyweight radius at $s = 0$,
- $r_f(s)$ denote the incremental radial change produced by the ramp.

The total effective flyweight radius is therefore

$$r_{f,\text{tot}}(s) = r_{f,0} + r_f(s).$$

The radial centrifugal force is

$$F_{p,\text{rad}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big). \tag{8.6}$$

Radial-to-Axial Projection The ramp redirects radial force into axial force. As shown in Figure 19, the axial component of a force acting along the ramp is obtained by

$$F_{p,\text{axial component}} = F_{p,\text{rad}} \tan \alpha_p(s). \tag{8.7}$$

From Equation (8.3), the ramp angle satisfies

$$\tan \alpha_p(s) = \frac{dr_f}{ds}.$$

Substituting gives

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds}. \tag{8.8}$$

Thus the axial gain of centrifugal force is governed by the local ramp slope, while the centrifugal magnitude depends on the total flyweight radius.

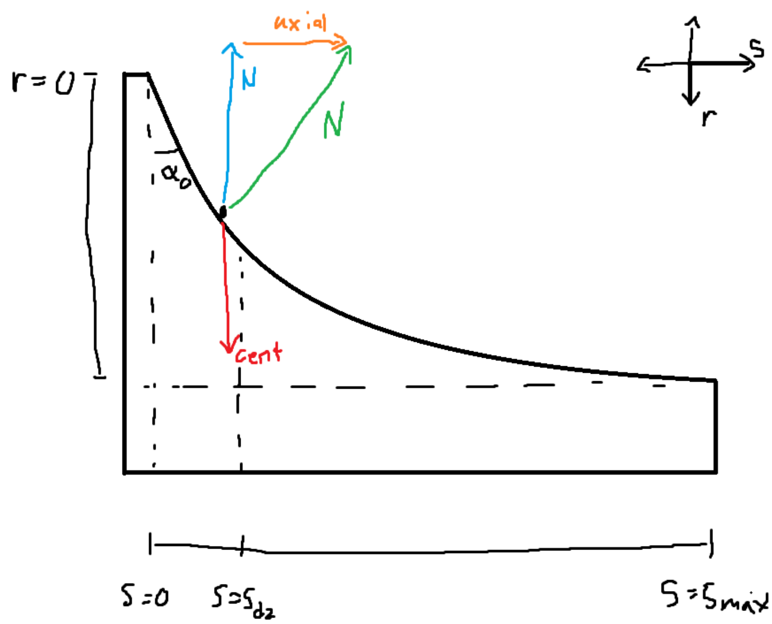

Figure 19: Primary ramp geometry. Radial centrifugal force is redirected into axial force through the ramp. The projection introduces a factor of $\tan \alpha_p$, which equals dr_f/ds by definition of the ramp slope.

Primary Compression Spring From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let k_p denote primary stiffness and $x_{p,0}$ preload.

$$F_{p,\text{spring}}(s) = k_p(x_{p,0} + s). \tag{8.9}$$

Net Primary Axial Force The net axial force generated by the primary mechanism is the difference between the centrifugal axial component and the spring force:

Primary Axial Force **(8.10)**

$$F_{p,\mathrm{ax}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds} - k_p(x_{p,0} + s).$$

8.3 Secondary Axial Force Model

The secondary pulley is *torque-reactive*: its axial clamping force is generated in response to torque transmitted through the CVT. Unlike the primary, which produces clamping based on rotational speed, the secondary produces clamping primarily when torque flows through it.

The mechanism consists of a helical ramp between the two sheaves and a single spring located between them. This spring serves two roles:

- It acts axially in compression as the sheaves separate.
- It resists relative rotation torsionally as the helix turns.

Thus, axial motion and rotational motion of the helix both load the same spring.

Figure 20: Secondary CVT assembly showing the helix mechanism and spring. Torque τ_s transmitted through the belt produces tangential force at the helix ramps, which is redirected into axial clamping force.

Physical Mechanism (Screw Analogy) The secondary helix behaves mechanically like a screw.

If you try to spin a screw in free space, nothing happens axially. If you spin a screw into a block that is not restrained, the block simply rotates with the screw—again producing no axial force. Axial force only develops when there is resistance to rotation on one side while torque is applied on the other.

In other words, a screw converts *reacted torque* into axial force.

The secondary helix operates on the same principle. The belt applies a driving torque to the secondary pulley. If the driven shaft were free, the entire assembly would rotate without developing tangential reaction force at the helix. However, when the driven load resists rotation, torque is transmitted through the helix ramps.

Those ramps are inclined surfaces—just like screw threads. As torque attempts to rotate one helix face relative to the other, contact forces develop along the inclined surfaces. Because the surfaces are angled, this tangential reaction force has an axial component.

The result is axial clamping force.

The key distinction from a conventional screw is direction: in a normal screw, torque pulls two components together. In the CVT helix, the orientation of the ramp is such that torque causes the movable sheave to be pushed outward, increasing belt clamping.

Thus, the secondary does not clamp because it spins. It clamps because torque is being transmitted and resisted.

Definition of Transmitted Torque Let τ_s denote the torque transmitted through the secondary shaft. This is the torque carried from the belt, through the helix mechanism, and into the driven load.

For the present derivation, τ_s is treated as a known input. Its value will be computed later in **??**.

Torsional Spring Torque As the helix rotates relative to the opposing sheave, the spring is twisted torsionally.

From Theory 7 (Hooke's Law (Torsional)),

$$\tau = k\Delta\theta.$$

Let $k_{s,\theta}$ denote torsional stiffness and $\theta_{s,0}$ preload.

$$\tau_{s,\text{spring}}(s) = k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big). \tag{8.11}$$

This torsional component of the same spring contributes to helix loading even when no torque is transmitted ($\tau_s = 0$), providing baseline clamping.

Total Torque Applied to Helix

$$\tau_{s,\text{total}} = \tau_s + \tau_{s,\text{spring}}(s). \tag{8.12}$$

Both contributions act in the same rotational direction at the helix. The transmitted torque loads the helix through belt force, and the torsional spring resists relative rotation. Both therefore generate axial clamping through the same geometry.

Torque-to-Axial Conversion Through the Helix From Theory 3 (Torque Definition),

$$\tau = F_t r_h \quad \Rightarrow \quad F_t = \frac{\tau_{s,\text{total}}}{r_h}.$$

The tangential force F_t acts along two symmetric helix ramp faces. Each ramp carries half of the total tangential load, introducing the factor of $1/2$ in the axial projection below.

From helix geometry (see Figure 21),

$$F_{s,\text{helix}} = \frac{F_t}{2\tan\alpha_s(s)}. \tag{8.13}$$

Substituting F_t and using

$$\tan\alpha_s(s) = \frac{1}{r_h\,\dfrac{d\theta_s}{ds}}$$

gives

$$F_{s,\text{helix}}(s, \tau_s) = \frac{\tau_{s,\text{total}}}{2} \frac{d\theta_s}{ds}. \tag{8.14}$$

Notably, the helix radius r_h cancels: torque is converted into axial force according to the geometric rotation-per-travel relationship $d\theta_s/ds$, consistent with screw mechanics.

TODO: Remark on less friction at bigger radius if being considered

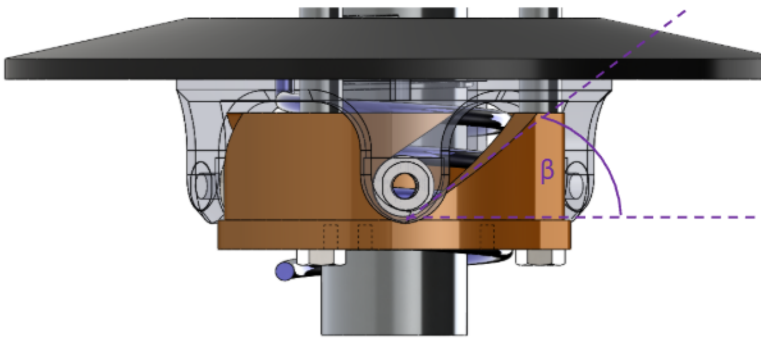

Figure 21: Helix free-body concept. Tangential force from shaft torque is redirected into axial force along the inclined ramps. Two symmetric ramp faces share the tangential load, producing the 1/2 factor.

Secondary Compression Spring The same spring also acts axially between the sheaves. At $s = 0$, the spring is pre-compressed by an amount $x_{s,0}$, producing a baseline separating force that assists upshift.

As the sheaves move apart (increasing s), the spring extends and its compression decreases.

From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let $k_{s,x}$ denote axial stiffness. Because increasing s increases compression, the effective compression is

$$\Delta x = x_{s,0} + s.$$

Thus,

$$F_{s,\text{comp}}(s) = k_{s,x}(x_{s,0} + s). \tag{8.15}$$

This axial spring force assists upshift at small s, but weakens progressively as the sheaves separate.

Net Secondary Axial Force The total secondary axial force is the sum of:

- Helix-generated torque-reactive clamping,
- Torsional spring contribution (through the helix),
- Direct axial compression spring force.

Secondary Axial Force **(8.16)**

$$F_{s,\mathrm{ax}}(s,\tau_s) = \underbrace{\frac{\tau_s + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)}{2}\frac{d\theta_s}{ds}}_{\text{torque-reactive helix (including torsional spring)}} + \underbrace{k_{s,x}(x_{s,0} + s)}_{\text{axial spring}}.$$

8.4 Belt Centrifugal Force on the Wrap

In addition to the actuator and spring forces derived previously, the belt itself generates centrifugal loading as it rotates around each pulley. This section derives the resulting contribution to axial clamping force.

Throughout this sub-section, let

$$j \in \{p, s\}$$

denote either pulley, where $j = p$ corresponds to the primary and $j = s$ corresponds to the secondary.

Centrifugal Force Formulation From Theory 5 (Centrifugal Force), a mass m rotating at angular velocity ω at radius r experiences centrifugal force

$$F_c = m\omega^2 r.$$

To apply this to the belt wrapped around a pulley, we require:

1. The radius at which this mass rotates.
2. The mass of belt material in contact with the pulley.
3. The angular velocity of the belt mass.

Each of these quantities is derived below.

8.5 Belt Centrifugal Force on the Wrap

In addition to the actuator and spring forces derived previously, the belt itself generates centrifugal loading as it rotates around each pulley. This section derives the resulting contribution to axial clamping force.

Throughout this section, let

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From Theory 5 (Centrifugal Force), a mass m rotating at angular velocity ω at radius r experiences centrifugal force

$$F_c = m\omega^2 r.$$

To apply this to the belt wrapped around pulley j, three quantities must be defined:

1. the radius at which the belt mass rotates,
2. the mass of belt material carried around the wrap,
3. the angular velocity of that belt mass.

Each is developed below.

1. Radius at Which the Belt Mass Rotates Centrifugal force acts at the center of mass of the rotating belt element, not at the outer or inner belt surface. The location of this center of mass therefore determines the radius that must be used in the centrifugal force calculation.

The belt cross-section is approximated as a trapezoid (see Figure 22) with:

- radial thickness h_b,
- outer width b_{out},
- inner width b_{in}.

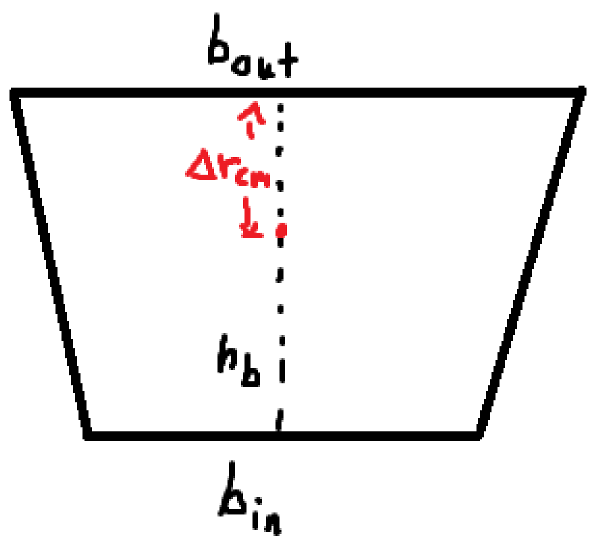

Figure 22: Trapezoidal belt cross-section. The outer width b_{out} corresponds to the surface away from the pulley axis; the inner width b_{in} corresponds to the surface closer to the axis. The radial thickness is h_b.

The cross-sectional area of the trapezoid is

$$A_b = \frac{h_b}{2}\left(b_{\text{out}} + b_{\text{in}}\right). \tag{8.17}$$

Let y denote radial distance measured inward from the outer surface, so that $y = 0$ at the outer surface and $y = h_b$ at the inner surface. For a trapezoid with linear sidewalls, the local width varies linearly with y:

$$b(y) = b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y. \tag{8.18}$$

The centroid location Δr_{cm}, measured inward from the outer surface, follows from the area-centroid definition:

$$\Delta r_{\text{cm}} = \frac{1}{A_b} \int_0^{h_b} y\, b(y)\ \mathrm{d}y. \tag{8.19}$$

Substituting Equation (8.18) into Equation (8.19) gives

$$\begin{aligned}
\int_0^{h_b} y\, b(y)\ \mathrm{d}y &= \int_0^{h_b} y \left(b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y \right) \mathrm{d}y \\
&= \int_0^{h_b} b_{\text{out}} y\ \mathrm{d}y + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \int_0^{h_b} y^2\ \mathrm{d}y \\
&= \frac{h_b^2}{2} b_{\text{out}} + \frac{h_b^2}{3} \left(b_{\text{in}} - b_{\text{out}} \right) \\
&= \frac{h_b^2}{6} \left(b_{\text{out}} + 2 b_{\text{in}} \right). \qquad (8.20)
\end{aligned}$$

Dividing by the area from Equation (8.17) yields

$$\Delta r_{\text{cm}} = h_b \frac{b_{\text{out}} + 2 b_{\text{in}}}{3 \left(b_{\text{out}} + b_{\text{in}} \right)}. \tag{8.21}$$

For pulley j, let $r_{j,\text{out}}(s)$ denote the outer belt radius. The radius at which the belt mass rotates is therefore

$$r_{j,\text{cm}}(s) = r_{j,\text{out}}(s) - \Delta r_{\text{cm}}. \tag{8.22}$$

2. Mass of Belt Material on the Wrap To determine the centrifugal loading on pulley j, we next determine the amount of belt mass associated with an infinitesimal portion of its wrap.

Let θ denote the angular coordinate measured from the center of the wrap, so that the wrapped region on pulley j spans

$$\theta \in \left[-\frac{\phi_j}{2}, \frac{\phi_j}{2} \right],$$

where $\phi_j(s)$ is the total wrap angle on pulley j (see Figure 23).

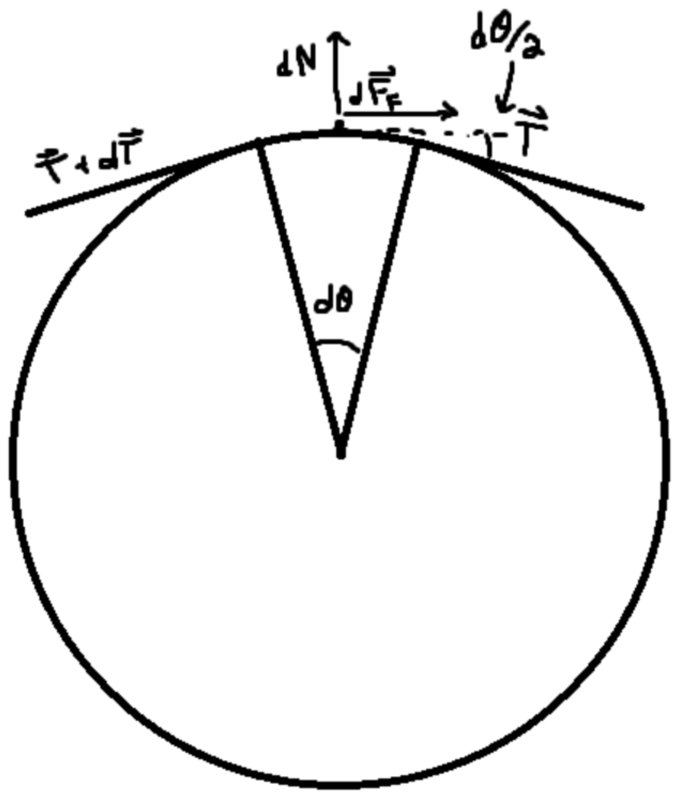

Figure 23: Belt wrapped around pulley j. A differential element at angle θ has arc length $\mathrm{d}\ell = r_{j,\mathrm{cm}}\,\mathrm{d}\theta$ and experiences radially outward centrifugal force.

A differential belt element subtending angle $\mathrm{d}\theta$ at the mass-centroid radius $r_{j,\mathrm{cm}}$ has arc length

$$\mathrm{d}\ell = r_{j,\mathrm{cm}}\,\mathrm{d}\theta.$$

Since the belt has cross-sectional area A_b, the differential belt element has volume

$$\mathrm{d}V_j = A_b\,\mathrm{d}\ell. \tag{8.23}$$

Substituting $\mathrm{d}\ell = r_{j,\mathrm{cm}}\,\mathrm{d}\theta$ gives

$$\mathrm{d}V_j = A_b\,r_{j,\mathrm{cm}}\,\mathrm{d}\theta. \tag{8.24}$$

Let ρ_b denote the belt material density (kg/m^3). The corresponding differential mass is then

$$\begin{aligned}\mathrm{d}m_j &= \rho_b\,\mathrm{d}V_j \\ &= \rho_b A_b\,r_{j,\mathrm{cm}}\,\mathrm{d}\theta. \end{aligned} \tag{8.25}$$

This gives the amount of belt material associated with an infinitesimal portion of the wrap on pulley j.

3. Angular Velocity of the Belt Mass The rotating belt mass on pulley j must also be assigned an angular speed. Let $\hat{v}_b$ denote the modeled belt transport speed introduced in the system state. The corresponding angular velocity of the belt mass centroid on pulley j is then

$$\omega_{b,j} = \frac{\hat{v}_b}{r_{j,\mathrm{cm}}(s)}. \tag{8.26}$$

This relation expresses the fact that the same belt transport speed corresponds to different local angular velocities depending on the radius of motion.

Distributed Centrifugal Load Along the Wrap With the radius, differential mass, and angular velocity now defined, the centrifugal force on the differential belt element follows from Theory 5 (Centrifugal Force):

$$\begin{aligned} \mathrm{d}F_{c,j} &= \mathrm{d}m_j\, \omega_{b,j}^2\, r_{j,\mathrm{cm}} \\ &= (\rho_b A_b\, r_{j,\mathrm{cm}}\, \mathrm{d}\theta) \left(\frac{\hat{v}_b}{r_{j,\mathrm{cm}}}\right)^2 r_{j,\mathrm{cm}} \\ &= \rho_b A_b\, \hat{v}_b^2\, \mathrm{d}\theta. \end{aligned} \tag{8.27}$$

This force acts radially outward at each point along the wrap.

Total Radial Force Over the Wrap Integrating Equation (8.27) over the full wrap angle gives the total radially-directed centrifugal force on pulley j:

$$\begin{aligned} F_{c,j,\mathrm{rad}} &= \int_{-\phi_j/2}^{\phi_j/2} \rho_b A_b\, \hat{v}_b^2\, \mathrm{d}\theta \\ &= \rho_b A_b\, \hat{v}_b^2 \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}\theta \\ &= \rho_b A_b\, \hat{v}_b^2\, \phi_j(s). \end{aligned} \tag{8.28}$$

Radial-to-Axial Conversion Through the Sheave Cone The radial centrifugal force acts outward against the conical sheave faces. Because the sheaves are inclined at cone half-angle β, radial loading cannot occur without simultaneous axial motion of the sheaves. Thus, a purely radial force applied to the belt generates an axial reaction component that contributes to clamping.

From the pulley geometry derived previously (see Equation (7.2)), the relationship between radial and axial displacement in the active shift region is

$$\frac{\mathrm{d}r}{\mathrm{d}s} = \frac{1}{2\tan\beta}.$$

For an infinitesimal displacement,

$$\delta W = F_{\text{rad}}\,\mathrm{d}r = F_{\text{ax}}\,\mathrm{d}s.$$

Substituting $\mathrm{d}r = \dfrac{1}{2\tan\beta}\,\mathrm{d}s$ gives

$$F_{\text{rad}}\,\frac{1}{2\tan\beta}\,\mathrm{d}s = F_{\text{ax}}\,\mathrm{d}s.$$

Canceling $\mathrm{d}s$ yields

$$F_{\text{ax}} = \frac{F_{\text{rad}}}{2\tan\beta}.$$

Thus, the axial clamping contribution from belt centrifugal force on pulley j is

$$F_{c,j} = \frac{\rho_b A_b\,\hat{v}_b^2\,\phi_j(s)}{2\tan\beta}. \tag{8.29}$$

8.6 Axial Shift Dynamics

Earlier (see Equation (8.1)), the evolution of the shift coordinate was established as

$$m\,\ddot{s} = F_{p,\text{ax}} - F_{s,\text{ax}}, \tag{8.30}$$

where $F_{p,\text{ax}}$ and $F_{s,\text{ax}}$ are the axial forces generated by the primary and secondary mechanisms, respectively.

We now complete this model by:

1. Expanding the force terms to include centrifugal belt loading,
2. Replacing the lumped mass m with the equivalent inertia associated with coordinated shift motion.

Equivalent Inertia Associated with s Newton's second law applies to physical masses in translation. Since s parameterizes coordinated motion of several bodies, we construct an equivalent mass m_{eq} such that the total kinetic energy associated with shift can be written in the form of Theory 4 (Kinetic Energy (Translational)):

$$T_{\text{shift}} = \frac{1}{2}\,m_{\text{eq}}\,\dot{s}^2.$$

The movable primary sheave translates with speed $\dot{s}$ in the positive coordinate direction. The movable secondary sheave translates with equal magnitude speed (but opposite physical direction), so its kinetic energy also contributes proportionally to $\dot{s}^2$.

Let $m_{p,m}$ and $m_{s,m}$ denote the masses of the movable primary and secondary sheaves.

Belt Mass Contribution The total belt mass is

$$m_b = \rho_b A_b L_b, \tag{8.31}$$

where A_b is the trapezoidal cross-sectional area from Equation (8.17) and L_b is the constant belt length.

The belt moves continuously with a single belt speed, but the axial component of that motion associated with a change in s is determined by geometry. From the axial-radial coupling in Equation (7.28), a change in sheave separation s is shared between the two conical faces supporting the belt wedge. Thus, the belt's axial velocity relative to the pulley axis is

$$\dot{s}_b = \frac{\dot{s}}{2}. \tag{8.32}$$

The kinetic energy associated with this axial motion is therefore

$$T_b = \frac{1}{2} m_b \left(\frac{\dot{s}}{2}\right)^2 = \frac{1}{2}\left(\frac{m_b}{4}\right)\dot{s}^2.$$

Expressed in the generalized coordinate s, the belt contributes an effective mass of $m_b/4$.

Total Equivalent Mass The total inertia associated with shift becomes

$$m_{\text{eq}} = m_{p,m} + m_{s,m} + \frac{m_b}{4} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}. \tag{8.33}$$

Net Axial Force in the s Coordinate Collecting the axial force contributions derived previously:

- $F_{p,\text{ax}}(s, \omega_p)$ from Equation (8.10),
- $F_{s,\text{ax}}(s, \tau_s)$ from Equation (8.16),
- $F_{c,p}(s, \hat{v}_b)$ from Equation (8.29) with $j = p$,
- $F_{c,s}(s, \hat{v}_b)$ from Equation (8.29) with $j = s$.

With the sign convention defined above, the generalized force in the positive s direction is

$$\sum F_s = \Big(F_{p,\text{ax}} + F_{c,p}\Big) - \Big(F_{s,\text{ax}} + F_{c,s}\Big). \tag{8.34}$$

Shift Equation of Motion Applying Newton's second law in the generalized coordinate,

$$\sum F_s = m_{\text{eq}}\, \ddot{s},$$

and substituting Equation (8.33) and Equation (8.34), we obtain

Axial Shift Acceleration **(8.35)**

$$\ddot{s} = \frac{\Big(F_{p,\mathrm{ax}}(s,\omega_p) + F_{c,p}(s,v_b)\Big) - \Big(F_{s,\mathrm{ax}}(s,\tau_s) + F_{c,s}(s,v_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}.$$

Equation (8.35) provides the axial shift acceleration as a function of the current state variables $(s, \dot{s}, \omega_p, \omega_s)$ and the reacted secondary torque τ_s. The remaining coupling between axial and rotational dynamics enters through τ_s, which will be determined later using Theory 2 (Newton's Second Law (Rotation)).

8.7 Section Summary and Forward Outlook

In this section we established a complete axial force model for the CVT mechanism and derived a closed-form equation of motion for the generalized shift coordinate $s(t)$.

At this stage, the axial subsystem is formally closed with respect to the state variables of the model, but it still depends on one remaining coupling quantity that is not determined internally by the axial force balance: the reacted secondary torque τ_s. Although τ_s appears explicitly in the secondary axial force model, its value is determined by the rotational dynamics of the drivetrain and by torque transmission through the belt. Thus, aside from the shared state variables, the axial and rotational subsystems are coupled through this torque term.

The next step is to determine how torque is transmitted across the CVT as a function of pulley radii and speed ratio. This will establish the relationship between τ_s, input torque at the primary, and the evolving ratio $R = r_s/r_p$. Once this transmission relationship is obtained, the axial and rotational equations can be combined into a unified, state-space representation of the full CVT dynamics.

With the axial force structure now complete, attention turns to the torque transmission mechanism that closes the remaining dynamical loop.

9 Torque Transmission and Coupling

The axial force models developed in Section 8 depend on the torque transmitted through the CVT. However, this torque has not yet been defined as a system quantity.

The engine applies torque at the primary pulley, while the vehicle load resists motion at the secondary. These components do not interact directly; their only mechanical connection is through the belt. To determine how torque is transmitted between them, the belt must be treated as the medium through which power flows.

We proceed by first defining the torque path through the system, then examining how power is carried by the belt, and finally introducing a single coupling torque that characterizes the CVT interface.

9.1 Torque Path Through the Belt

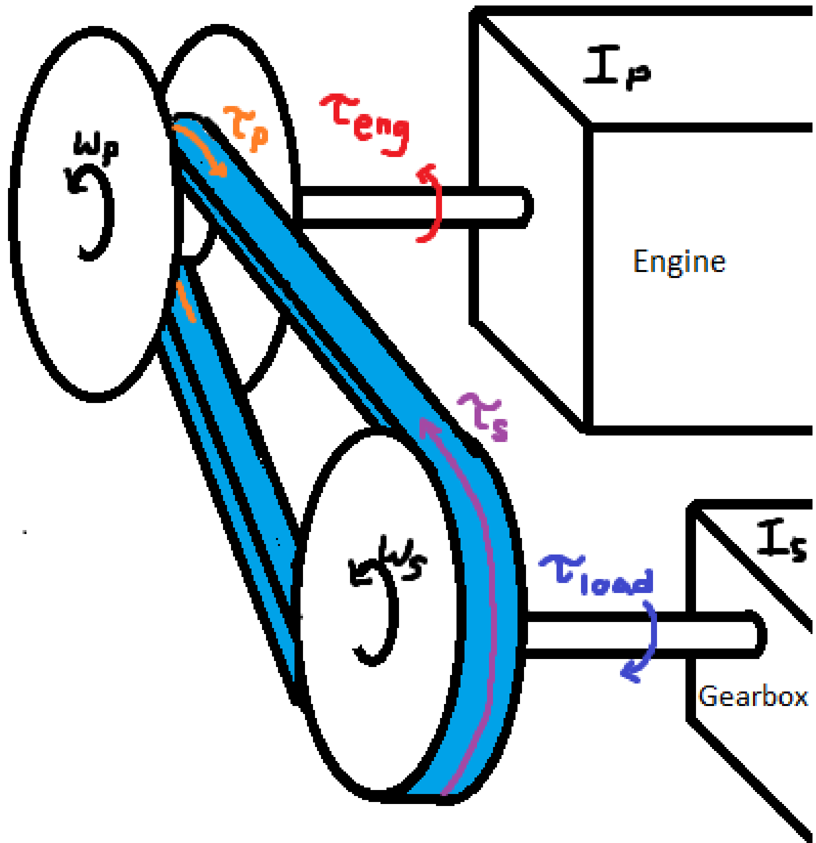

Figure 24: Torque flow through the CVT system. The primary pulley applies torque τ_p to the belt, while the belt applies torque τ_s to the secondary pulley.

Let

$$\tau_p: \text{ torque applied by the primary pulley onto the belt,} \tag{9.1}$$

$$\tau_s: \text{ torque applied by the belt onto the secondary pulley.} \tag{9.2}$$

These quantities represent the torque entering and leaving the belt transmission path, respectively.

9.2 Belt Power Flow and Torque Transmission

Since the belt is the only connection between the pulleys, it must carry the mechanical power transmitted through the system.

Under Assumption 8 (Constant Belt Length), the belt moves along its path with a single transport speed $v_b(t)$. Since the belt is modeled as having constant length with no longitudinal deformation, different tangential speeds at different points along the belt would imply local stretching or compression, which this model does not permit.

Torque transmission arises from tangential forces acting between the belt and the pulley surfaces. Let F_t denote the effective tangential force transmitted through the belt.

The mechanical power carried by the belt is therefore

$$P = F_t\, v_b. \tag{9.3}$$

Under Assumption 13 (Ideal CVT Power Transmission), this power is conserved through the transmission, so that the power entering the belt at the primary equals the power leaving at the secondary.

The same tangential force F_t acts at the effective contact radii of each pulley. The torques at the primary and secondary are therefore

$$\tau_p = r_{p,\text{eff}}\, F_t, \tag{9.4}$$

$$\tau_s = r_{s,\text{eff}}\, F_t. \tag{9.5}$$

Since both torques arise from the same transmitted force, F_t can be eliminated directly. From the primary expression,

$$F_t = \frac{\tau_p}{r_{p,\text{eff}}}, \tag{9.6}$$

and substituting into the secondary torque gives

$$\tau_s = r_{s,\text{eff}} \left(\frac{\tau_p}{r_{p,\text{eff}}} \right) = \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}}\, \tau_p. \tag{9.7}$$

Recalling the geometric definition of the CVT ratio

$$R = \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}}, \tag{9.8}$$

it follows directly that

$$\tau_s = R\, \tau_p. \tag{9.9}$$

9.3 Definition of Coupling Torque

The torques τ_p and τ_s both arise from the same transmitted belt force. It is therefore natural to define a single *coupling torque* that characterizes the interaction between the pulleys:

$$\tau_c \equiv \tau_p. \tag{9.10}$$

The corresponding secondary torque is then

$$\tau_s = R\, \tau_c. \tag{9.11}$$

The transmitted torque is thus not a property of either pulley in isolation, but of the belt–pulley interaction as a whole. The quantities τ_p and τ_s represent the primary-side and secondary-side expressions of this same coupling torque.

9.4 Transmission Regimes

The coupling torque τ_c defined above is the torque actually transmitted through the belt. It is not prescribed externally. Instead, its value depends on the contact state of the transmission.

No-slip torque. If the belt remains adhered at the pulley contacts, then the transmission behaves as a sticking constraint. In that case, the transmitted torque is whatever value is required to preserve that adhered motion.

We denote this required torque by

$$\tau_{\mathrm{ns}}, \tag{9.12}$$

where the subscript "ns" denotes *no slip*. Thus, in a sticking state,

$$\tau_c = \tau_{\mathrm{ns}}. \tag{9.13}$$

Traction limits. Adherence cannot be maintained for arbitrary transmitted torque. The belt–pulley contact can only sustain a finite tangential traction, and therefore only a finite coupling torque.

We therefore introduce directional traction bounds and write the admissible sticking interval as

$$\tau_- \leq \tau_{\mathrm{ns}} \leq \tau_+. \tag{9.14}$$

If Equation (9.14) holds, then the no-slip torque lies within the available traction limits, so sticking is physically admissible.

If instead τ_{ns} lies outside this interval, then the torque required to maintain adherence cannot be supported by the contact, and slip must occur.

Why a slip metric is still needed. The traction condition alone is not enough to determine the regime. Once slipping has begun, the transmission does not return immediately to stick simply because τ_{ns} later re-enters the admissible interval. Stick is only recovered once the existing relative motion has decayed to zero.

For this reason, one further quantity is needed: a measure of the relative slip velocity.

Under Assumption 14 (Single Active Slip Interface), the model permits slip at only one pulley contact at a time. A single scalar mismatch is therefore sufficient:

$$v_\Delta = r_{p,\mathrm{eff}}\omega_p - r_{s,\mathrm{eff}}\omega_s. \tag{9.15}$$

This quantity compares the two tangential surface speeds that the pulleys would impose on the belt if each were enforcing adherence.

If

$$v_\Delta = 0, \tag{9.16}$$

then the pulley motions are compatible with a common adhered belt motion, so a sticking state is kinematically possible.

If instead

$$v_\Delta \neq 0, \tag{9.17}$$

then the pulleys are demanding different tangential belt speeds.

The sign of v_Δ indicates the slip direction: if $v_\Delta > 0$, the primary tends to drive the belt faster than the secondary is accepting it; if $v_\Delta < 0$, the opposite is true.

Stick and slip states. The transmission can therefore be in stick only when two conditions are satisfied simultaneously:

1. the required no-slip torque lies within the admissible traction interval, and
2. the relative slip velocity has vanished.

This gives the regime structure

$$\tau_c = \begin{cases} \tau_{\text{ns}}, & v_\Delta = 0 \text{ and } \tau_- \leq \tau_{\text{ns}} \leq \tau_+, \\ \text{traction-limited value}, & \text{otherwise.} \end{cases} \tag{9.18}$$

Physical interpretation. If the required no-slip torque leaves the admissible interval, the contact can no longer maintain adherence and slip begins.

Once slip is present, the transmission remains on a slip branch until the existing mismatch has been driven back to zero. This is why the slip metric is needed in addition to the traction limits: the traction limits determine whether sticking is admissible, while v_Δ determines whether slipping is still occurring and, if so, in which direction.

The next section derives the no-slip torque τ_{ns}. The following section derives the traction limits τ_- and τ_+ and thereby makes the traction-limited branch explicit.

10 Derivation of the No-Slip Coupling Torque

In Section 9.4, the coupling torque τ_c was introduced as an internal torque whose value depends on the active transmission regime. The purpose of this section is to derive the explicit form of τ_c on the sticking branch, i.e. the no-slip torque τ_{ns}.

We proceed in four steps:

1. Write the rotational equations of motion for the primary and secondary pulleys.
2. Recall the external torque and inertia models on each side.
3. Impose the no-slip kinematic constraint.
4. Eliminate angular accelerations to obtain the no-slip coupling torque.

The result is the sticking-branch torque law used later in the regime model.

10.1 System Definition and Torque Path

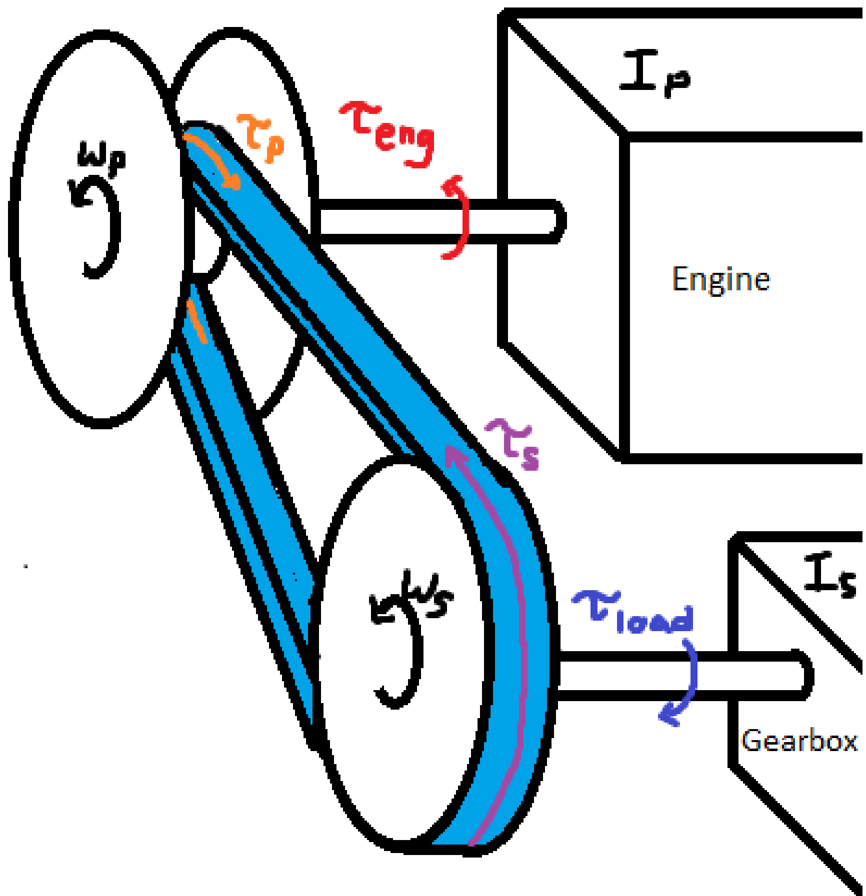

Figure 25: Torque transmission schematic. Engine torque τ_{eng} drives the primary pulley (angular velocity ω_p). Coupling torque τ_c is transmitted through the belt to the secondary pulley (angular velocity ω_s). The secondary delivers torque to the downstream drivetrain, which resists motion with torque τ_{load}.

We model two rotating bodies:

- the primary side, rigidly connected to the engine,
- the secondary side, driving the downstream drivetrain and vehicle.

The external torques and effective inertias used below are

- $\tau_{\text{eng}}(t)$: engine output torque applied to the primary pulley,
- $\tau_{\text{load}}(t)$: resisting torque reflected to the secondary pulley,
- I_p : total rotational inertia of the engine + primary CVT pulley,
- I_s : total effective rotational inertia about the secondary pulley.

Their explicit models are recalled in the following subsections. The remaining internal unknown is the transmitted coupling torque $\tau_c(t)$.

10.2 Rotational Equations of Motion

Applying Theory 2 (Newton's Second Law (Rotation)) to the primary side gives

Primary Pulley Rotational Dynamics **(10.1)**

$$\tau_{\text{eng}}(t) - \tau_c(t) = I_p\,\dot{\omega}_p(t)$$

Using the secondary-side no-slip torque mapping established in **??**, the secondary rotational equation on the sticking branch is

Secondary Pulley Rotational Dynamics **(10.2)**

$$R(t)\tau_c(t) - \tau_{\text{load}}(t) = I_s\,\dot{\omega}_s(t)$$

At this point, Equation (10.1) and Equation (10.2) provide two equations for the three internal unknowns

$$\dot{\omega}_p(t), \qquad \dot{\omega}_s(t), \qquad \tau_c(t).$$

The remaining quantities appearing in these equations — namely the applied torques $\tau_{\text{eng}}(t)$ and $\tau_{\text{load}}(t)$, the effective inertias I_p and I_s, and the geometric ratio $R(t)$ — are treated as known inputs, parameters, or previously defined geometric quantities. Their explicit forms are recalled in the following subsections.

Thus, the rotational system is not yet closed: two equations are available for three unknowns. A third relation is therefore required to determine the coupling torque. For the no-slip branch, that final relation will come from the sticking kinematic constraint introduced after the torque and inertia models are stated.

10.3 Engine Torque Model

The engine provides the driving torque $\tau_{\text{eng}}(t)$ acting on the primary pulley.

In this model, engine output torque is represented as a user-supplied function of primary angular velocity. Under the full-throttle assumption (Assumption 16 (Full-Throttle Engine Model)),

$$\tau_{\text{eng}}(t) = \tau_{\text{eng}}\big(\omega_p(t)\big), \tag{10.3}$$

so torque depends only on instantaneous primary speed.

This representation is consistent with standard engine characterization, where torque is measured as a function of crankshaft speed under steady full-load conditions.

Admissibility Constraints The supplied torque function must satisfy:

- It is defined over a bounded operating domain $\omega_{\min} \le \omega_p \le \omega_{\max}$.
- It is finite and real-valued over this domain.
- Outside this domain, torque is either saturated or ω_p is constrained to remain within bounds.

No specific functional form is required. The dynamical model only requires the mapping $\omega_p \mapsto \tau_{\text{eng}}$ to evaluate Equation (10.1).

Example: Kohler CH440 Engine Curve As an illustrative example, Figure 26 shows a representative torque and power curve for the Kohler CH440 engine.

SAE Baja Restricted Kohler CH440							
Engine Speed (rpm)	3600	3400	3200	3000	2800	2600	2400
Torque Net Corrected (ft-lb)	13.5	14.5	15.4	16.6	17.4	18.1	18.5
Power Net Corrected (HP)	9.3	9.4	9.4	9.5	9.3	9.0	8.5

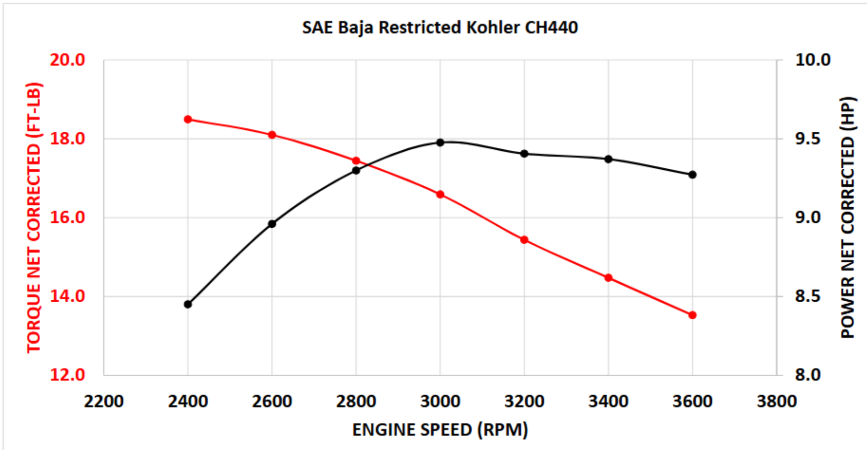

Figure 26: Representative torque and power curves for the Kohler CH440 engine. Torque is supplied as a function of angular velocity and serves as the input mapping in Equation (10.3).

In simulation, the discrete curve is interpolated to provide a continuous admissible torque function over the operating range.

10.4 Load Torque Model

The resisting torque $\tau_{\text{load}}(t)$ in Equation (10.2) represents the longitudinal road load reflected to the secondary pulley.

We proceed in three stages:

1. Express vehicle speed in terms of secondary rotation.
2. Derive the total longitudinal road force.
3. Reflect that force back to the secondary shaft.

Kinematic Relations Let r_w denote effective wheel rolling radius and $G > 0$ the fixed gear ratio defined by

$$\omega_s = G\,\omega_w.$$

Vehicle speed is therefore

$$v(t) = r_w \omega_w(t) = \frac{r_w}{G}\,\omega_s(t). \tag{10.4}$$

Longitudinal Road Force Using Theory 1 (Newton's Second Law (Translation)), the total longitudinal resistive force on the vehicle is modeled as

$$F_{\text{road}} = F_{\text{rr}} + F_{\text{grade}} + F_{\text{drag}}. \tag{10.5}$$

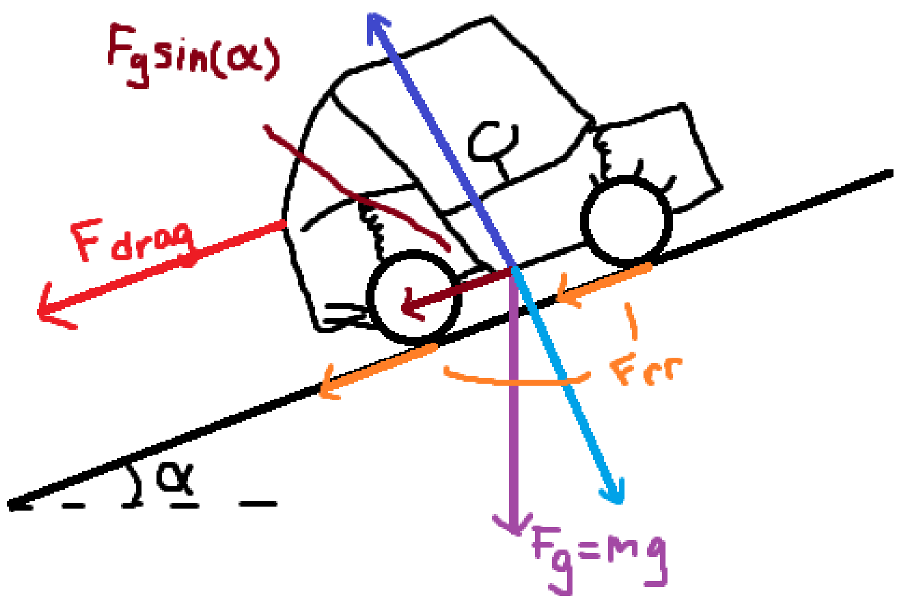

Figure 27: Free-body diagram of a vehicle on an incline. The weight mg resolves into $mg \sin\alpha$ along the slope and $mg\cos\alpha$ normal to the surface. Rolling resistance and aerodynamic drag oppose motion.

Rolling Resistance From Theory 14 (Rolling Resistance),

$$F_{\text{rr}} = C_{\text{rr}}\, N. \tag{10.6}$$

From the incline geometry,

$$N = mg\cos\alpha. \tag{10.7}$$

Thus

$$F_{\text{rr}}(v, \alpha) = C_{\text{rr}} mg \cos\alpha \;\text{sgn}(v). \tag{10.8}$$

Grade Resistance From Theory 15 (Gravitational Force), the gravitational component along the slope is

$$F_{\text{grade}}(\alpha) = mg \sin\alpha. \tag{10.9}$$

Aerodynamic Drag From Theory 13 (Air Resistance (Quadratic Drag)),

$$F_{\text{drag}}(v) = \frac{1}{2}\rho C_d A_f \, v^2 \, \text{sgn}(v). \tag{10.10}$$

Total Road Force Substituting the above components into Equation (10.5) yields

$$\begin{aligned} F_{\text{road}}(t) = {} & C_{\text{rr}} mg \cos\alpha \, \text{sgn}(v(t)) \\ & + mg \sin\alpha \\ & + \frac{1}{2}\rho C_d A_f \, v(t)^2 \, \text{sgn}(v(t)) \, . \end{aligned} \tag{10.11}$$

Using Equation (10.4),

$$v(t) = \frac{r_w}{G} \, \omega_s(t),$$

the road force is now fully expressed in terms of the secondary state variable $\omega_s(t)$.

Reflection to Secondary Torque Wheel torque is

$$\tau_{\text{road},w}(t) = r_w F_{\text{road}}(t). \tag{10.12}$$

Assuming ideal power transmission (Assumption 13 (Ideal CVT Power Transmission)), the resisting torque reflected to the secondary pulley is

$$\tau_{\text{load}}(t) = \frac{1}{G} \, \tau_{\text{road},w}(t) = \frac{r_w}{G} F_{\text{road}}(t). \tag{10.13}$$

Substituting Equation (10.11) gives

$$\boxed{\tau_{\text{load}}(t) = \frac{r_w}{G} \left[C_{\text{rr}} mg \cos\alpha \, \text{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big) + mg \sin\alpha + \frac{1}{2}\rho C_d A_f \left(\frac{r_w}{G}\omega_s(t)\right)^2 \text{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big) \right].} \tag{10.14}$$

10.5 Inertia Definitions

The rotational equations Equation (10.1)–Equation (10.2) require the effective rotational inertia about each pulley axis.

Primary Pulley Inertia The primary pulley is rigidly connected to the engine. All components on this side rotate with angular velocity ω_p, so no reflection is required.

The effective inertia is therefore the direct sum

$$I_p = I_{\text{eng}} + I_{\text{prim}}, \tag{10.15}$$

where

- I_{eng} is the engine rotational inertia,
- I_{prim} is the primary CVT pulley inertia.

Secondary Pulley Inertia On the secondary side, all downstream components must be expressed as an equivalent inertia about the secondary axis.

Let

- I_{sec} : secondary CVT pulley inertia,
- I_{gb} : gearbox + intermediate shaft inertia,
- I_{wheel} : wheel + hub inertia about the wheel axis,
- m : total vehicle mass,
- r_w : effective wheel rolling radius,
- G : fixed gear ratio defined by $\omega_s = G\,\omega_w$.

Vehicle Mass as Equivalent Rotational Inertia The vehicle's translational kinetic energy is

$$T_{\text{trans}} = \frac{1}{2}mv^2,$$

with $v = r_w\omega_w$.

Substituting,

$$T_{\text{trans}} = \frac{1}{2}m(r_w\omega_w)^2 = \frac{1}{2}(mr_w^2)\,\omega_w^2.$$

Thus the vehicle mass behaves as an effective rotational inertia

$$I_{\text{veh,wheel}} = mr_w^2$$

about the wheel axis.

Reflection to the Secondary Axis Because $\omega_s = G\,\omega_w$, we have $\omega_w = \omega_s/G$.

Expressing the downstream kinetic energy in terms of ω_s:

$$T = \frac{1}{2}I_{\text{wheel}}\left(\frac{\omega_s}{G}\right)^2 + \frac{1}{2}(mr_w^2)\left(\frac{\omega_s}{G}\right)^2.$$

Factoring,

$$T = \frac{1}{2}\left(\frac{I_{\text{wheel}} + m r_w^2}{G^2}\right)\omega_s^2.$$

Therefore, inertias at the wheel reflect to the secondary pulley as

$$I_{\text{reflected}} = \frac{I_{\text{wheel}} + m r_w^2}{G^2}.$$

Total Secondary Inertia The total effective inertia about the secondary pulley is

$$I_s = I_{\text{sec}} + I_{\text{gb}} + \frac{I_{\text{wheel}} + m r_w^2}{G^2}. \tag{10.16}$$

Equation Equation (10.16) defines the total inertia opposing angular acceleration of the secondary pulley.

10.6 No-Slip Ratio Kinematics

As established in Section 9.4, the sticking branch is characterized by the condition

$$v_\Delta = 0,$$

where v_Δ was defined in Equation (9.15) as

$$v_\Delta = r_{p,\text{eff}}(t)\,\omega_p(t) - r_{s,\text{eff}}(t)\,\omega_s(t).$$

Setting this equal to zero gives

$$r_{p,\text{eff}}(t)\,\omega_p(t) = r_{s,\text{eff}}(t)\,\omega_s(t). \tag{10.17}$$

Using the geometric CVT ratio

$$R(t) = \frac{r_{s,\text{eff}}(t)}{r_{p,\text{eff}}(t)},$$

Equation (10.17) becomes

$$\omega_p(t) = R(t)\,\omega_s(t). \tag{10.18}$$

This is the defining kinematic relation of the sticking branch. If Equation (10.18) does not hold, the present derivation no longer applies and the transmission must instead be described by the slip branch.

Differentiating Equation (10.18) with respect to time gives

$$\dot{\omega}_p(t) = R(t)\,\dot{\omega}_s(t) + \omega_s(t)\,\dot{R}(t), \tag{10.19}$$

which is the corresponding acceleration-level compatibility condition.

10.7 Eliminating Angular Accelerations

We now combine the primary rotational dynamics Equation (10.1), the secondary rotational dynamics Equation (10.2), and the no-slip compatibility relation Equation (10.19) to solve for the coupling torque on the sticking branch.

From Equation (10.1),

$$\tau_c = \tau_{\text{eng}} - I_p\,\dot{\omega}_p.$$

Substituting Equation (10.19) gives

$$\tau_c = \tau_{\text{eng}} - I_p\big(R\dot{\omega}_s + \omega_s\dot{R}\big). \tag{10.20}$$

From Equation (10.2),

$$\dot{\omega}_s = \frac{R\tau_c - \tau_{\text{load}}}{I_s}.$$

Substituting this into Equation (10.20) gives

$$\begin{aligned}
\tau_c &= \tau_{\text{eng}} - I_p\left[R\left(\frac{R\tau_c - \tau_{\text{load}}}{I_s}\right) + \omega_s\dot{R}\right] \\
&= \tau_{\text{eng}} - \frac{I_p}{I_s}\big(R^2\tau_c - R\tau_{\text{load}}\big) - I_p\omega_s\dot{R}. \qquad (10.21)
\end{aligned}$$

Collecting terms yields

$$\tau_c\left(1 + \frac{I_p}{I_s}R^2\right) = \tau_{\text{eng}} + \frac{I_p}{I_s}R\tau_{\text{load}} - I_p\omega_s\dot{R}. \tag{10.22}$$

Because this derivation has imposed the sticking constraint Equation (10.18), the resulting solution is precisely the no-slip coupling torque.

10.8 Closed-Form No-Slip Coupling Torque

The no-slip coupling torque is therefore

No-Slip Coupling Torque **(10.23)**

$$\tau_{\text{ns}}(t) = \frac{\tau_{\text{eng}}(t) + \dfrac{I_p}{I_s}R(t)\,\tau_{\text{load}}(t) - I_p\,\omega_s(t)\,\dot{R}(t)}{1 + \dfrac{I_p}{I_s}R(t)^2}.$$

Equation Equation (10.23) is the torque required to maintain the no-slip constraint Equation (10.18), or equivalently the condition $v_\Delta = 0$.

10.9 Section Summary

This section derived the no-slip coupling torque τ_{ns} by specializing the rotational dynamics to the sticking branch.

The derivation used three ingredients:

- the primary and secondary rotational equations of motion written in terms of the internal coupling torque τ_c,
- the no-slip kinematic constraint $\omega_p = R\,\omega_s$ and its time derivative $\dot{\omega}_p = R\dot{\omega}_s + \omega_s \dot{R}$,
- the no-slip torque mapping $\tau_{s,\mathrm{ns}} = R\,\tau_c$.

Together, these relations close the sticking branch and yield the explicit result Equation (10.23).

11 Traction Limits and Slip Conditions

In Section 9.4, the sticking branch was defined by the requirement that the no-slip coupling torque remain within an admissible traction interval. In Section 10, the corresponding no-slip torque τ_{ns} was derived explicitly.

The remaining task is therefore to determine the traction bounds themselves.

Because belt–pulley friction is finite, the contact cannot sustain arbitrary tangential traction. For any instantaneous clamping state, only a bounded range of transmitted coupling torque can be supported before adherence is lost. Once the demanded torque leaves that admissible range, the no-slip branch is no longer physically realizable and the transmission must enter slip.

Accordingly, the objective of this section is to derive bounds of the form

$$\tau_- \le \tau \le \tau_+, \tag{11.1}$$

where τ_- and τ_+ are the negative- and positive-direction traction limits, respectively.

These bounds will later be compared against the no-slip torque τ_{ns} to determine whether the transmission remains in stick or transitions to slip.

11.1 Tight–Slack Tension Difference and Torque Capacity

The same traction-capacity argument applies to both pulleys. Accordingly, in this section we use the subscript

$$j \in \{p, s\},$$

where $j = p$ denotes the primary pulley and $j = s$ denotes the secondary pulley.

Consider the wrapped pulley shown in Figure 28. If torque is transmitted from pulley j to the belt, the belt tension cannot be the same on both spans. One side must carry a higher tension than the other. For a given transmission direction, we therefore define

$$\Delta T_j \;\equiv\; T_{j,\mathrm{tight}} - T_{j,\mathrm{slack}}, \tag{11.2}$$

where $T_{j,\mathrm{tight}}$ and $T_{j,\mathrm{slack}}$ are the belt tensions on the two straight spans shown in the figure.

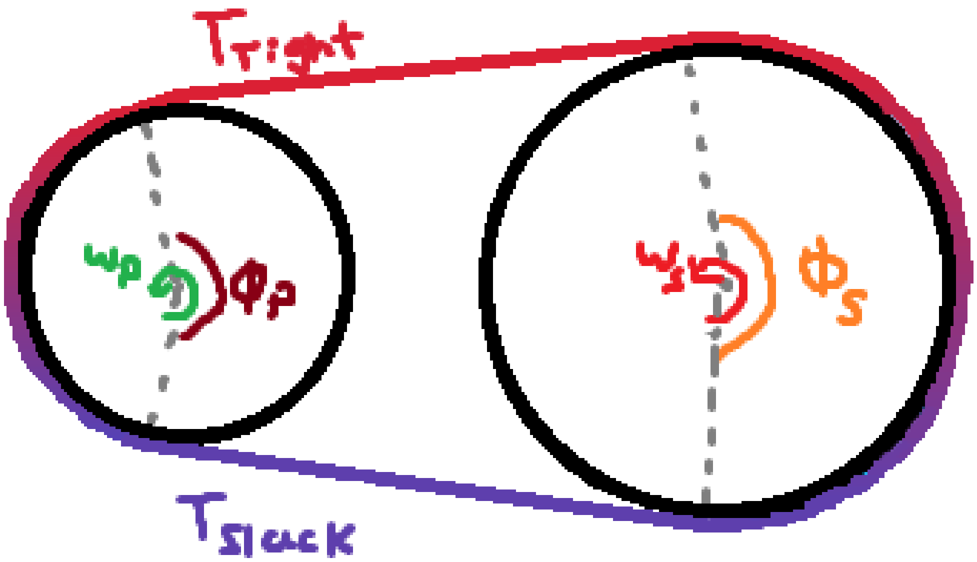

Figure 28: Torque transmission requires a tight–slack tension difference across the wrap.

This tension difference arises because the pulley applies tangential traction to the belt along the wrap angle ϕ_j. The cumulative effect of this traction changes the belt tension progressively from the slack side to the tight side.

At effective contact radius $r_{j,\mathrm{eff}}$, the transmitted torque is

$$\tau_j = r_{j,\mathrm{eff}}\,\Delta T_j, \qquad |\tau_j| = r_{j,\mathrm{eff}}\,|\Delta T_j|. \tag{11.3}$$

Thus, limiting the transmitted torque is equivalent to limiting the admissible tension difference ΔT_j.

Under Assumption 4 (Idealized Belt–Pulley Friction Law) and Assumption 5 (Constant Effective Friction Coefficient), the belt–pulley interface obeys a Coulomb friction law with constant coefficient μ. The tangential traction at any point along the wrap is therefore bounded by the local normal load:

$$|F_f| \leq \mu N.$$

The largest admissible tension difference for a given pulley state and transmission direction occurs when friction is fully mobilized everywhere along the wrap, i.e. at impending slip. Beyond this point, additional tension rise cannot be supported and sliding must occur.

We therefore define a maximum supported tension difference $\Delta T_{j,\max}$, corresponding to the friction-saturated state, and the associated torque capacity

$$\tau_{j,\max} = r_{j,\mathrm{eff}}\,\Delta T_{j,\max}. \tag{11.4}$$

The next subsection computes $\Delta T_{j,\max}$ by examining the incremental change in belt tension along the wrap.

Remark (Friction)

TODO: Discuss other options for modelling friction of the rubber belt, reference that paper. Appendix?

11.2 Differential Belt Element and Tangential Force Balance

To determine the maximum admissible tension difference $\Delta T_{j,\max}$ introduced in Section 11.1, we now examine how belt tension changes *locally* as the belt passes around a generic pulley $j \in \{p, s\}$.

Specifically, we seek an expression for the incremental tension change $\mathrm{d}T_j$ over a differential wrap angle $\mathrm{d}\theta$. Once $\mathrm{d}T_j$ is related to the available friction at the interface, integration over the total wrap angle ϕ_j will yield $\Delta T_{j,\max}$.

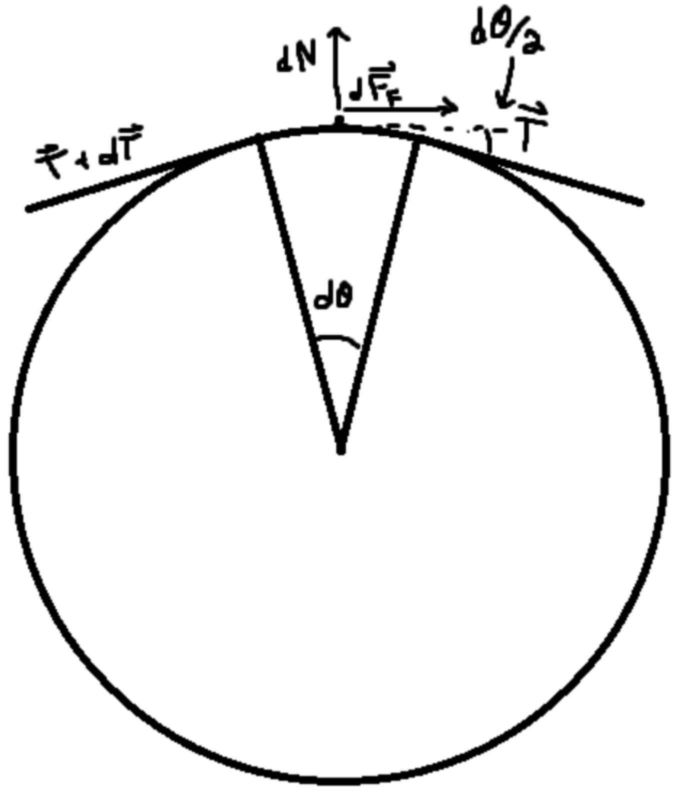

Figure 29: Free-body diagram of a differential belt element spanning $\mathrm{d}\theta$.

Consider the differential belt element shown in Figure 29, spanning an infinitesimal wrap angle $\mathrm{d}\theta$ on pulley j. The belt tension entering the element is T_j, and leaving is $T_j + \mathrm{d}T_j$.

The pulley applies to this element:

- a differential normal force $\mathrm{d}N_j$ (radial),
- a differential tangential friction force $\mathrm{d}F_{f,j}$.

The positive tangential direction is taken along the midpoint tangent of the element, consistent with increasing wrap angle and with increasing belt tension from the slack side toward the tight side.

Tangential Newton's Second Law Applying Theory 1 (Newton's Second Law (Translation)) in the tangential direction gives

$$\sum F_t = \mathrm{d}m_j\, a_{t,j}, \tag{11.5}$$

where $a_{t,j}$ is the tangential acceleration of the belt material at the contact point on pulley j.

The tangential components of the end tensions follow directly from geometry:

$$(T_j)_t = T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right), \tag{11.6}$$

$$(T_j + \mathrm{d}T_j)_t = (T_j + \mathrm{d}T_j)\cos\left(\frac{\mathrm{d}\theta}{2}\right). \tag{11.7}$$

The left-end tension acts in the positive tangential direction, while the right-end tension acts in the negative tangential direction. Including the differential friction force, the tangential force balance becomes

$$T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) - (T_j + \mathrm{d}T_j)\cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_{f,j} = \mathrm{d}m_j\, a_{t,j}. \tag{11.8}$$

Simplifying the first two terms exactly gives

$$-\,\mathrm{d}T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_{f,j} = \mathrm{d}m_j\, a_{t,j}. \tag{11.9}$$

Rearranging,

$$\mathrm{d}T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) = \mathrm{d}F_{f,j} - \mathrm{d}m_j\, a_{t,j}. \tag{11.10}$$

Belt mass element From **??**, the belt mass associated with the differential contact element on pulley j is

$$\mathrm{d}m_j = \rho_b A_b r_{j,\mathrm{cm}}\, \mathrm{d}\theta. \tag{11.11}$$

Thus,

$$\mathrm{d}m_j\, a_{t,j} = \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\, \mathrm{d}\theta. \tag{11.12}$$

Substituting into Equation (11.10) gives

$$\mathrm{d}T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) = \mathrm{d}F_{f,j} - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\, \mathrm{d}\theta. \tag{11.13}$$

Friction limit and impending slip As established in Section 11.1, the maximum admissible tension rise occurs when friction is fully mobilized along the wrap, i.e. at impending slip. At the differential level, this corresponds to

$$\mathrm{d}F_{f,j} = \mu \, \mathrm{d}N_j. \tag{11.14}$$

Substituting Equation (11.14) into Equation (11.13) yields the maximum local tension increase supported without sliding:

$$\mathrm{d}T_{j,\max} = \frac{\mu \, \mathrm{d}N_j - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j} \, \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{11.15}$$

11.3 Integration Across the Wrap

The tight–slack tension difference introduced in Section 11.1 is obtained by accumulating the incremental tension changes $\mathrm{d}T_j$ along the contact arc of pulley $j \in \{p, s\}$. Since the belt tension evolves continuously around the wrap,

$$\Delta T_j = T_{j,\mathrm{tight}} - T_{j,\mathrm{slack}} = \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}T_j. \tag{11.16}$$

Accordingly, the maximum admissible tension difference is obtained by integrating the local maximum increment $\mathrm{d}T_{j,\max}$ from Equation (11.15):

$$\Delta T_{j,\max} = \int_{-\phi_j/2}^{\phi_j/2} \frac{\mu \, \mathrm{d}N_j - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j} \, \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{11.17}$$

In the differential limit $\mathrm{d}\theta \to 0$, $\cos(\mathrm{d}\theta/2) \to 1$, so this reduces to

$$\Delta T_{j,\max} = \int_{-\phi_j/2}^{\phi_j/2} \mu \, \mathrm{d}N_j - \int_{-\phi_j/2}^{\phi_j/2} \rho_b A_b r_{j,\mathrm{cm}} a_{t,j} \, \mathrm{d}\theta. \tag{11.18}$$

The first integral is the total normal load carried over the wrap of pulley j, which we denote by

$$N_{\phi,j} \equiv \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}N_j. \tag{11.19}$$

The explicit relationship between $N_{\phi,j}$ and the axial clamping force is derived in Section 11.5.

From Assumption 9 (Uniform Contact Radius Around Wrap), $r_{j,\mathrm{cm}}$ and $a_{t,j}$ are uniform over the wrap, so the second integral simplifies to $\rho_b A_b r_{j,\mathrm{cm}} a_{t,j} \phi_j$. Hence,

$$\Delta T_{j,\max} = \mu N_{\phi,j} - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j} \phi_j. \tag{11.20}$$

Thus, the maximum admissible tension difference for pulley j depends on two remaining quantities: the total wrap normal load $N_{\phi,j}$ and the tangential belt acceleration $a_{t,j}$.

11.4 Tangential Acceleration of a Belt Element on a Circular Wrap

In Equation (11.20), Newton's second law is applied *along the local direction of belt motion*, so we require an explicit expression for the tangential acceleration $a_{t,j}$ of the belt material on pulley j. Since the inertial term acts on the belt element mass, the relevant kinematics are those of the belt element centroid, located at radius $r_{j,\mathrm{cm}}(t)$.

Goal For a belt element of differential mass $\mathrm{d}m_j$ moving along the wrap of pulley j, the tangential force balance uses

$$\sum F_t = \mathrm{d}m_j \, a_{t,j}.$$

Our goal is therefore to compute $a_{t,j}$ from the belt kinematics when both the local angular speed and the centroid radius may vary:

$$\omega_j(t) = \dot{\theta}_j(t), \qquad \dot{\omega}_j(t), \qquad r_{j,\mathrm{cm}}(t), \qquad \dot{r}_{j,\mathrm{cm}}(t).$$

Coordinate system and unit vectors Because we care about forces *along the belt*, we choose a basis aligned with the local wrap geometry on pulley j:

- $\mathbf{e}_{r,j}(t)$ points radially outward from the pulley center to the belt point,
- $\mathbf{e}_{\theta,j}(t)$ is the unit tangent, obtained by rotating $\mathbf{e}_{r,j}$ by $+90°$ in the direction of increasing θ_j.

The tangential direction of belt motion is therefore exactly $\mathbf{e}_{\theta,j}$.

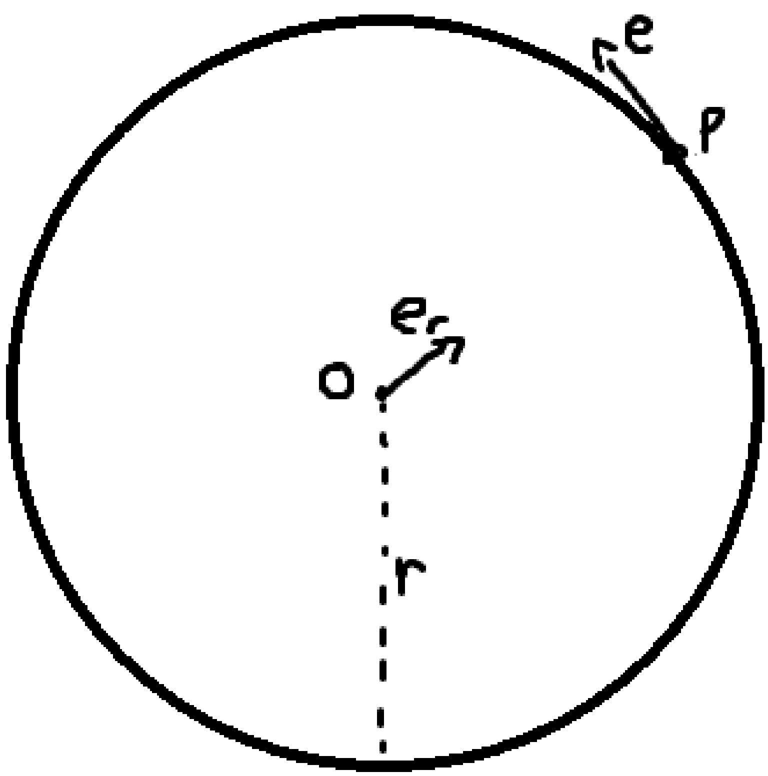

Figure 30: Definition of the radial and tangential unit vectors.

Tangential acceleration as a component of a Tangential acceleration is the component of the acceleration vector along $\mathbf{e}_{\theta,j}$:

$$a_{t,j} \equiv \mathbf{a}_j \cdot \mathbf{e}_{\theta,j}. \tag{11.21}$$

We therefore compute the full acceleration $\mathbf{a}_j = \dot{\mathbf{v}}_j$ and then read off its $\mathbf{e}_{\theta,j}$ component.

Step 1: position and velocity The position of the belt element centroid relative to the pulley center is

$$\mathbf{r}_j(t) = r_{j,\mathrm{cm}}(t)\, \mathbf{e}_{r,j}(t). \tag{11.22}$$

Differentiating Equation (11.22) gives

$$\mathbf{v}_j = \dot{\mathbf{r}}_j = \dot{r}_{j,\mathrm{cm}}\, \mathbf{e}_{r,j} + r_{j,\mathrm{cm}}\, \dot{\mathbf{e}}_{r,j}. \tag{11.23}$$

Thus, even if $r_{j,\mathrm{cm}}$ were constant, the velocity would still depend on the rotation of the local basis.

Step 2: time derivatives of the unit vectors Over a small angular change $\mathrm{d}\theta_j$, the unit vectors rotate rigidly by the same angle. Hence

$$\frac{\mathrm{d}\mathbf{e}_{r,j}}{\mathrm{d}\theta_j} = \mathbf{e}_{\theta,j}, \qquad \frac{\mathrm{d}\mathbf{e}_{\theta,j}}{\mathrm{d}\theta_j} = -\mathbf{e}_{r,j}. \tag{11.24}$$

Using the chain rule with $\omega_j = \dot{\theta}_j$, these become

$$\dot{\mathbf{e}}_{r,j} = \omega_j\, \mathbf{e}_{\theta,j}, \qquad \dot{\mathbf{e}}_{\theta,j} = -\omega_j\, \mathbf{e}_{r,j}. \tag{11.25}$$

Step 3: velocity in radial–tangential form Substituting Equation (11.25) into Equation (11.23) gives

$$\mathbf{v}_j = \dot{r}_{j,\mathrm{cm}}\, \mathbf{e}_{r,j} + r_{j,\mathrm{cm}}\omega_j\, \mathbf{e}_{\theta,j}. \tag{11.26}$$

Thus, $\dot{r}_{j,\mathrm{cm}}$ is the radial speed and $r_{j,\mathrm{cm}}\omega_j$ is the tangential speed.

Step 4: acceleration and tangential component Differentiating Equation (11.26) gives

$$\begin{aligned} \mathbf{a}_j = \dot{\mathbf{v}}_j &= \frac{\mathrm{d}}{\mathrm{d}t}\big(\dot{r}_{j,\mathrm{cm}}\, \mathbf{e}_{r,j}\big) + \frac{\mathrm{d}}{\mathrm{d}t}\big(r_{j,\mathrm{cm}}\omega_j\, \mathbf{e}_{\theta,j}\big) \\ &= \ddot{r}_{j,\mathrm{cm}}\, \mathbf{e}_{r,j} + \dot{r}_{j,\mathrm{cm}}\, \dot{\mathbf{e}}_{r,j} + (\dot{r}_{j,\mathrm{cm}}\, \omega_j + r_{j,\mathrm{cm}}\, \dot{\omega}_j)\mathbf{e}_{\theta,j} + r_{j,\mathrm{cm}}\omega_j\, \dot{\mathbf{e}}_{\theta,j}. \end{aligned} \tag{11.27}$$

Substituting the unit-vector derivatives from Equation (11.25) and collecting like terms yields

$$\begin{aligned} \mathbf{a}_j &= \ddot{r}_{j,\mathrm{cm}}\, \mathbf{e}_{r,j} + \dot{r}_{j,\mathrm{cm}}(\omega_j\, \mathbf{e}_{\theta,j}) + (\dot{r}_{j,\mathrm{cm}}\, \omega_j + r_{j,\mathrm{cm}}\, \dot{\omega}_j)\mathbf{e}_{\theta,j} + r_{j,\mathrm{cm}}\omega_j(-\omega_j\, \mathbf{e}_{r,j}) \\ &= \big(\ddot{r}_{j,\mathrm{cm}} - r_{j,\mathrm{cm}}\omega_j^2\big)\mathbf{e}_{r,j} + \big(r_{j,\mathrm{cm}}\dot{\omega}_j + 2\dot{r}_{j,\mathrm{cm}}\, \omega_j\big)\mathbf{e}_{\theta,j}. \end{aligned} \tag{11.28}$$

By Equation (11.21), the tangential acceleration is therefore

$$a_{t,j} = r_{j,\mathrm{cm}}\dot{\omega}_j + 2\dot{r}_{j,\mathrm{cm}}\, \omega_j. \tag{11.29}$$

Origin of the factor of 2 The two $\dot{r}_{j,\mathrm{cm}}\,\omega_j$ contributions in Equation (11.28) arise from two distinct effects:

- $\dot{r}_{j,\mathrm{cm}}\,\dot{\mathbf{e}}_{r,j}$: the radial velocity $\dot{r}_{j,\mathrm{cm}}\,\mathbf{e}_{r,j}$ acquires a tangential contribution because the radial direction itself rotates at rate ω_j,
- $\frac{\mathrm{d}}{\mathrm{d}t}(r_{j,\mathrm{cm}}\omega_j)$: the tangential speed changes when the centroid radius changes.

Together, these give the total term $2\dot{r}_{j,\mathrm{cm}}\,\omega_j$.

11.5 Normal Force from Axial Clamping

The traction bound Equation (11.20) depends on the total normal load over the wrap of pulley j,

$$N_{\phi,j} \equiv \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}N_j.$$

We now express $N_{\phi,j}$ in terms of the axial clamping force generated by that pulley.

Axial clamp produces radial normal load The belt is pressed between two opposing conical sheaves. On pulley j, an axial clamping force $F_{\mathrm{ax},j}$ pushes the sheaves together. Because the sheave faces are inclined at cone half-angle β, this axial force generates a radial normal load on the belt.

From the conical geometry relation established in Equation (7.2),

$$\frac{\mathrm{d}r_j}{\mathrm{d}s} = \frac{1}{2\tan\beta}.$$

To relate the axial and radial forces consistently, we use virtual work. The incremental work done by the axial clamp must equal the incremental work associated with radial compression of the belt:

$$F_{\mathrm{ax},j}\ \mathrm{d}s = F_{\mathrm{rad},j}\ \mathrm{d}r_j.$$

Substituting $\mathrm{d}r_j = (\mathrm{d}r_j/\,\mathrm{d}s)\ \mathrm{d}s$ gives

$$F_{\mathrm{rad},j} = \frac{F_{\mathrm{ax},j}}{\mathrm{d}r_j/\,\mathrm{d}s} = 2F_{\mathrm{ax},j}\tan\beta. \tag{11.30}$$

Thus, the axial clamp force on pulley j produces a total radial normal load

$$F_{\mathrm{rad},j} = 2F_{\mathrm{ax},j}\tan\beta. \tag{11.31}$$

Identification with the wrap-normal load The radial load $F_{\mathrm{rad},j}$ derived above represents the total inward force applied by the two sheave faces of pulley j onto the belt. That load is carried by the distributed belt–pulley contact forces $\mathrm{d}N_j$ along the wrap.

Therefore, the total normal load available for traction on pulley j is

$$\boxed{N_{\phi,j} = F_{\mathrm{rad},j} = 2F_{\mathrm{ax},j}\tan\beta.} \tag{11.32}$$

This result states that the total wrap normal load, and hence the normal load available for traction, is set directly by the axial clamping force through the cone geometry.

11.6 Torque Capacity

From Section 11.1, the maximum transmissible torque on pulley j is determined by the maximum admissible tension difference across the wrap:

$$\tau_{j,\max} = r_{j,\text{eff}}\, \Delta T_{j,\max}. \tag{11.33}$$

Using the integrated result from Equation (11.20), together with the wrap-normal relation Equation (11.32) and the belt tangential acceleration from Equation (11.29),

$$\Delta T_{j,\max} = \mu N_{\phi,j} - \rho_b A_b r_{j,\text{cm}} a_{t,j} \phi_j, \qquad N_{\phi,j} = 2F_{\text{ax},j} \tan\beta, \qquad a_{t,j} = r_{j,\text{cm}}\dot{\omega}_j + 2\dot{r}_{j,\text{cm}}\,\omega_j,$$

the explicit torque-capacity expression becomes

Pulley Torque Capacity **(11.34)**

$$\tau_{j,\max} = r_{j,\text{eff}} \left[2\mu F_{\text{ax},j} \tan\beta - \rho_b A_b r_{j,\text{cm}} \phi_j \left(r_{j,\text{cm}}\dot{\omega}_j + 2\dot{r}_{j,\text{cm}}\,\omega_j\right)\right].$$

The remaining task is to substitute the appropriate rotational acceleration $\dot{\omega}_j$ and axial clamp force $F_{\text{ax},j}$ for each pulley to obtain explicit bounds on the coupling torque.

Remark (Why this differs from the classical capstan exponential)

In the classical capstan problem, the normal load is generated *by the belt tension* itself (a larger tension implies a proportionally larger contact normal), which leads to the exponential relation $T_1/T_2 = e^{\mu\phi}$.
Here, the normal load is imposed externally by the clamping mechanism: N_ϕ is determined by F_{ax} through Equation (11.32). Therefore, the available traction scales *linearly* with clamping force, not exponentially with belt tension. Centrifugal loading and ratio change enter additionally through the inertial term in Equation (11.34).

Remark (Torque-Reactive Clamping and Implicit Capacity at the Secondary)

For the secondary pulley, the specific axial clamp force F_{ax} used in this model depends on the transmitted torque τ_s (see Equation (8.16)). However, the traction capacity $\tau_{\max}$ derived above depends on F_{ax} through Equation (11.32) and Equation (11.34).
Consequently, at the secondary interface the admissible torque satisfies an implicit relation of the form

$$|\tau_s| \le \tau_{\max}\big(F_{\text{ax}}(\tau_s)\big).$$

In other words, the transmitted torque influences the clamp force, the clamp force determines the traction capacity, and that capacity bounds the transmitted torque. This circular dependence arises only because the secondary mechanism is torque-reactive.
For the specific secondary model adopted in this work, this implicit inequality can be reduced to a closed-form solution (see Appendix D). The explicit algebra is omitted here to preserve clarity, as it depends on the detailed functional form of $F_{\text{ax}}(s, \tau_s)$ and is not required to understand the main traction derivation.

11.7 Section Conclusion and Transition

This section derived a traction-limited torque bound for a single pulley.

Equation Equation (11.34) gives the maximum torque that can be transmitted at the current operating state. It depends only on:

- axial clamping force F_{ax},
- cone angle β,
- wrap angle ϕ,
- belt mass and kinematics.

No exponential amplification occurs because the normal force is externally imposed and bounded.

12 Pulley-Specific Traction Bounds and Coupling Torque Closure

The generic capacity law Equation (11.34) gives the maximum transmissible torque for a pulley j in terms of its local clamping force, geometry, and belt kinematics. To obtain usable limits on the coupling torque, this expression must now be specialized to the primary and secondary pulleys.

For each pulley, the procedure is the same:

1. identify the torque transmitted at that pulley,
2. substitute the appropriate rotational acceleration,
3. substitute the pulley-specific axial clamping-force model,
4. solve the resulting implicit inequality for the admissible torque range.

On the primary side this is relatively direct, since the axial clamp force does not depend explicitly on the transmitted torque. On the secondary side, the torque-reactive helix introduces an additional dependence on transmitted torque, so the algebra is more involved.

The resulting admissible interval for the coupling torque is obtained by intersecting the bounds contributed by the two pulleys.

12.1 Primary Pulley Traction Bounds

On the primary pulley, the transmitted torque at the belt contact is the coupling torque itself. Thus, applying Equation (11.34) with $j = p$ gives

$$\tau_{p,\max} = r_{p,\text{eff}} \left[2\mu F_{\text{ax},p} \tan\beta - \rho_b A_b r_{p,\text{cm}} \phi_p \left(r_{p,\text{cm}} \dot{\omega}_p + 2\dot{r}_{p,\text{cm}}\, \omega_p \right) \right]. \tag{12.1}$$

Therefore, admissible transmission at the primary requires

$$|\tau_c| \leq \tau_{p,\max}. \tag{12.2}$$

From the primary rotational dynamics Equation (10.1),

$$\dot{\omega}_p = \frac{\tau_{\text{eng}} - \tau_c}{I_p}. \tag{12.3}$$

Substituting Equation (12.3) into Equation (12.1) gives

$$|\tau_c| \le r_{p,\text{eff}} \left[2\mu F_{\text{ax},p} \tan\beta - \rho_b A_b r_{p,\text{cm}} \phi_p \left(r_{p,\text{cm}} \frac{\tau_{\text{eng}} - \tau_c}{I_p} + 2\dot{r}_{p,\text{cm}}\, \omega_p \right) \right]. \tag{12.4}$$

Since τ_c appears on both sides, the positive and negative directions must be treated separately.

Positive-direction bound $\tau_c \ge 0$ For $\tau_c \ge 0$, $|\tau_c| = \tau_c$, so

$$\tau_c \le r_{p,\text{eff}} \left[2\mu F_{\text{ax},p} \tan\beta - \rho_b A_b r_{p,\text{cm}} \phi_p \left(r_{p,\text{cm}} \frac{\tau_{\text{eng}} - \tau_c}{I_p} + 2\dot{r}_{p,\text{cm}}\, \omega_p \right) \right]$$

$$\frac{\tau_c}{r_{p,\text{eff}}} - \rho_b A_b r_{p,\text{cm}} \phi_p \frac{r_{p,\text{cm}}}{I_p} \tau_c \le 2\mu F_{\text{ax},p} \tan\beta - \rho_b A_b r_{p,\text{cm}} \phi_p \left(\frac{r_{p,\text{cm}}}{I_p} \tau_{\text{eng}} + 2\dot{r}_{p,\text{cm}}\, \omega_p \right). \tag{12.5}$$

Hence,

$$\tau_c \le \tau_{p,+}, \qquad \tau_{p,+} = \frac{2\mu F_{\text{ax},p} \tan\beta - \rho_b A_b r_{p,\text{cm}} \phi_p \left(\frac{r_{p,\text{cm}}}{I_p} \tau_{\text{eng}} + 2\dot{r}_{p,\text{cm}}\, \omega_p \right)}{\frac{1}{r_{p,\text{eff}}} - \rho_b A_b r_{p,\text{cm}} \phi_p \frac{r_{p,\text{cm}}}{I_p}}. \tag{12.6}$$

Negative-direction bound $\tau_c \le 0$ For $\tau_c \le 0$, $|\tau_c| = -\tau_c$, so

$$-\tau_c \le r_{p,\text{eff}} \left[2\mu F_{\text{ax},p} \tan\beta - \rho_b A_b r_{p,\text{cm}} \phi_p \left(r_{p,\text{cm}} \frac{\tau_{\text{eng}} - \tau_c}{I_p} + 2\dot{r}_{p,\text{cm}}\, \omega_p \right) \right]$$

$$-\frac{\tau_c}{r_{p,\text{eff}}} - \rho_b A_b r_{p,\text{cm}} \phi_p \frac{r_{p,\text{cm}}}{I_p} \tau_c \le 2\mu F_{\text{ax},p} \tan\beta - \rho_b A_b r_{p,\text{cm}} \phi_p \left(\frac{r_{p,\text{cm}}}{I_p} \tau_{\text{eng}} + 2\dot{r}_{p,\text{cm}}\, \omega_p \right). \tag{12.7}$$

Hence,

$$\tau_c \ge \tau_{p,-}, \qquad \tau_{p,-} = -\frac{2\mu F_{\text{ax},p} \tan\beta - \rho_b A_b r_{p,\text{cm}} \phi_p \left(\frac{r_{p,\text{cm}}}{I_p} \tau_{\text{eng}} + 2\dot{r}_{p,\text{cm}}\, \omega_p \right)}{\frac{1}{r_{p,\text{eff}}} + \rho_b A_b r_{p,\text{cm}} \phi_p \frac{r_{p,\text{cm}}}{I_p}}. \tag{12.8}$$

Therefore, the primary pulley contributes the admissible interval

$$\boxed{\tau_{p,-} \le \tau_c \le \tau_{p,+}.} \tag{12.9}$$

Finally, substituting the primary axial-force model Equation (8.10) into Equation (12.6) and Equation (12.8), with

$$F_{\text{ax},p} = m_f \omega_p^2\, r_f(s)\, \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p (x_{p,0} + s),$$

gives the explicit primary traction bounds

Primary Traction Upper Bound **(12.10)**

$$\tau_{p,+} = \frac{2\mu \tan\beta \left[m_f \omega_p^2 r_f(s) \, \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p \big(x_{p,0} + s\big) \right] - \rho_b A_b r_{p,\mathrm{cm}} \phi_p \left(\frac{r_{p,\mathrm{cm}}}{I_p} \tau_{\mathrm{eng}} + 2\dot{r}_{p,\mathrm{cm}} \, \omega_p \right)}{\frac{1}{r_{p,\mathrm{eff}}} - \rho_b A_b r_{p,\mathrm{cm}} \phi_p \frac{r_{p,\mathrm{cm}}}{I_p}},$$

and

Primary Traction Lower Bound **(12.11)**

$$\tau_{p,-} = -\frac{2\mu \tan\beta \left[m_f \omega_p^2 r_f(s) \, \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p \big(x_{p,0} + s\big) \right] - \rho_b A_b r_{p,\mathrm{cm}} \phi_p \left(\frac{r_{p,\mathrm{cm}}}{I_p} \tau_{\mathrm{eng}} + 2\dot{r}_{p,\mathrm{cm}} \, \omega_p \right)}{\frac{1}{r_{p,\mathrm{eff}}} + \rho_b A_b r_{p,\mathrm{cm}} \phi_p \frac{r_{p,\mathrm{cm}}}{I_p}}.$$

12.2 Secondary Pulley Traction Bounds

On the secondary pulley, the transmitted torque at the belt contact is τ_s. Applying Equation (11.34) with $j = s$ therefore gives

$$\tau_{s,\max} = r_{s,\mathrm{eff}} \left[2\mu F_{\mathrm{ax},s} \tan\beta - \rho_b A_b r_{s,\mathrm{cm}} \phi_s \left(r_{s,\mathrm{cm}} \dot{\omega}_s + 2\dot{r}_{s,\mathrm{cm}} \, \omega_s \right) \right]. \tag{12.12}$$

Accordingly, admissible transmission at the secondary requires

$$|\tau_s| \le \tau_{s,\max}. \tag{12.13}$$

From the secondary rotational dynamics Equation (10.2),

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\mathrm{load}}}{I_s}. \tag{12.14}$$

From the secondary axial-force model Equation (8.16),

$$F_{\mathrm{ax},s}(s, \tau_s) = \frac{\tau_s + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)}{2} \frac{\mathrm{d}\theta_s}{\mathrm{d}s} + k_{s,x}(x_{s,0} - s). \tag{12.15}$$

Substituting Equation (12.14) and Equation (12.15) into Equation (12.12) gives the implicit secondary traction inequality

$$\begin{aligned} |\tau_s| \le r_{s,\mathrm{eff}} \Bigg[& 2\mu \tan\beta \left(\frac{\tau_s + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)}{2} \frac{\mathrm{d}\theta_s}{\mathrm{d}s} + k_{s,x}(x_{s,0} - s) \right) \\ & - \rho_b A_b r_{s,\mathrm{cm}} \phi_s \left(r_{s,\mathrm{cm}} \frac{\tau_s - \tau_{\mathrm{load}}}{I_s} + 2\dot{r}_{s,\mathrm{cm}} \, \omega_s \right) \Bigg]. \end{aligned} \tag{12.16}$$

Since τ_s appears both through the acceleration term and through the torque-reactive axial clamping force, the positive and negative transmission directions must again be treated separately.

Positive-direction bound $\tau_s \geq 0$ For $\tau_s \geq 0$, $|\tau_s| = \tau_s$. Collecting the τ_s terms on the left gives

$$\begin{aligned} \frac{\tau_s}{r_{s,\text{eff}}} - \mu \tan\beta \frac{\mathrm{d}\theta_s}{\mathrm{d}s}\, \tau_s \rho_b A_b r_{s,\text{cm}} \phi_s \frac{r_{s,\text{cm}}}{I_s}\, \tau_s \; &\leq \mu \tan\beta \frac{\mathrm{d}\theta_s}{\mathrm{d}s} k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) \\ &\quad + 2\mu \tan\beta\, k_{s,x}(x_{s,0} - s) \\ &\quad + \rho_b A_b r_{s,\text{cm}} \phi_s \left(\frac{r_{s,\text{cm}}}{I_s} \tau_{\text{load}} - 2\dot{r}_{s,\text{cm}}\, \omega_s \right). \end{aligned} \tag{12.17}$$

Hence,

$$\tau_s \leq \tau_{s,+}, \tag{12.18}$$

with

Secondary Positive Traction Limit **(12.19)**

$$\tau_{s,+} = \frac{\mu \tan\beta \left[\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) + 2k_{s,x}(x_{s,0} - s) \right] + \rho_b A_b r_{s,\text{cm}} \phi_s \left(\frac{r_{s,\text{cm}}}{I_s} \tau_{\text{load}} - 2\dot{r}_{s,\text{cm}}\, \omega_s \right)}{\frac{1}{r_{s,\text{eff}}} - \mu \tan\beta \frac{\mathrm{d}\theta_s}{\mathrm{d}s} + \rho_b A_b r_{s,\text{cm}} \phi_s \frac{r_{s,\text{cm}}}{I_s}}.$$

Negative-direction bound $\tau_s \leq 0$ For $\tau_s \leq 0$, $|\tau_s| = -\tau_s$. Collecting the τ_s terms on the left gives

$$\begin{aligned} -\frac{\tau_s}{r_{s,\text{eff}}} - \mu \tan\beta \frac{\mathrm{d}\theta_s}{\mathrm{d}s}\, \tau_s + \rho_b A_b r_{s,\text{cm}} \phi_s \frac{r_{s,\text{cm}}}{I_s}\, \tau_s \; &\leq \mu \tan\beta \frac{\mathrm{d}\theta_s}{\mathrm{d}s} k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) \\ &\quad + 2\mu \tan\beta\, k_{s,x}(x_{s,0} - s) \\ &\quad + \rho_b A_b r_{s,\text{cm}} \phi_s \left(\frac{r_{s,\text{cm}}}{I_s} \tau_{\text{load}} - 2\dot{r}_{s,\text{cm}}\, \omega_s \right). \end{aligned} \tag{12.20}$$

Hence,

$$\tau_s \geq \tau_{s,-}, \tag{12.21}$$

with

Secondary Negative Traction Limit **(12.22)**

$$\tau_{s,-} = -\frac{\mu \tan\beta \left[\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) + 2k_{s,x}(x_{s,0} - s) \right] + \rho_b A_b r_{s,\text{cm}} \phi_s \left(\frac{r_{s,\text{cm}}}{I_s} \tau_{\text{load}} - 2\dot{r}_{s,\text{cm}}\, \omega_s \right)}{\frac{1}{r_{s,\text{eff}}} + \mu \tan\beta \frac{\mathrm{d}\theta_s}{\mathrm{d}s} - \rho_b A_b r_{s,\text{cm}} \phi_s \frac{r_{s,\text{cm}}}{I_s}}.$$

Therefore, the secondary pulley contributes the admissible interval

$$\boxed{\tau_{s,-} \leq \tau_s \leq \tau_{s,+}.} \tag{12.23}$$

12.3 Combined Traction-Limited Coupling Law

The previous two subsections produced admissible traction intervals for each pulley separately:

$$\tau_{p,-} \leq \tau_c \leq \tau_{p,+}, \qquad \tau_{s,-} \leq \tau_s \leq \tau_{s,+}.$$

Since the transmitted state must satisfy the traction limits at *both* pulleys simultaneously, the final admissible range is obtained by intersecting these two constraints.

Using the torque mapping

$$\tau_s = R\,\tau_c,$$

the secondary bounds may be written in terms of the coupling torque as

$$\frac{\tau_{s,-}}{R} \le \tau_c \le \frac{\tau_{s,+}}{R}.$$

Therefore, the overall admissible coupling-torque interval is

$$\tau_- \le \tau_c \le \tau_+, \tag{12.24}$$

with

$$\tau_- = \max\!\left(\tau_{p,-}, \frac{\tau_{s,-}}{R}\right), \qquad \tau_+ = \min\!\left(\tau_{p,+}, \frac{\tau_{s,+}}{R}\right). \tag{12.25}$$

These are the final negative- and positive-direction traction limits used in the regime law.

Stick and slip regimes As established in Section 9.4, the no-slip branch is admissible only when two conditions hold simultaneously:

1. the no-slip torque lies within the admissible interval, and
2. the relative slip metric satisfies $v_\Delta = 0$.

If both conditions are satisfied, the transmission remains in stick and the coupling torque is the no-slip torque τ_{ns}.

If instead the no-slip torque lies outside the admissible interval, or if a nonzero slip mismatch is already present, then the transmission is on a traction-limited slip branch. The active slip branch is selected by the sign of v_Δ, as introduced in Section 9.4.

This gives the final coupling law

Traction-Limited Coupling Torque Law **(12.26)**

$$\tau_c = \begin{cases} \tau_{\mathrm{ns}}, & v_\Delta = 0 \text{ and } \tau_- \le \tau_{\mathrm{ns}} \le \tau_+, \\ \tau_+, & v_\Delta > 0 \text{ or } \big(v_\Delta = 0 \text{ and } \tau_{\mathrm{ns}} > \tau_+\big), \\ \tau_-, & v_\Delta < 0 \text{ or } \big(v_\Delta = 0 \text{ and } \tau_{\mathrm{ns}} < \tau_-\big). \end{cases}$$

Interpretation The logic of Equation (12.26) is as follows:

- If the sticking solution is both kinematically compatible ($v_\Delta = 0$) and traction-admissible ($\tau_- \le \tau_{\mathrm{ns}} \le \tau_+$), then the transmission remains on the no-slip branch.
- If the demanded no-slip torque exceeds the positive traction limit, the contact saturates on the positive slip branch.

- If the demanded no-slip torque falls below the negative traction limit, the contact saturates on the negative slip branch.
- Once slip is present, the transmission remains on the corresponding slip branch until the mismatch v_Δ is driven back to zero.

Equation (12.26) therefore closes the coupling-torque model across both stick and slip regimes.

13 Complete CVT Dynamic Model

This section gathers the results of the previous derivations into one closed nonlinear dynamical model. No new physics is introduced here. The goal is simply to summarize, in one place,

1. the chosen state variables,
2. the algebraic quantities computed from those states,
3. the stick–slip torque closure,
4. and the final governing ODEs.

13.1 State Definition

The state vector is

$$\mathbf{x} = \big(\omega_p,\ \omega_s,\ s,\ \dot{s}\big), \tag{13.1}$$

where

- ω_p is the primary pulley angular velocity,
- ω_s is the secondary pulley angular velocity,
- s is the axial shift coordinate,
- $\dot{s}$ is the axial shift velocity.

13.2 Geometry and Ratio Closure

At each instant, the shift coordinate s determines the pulley geometry through the belt-length constraint and the conical kinematic relations derived earlier. In particular, the effective radii and wrap angles are functions of s, and the geometric CVT ratio is

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \tag{13.2}$$

Its time derivative is

$$\dot{R} = \dot{R}(s, \dot{s}), \tag{13.3}$$

as given by Equation (7.57).

Thus, once $(s, \dot{s})$ are known, the model can evaluate

$$r_{p,\text{eff}}, \quad r_{s,\text{eff}}, \quad r_{p,\text{cm}}, \quad r_{s,\text{cm}}, \quad \phi_p, \quad \phi_s, \quad R, \quad \dot{R},$$

together with any other geometry-dependent terms appearing in the force and traction models.

13.3 Torque Closure

The internal torque transmitted through the CVT is the coupling torque τ_c. This is the central algebraic unknown of the transmission model.

No-slip torque demand. If sticking is maintained, the required coupling torque is the no-slip torque derived in Equation (10.23):

$$\tau_{\text{ns}} = \frac{\tau_{\text{eng}} + \dfrac{I_p}{I_s} R\, \tau_{\text{load}} - I_p\, \omega_s\, \dot{R}}{1 + \dfrac{I_p}{I_s} R^2}. \tag{13.4}$$

Traction bounds. The pulley-specific traction limits derived in Section 12.1 and Section 12.2 combine to give the final admissible interval

$$\tau_- \le \tau_c \le \tau_+, \tag{13.5}$$

with

$$\tau_- = \max\!\left(\tau_{p,-}, \frac{\tau_{s,-}}{R}\right), \qquad \tau_+ = \min\!\left(\tau_{p,+}, \frac{\tau_{s,+}}{R}\right), \tag{13.6}$$

as given by Equation (12.25).

Slip metric and regime selection. The slip metric from Equation (9.15) is

$$v_\Delta = r_{p,\text{eff}}\, \omega_p - r_{s,\text{eff}}\, \omega_s. \tag{13.7}$$

Using τ_{ns}, τ_-, τ_+, and v_Δ, the actual coupling torque is selected by the stick–slip law Equation (12.26):

$$\tau_c = \begin{cases} \tau_{\text{ns}}, & v_\Delta = 0 \text{ and } \tau_- \le \tau_{\text{ns}} \le \tau_+, \\ \tau_+, & v_\Delta > 0 \text{ or } \big(v_\Delta = 0 \text{ and } \tau_{\text{ns}} > \tau_+\big), \\ \tau_-, & v_\Delta < 0 \text{ or } \big(v_\Delta = 0 \text{ and } \tau_{\text{ns}} < \tau_-\big). \end{cases} \tag{13.8}$$

13.4 Governing ODEs

With the coupling torque now closed, the rotational dynamics are

$$\dot{\omega}_p = \frac{1}{I_p}\left(\tau_{\text{eng}}(\omega_p) - \tau_c\right), \tag{13.9}$$

$$\dot{\omega}_s = \frac{1}{I_s}\left(R\, \tau_c - \tau_{\text{load}}(\omega_s)\right), \tag{13.10}$$

from Equation (10.1) and Equation (10.2).

The axial shift dynamics are

$$\dot{s} = \dot{s}, \tag{13.11}$$

$$\ddot{s} = \frac{\Big(F_{p,\text{ax}}(s, \omega_p) + F_{c,p}(s, \omega_p)\Big) - \Big(F_{s,\text{ax}}(s, R\tau_c) + F_{c,s}(s, \omega_s)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}, \tag{13.12}$$

from Equation (8.35), with the secondary axial-force term evaluated using $\tau_s = R\tau_c$.

13.5 Model Evaluation Sequence

Given the current state

$$\mathbf{x}(t) = \big(\omega_p, \omega_s, s, \dot{s}\big),$$

the model is evaluated in the following order:

1. Compute the geometry from s, including the effective radii, centroid radii, and wrap angles.
2. Compute $R(s)$ from Equation (13.2) and $\dot{R}(s, \dot{s})$ from Equation (13.3).
3. Evaluate the no-slip torque demand τ_{ns} from Equation (13.4).
4. Evaluate the pulley-specific traction bounds $\tau_{p,\pm}$ and $\tau_{s,\pm}$, then combine them into τ_- and τ_+ using Equation (13.6).
5. Evaluate the slip metric v_Δ from Equation (13.7), then select the actual coupling torque τ_c from Equation (13.8).
6. Evaluate the ODE right-hand sides Equation (13.9), Equation (13.10), Equation (13.11), and Equation (13.12).

13.6 Complete Nonlinear ODE System

The complete closed CVT model is therefore

$$\boxed{\begin{aligned}
\dot{\omega}_p &= \frac{1}{I_p}\left(\tau_{\text{eng}}(\omega_p) - \tau_c\right), \\
\dot{\omega}_s &= \frac{1}{I_s}\left(R(s)\,\tau_c - \tau_{\text{load}}(\omega_s)\right), \\
\frac{d}{dt}s &= \dot{s}, \\
\frac{d}{dt}\dot{s} &= \frac{\Big(F_{p,\text{ax}}(s,\omega_p) + F_{c,p}(s,\omega_p)\Big) - \Big(F_{s,\text{ax}}(s, R(s)\tau_c) + F_{c,s}(s,\omega_s)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}},
\end{aligned}} \tag{13.13}$$

with the algebraic closures

$$R = R(s), \qquad \dot{R} = \dot{R}(s,\dot{s}), \qquad \tau_c \text{ from Equation (13.8)},$$

In short, the model closes in the following way: the shift state determines the geometry, the geometry determines both the no-slip torque demand and the traction limits, the regime law selects the actual transmitted coupling torque, and that torque feeds back into both the rotational and axial dynamics.

14 Simulation Results

Because experimental drivetrain data for the present vehicle configuration was not available at the time of writing, the results presented here focus on establishing physical credibility of the model through three complementary strategies:

1. Verification that the dynamic model reproduces known quasi–static behaviour in appropriate limits.
2. Demonstration that the model produces trends consistent with established physical intuition of CVT operation.
3. Exploration of transient behaviours and force interactions that cannot be captured by steady–state formulations.

Accordingly, the results are structured to progressively build confidence in the formulation. We begin with a summary of simulation parameters, then verify steady–state force balance behaviour, examine the magnitude of the centrifugal belt correction, and finally present transient simulations of torque transmission and full vehicle launch behaviour.

14.1 Simulation Parameters

All simulations were performed using the nominal parameter set summarized in Table 6. Symbol definitions are centralized in Section 3.5; this results table reports the nominal values used in simulation together with their sources.

Table 6: Nominal simulation parameters grouped by subsystem. Parameter sources are indicated as CAD (measured from geometry), Spec (manufacturer specification), Lit. (literature value), or Est. (engineering estimate).

Subsystem	Symbol(s)	Parameter	Nominal value	Unit	Source
Primary clutch	m_f	Flyweight mass	–	kg	CAD / Spec
	$r_{f,0}$	Flyweight radius	–	m	CAD
	$r_f(s)$	Ramp geometry profile	profile curve	–	CAD
	k_p	Primary spring rate	–	N/m	Spec
	$x_{p,0}$	Primary spring preload	–	m	Spec
	β	Sheave half-angle	0.2007	rad	CAD
	$s_{\mathrm{dz}},\ s_{\max}$	Minimum / maximum shift position	–	m	CAD
Secondary clutch	$\theta_s(s)$	Helix angle profile	profile curve	–	CAD
	$k_{s,\theta}$	Torsional spring rate	–	N m/rad	Spec
	$\theta_{s,0}$	Torsional spring preload	–	rad	Spec
	$k_{s,x}$	Compression spring rate	–	N/m	Spec
	β	Sheave half-angle	0.2007	rad	CAD
Belt	ρ_b	Mass per unit length	–	kg/m	Lit.
	β	Belt V-angle	0.2007	rad	Spec
	μ	Friction coefficient	–	1	Est.
	L_b	Free belt length	0.9535	m	CAD
	$h_b,\ b_{\mathrm{out}},\ b_{\mathrm{in}}$	Cross–section geometry	$h_b = 0.01557$; others –	m	CAD
Vehicle	m	Total vehicle mass	–	kg	Measured
	r_w	Wheel radius	–	m	CAD
	G	Final drive ratio	–	1	Spec
	C_d	Drag coefficient	–	1	Est.
	A_f	Frontal area	–	m^2	CAD
	C_{rr}	Rolling resistance coefficient	–	1	Lit.
Engine	$\tau_{\mathrm{eng}}(\omega_p)$	Torque curve	lookup/fit curve	N m	Spec

Parameters derived directly from geometry (e.g. ramp profiles, helix geometry, belt path) are known with relatively high confidence. By contrast, friction coefficient and certain aerodynamic parameters are treated as estimated quantities and are therefore examined further in the sensitivity analysis presented later in this section.

14.2 Steady-State Force Balance Verification

Before examining transient behaviour, it is useful to verify that the dynamic model recovers the expected quasi–static behaviour of the CVT under steady operating conditions.

Physically, if engine speed is held constant and inertial effects vanish, the axial forces acting on the movable sheave should balance such that

$$F_{\text{net}}(s) = 0$$

at the equilibrium shift position.

Figure 31: Placeholder: axial force components acting on the primary clutch as a function of shift position at several fixed engine speeds. The curves represent centrifugal flyweight force, primary spring restoring force, and belt axial reaction. The equilibrium shift position occurs where the net axial force crosses zero.

Figure 31 illustrates the individual axial force contributions acting on the primary sheave. As engine speed increases, centrifugal flyweight force grows approximately quadratically, shifting the equilibrium point toward larger shift positions. This behaviour corresponds physically to the familiar engagement and upshift mechanism of the CVT.

From these force curves the equilibrium shift position can be extracted as the zero–crossing of the net force curve.

Figure 32: Placeholder: predicted equilibrium shift position as a function of engine angular velocity obtained from the steady-state force balance.

The resulting equilibrium shift curve shown in Figure 32 provides a useful consistency check against prior quasi–static formulations in the literature. In the limit of negligible inertia, the dynamic model converges to this same equilibrium condition, confirming that the formulation is internally consistent.

14.3 Magnitude of the Centrifugal Belt Correction

One of the central derivations of this work is the centrifugal belt force correction derived in **??**. Physically, this term arises because the rotating belt experiences radial inertia that tends to unseat it from the sheave groove.

This effect reduces the effective normal force acting on the belt and therefore lowers the maximum transmissible torque.

Figure 33: Placeholder: centrifugal belt force compared with total clamping force as a function of engine speed for both primary and secondary pulleys.

Figure 33 compares the centrifugal correction term with the total clamping force produced by the clutch mechanisms. At low engine speeds the centrifugal term is negligible. However, as engine speed increases the correction grows rapidly and becomes comparable in magnitude to the mechanical clamping force.

To quantify the impact of this effect on torque transmission, the maximum transmissible torque was computed both with and without the centrifugal correction.

Figure 34: Placeholder: predicted torque capacity of the belt drive with and without the centrifugal correction term. Neglecting the correction significantly overestimates the torque capacity at high engine speeds.

The difference between these curves represents the modelling error introduced by neglecting belt centrifugal effects. For the operating range considered here, the correction substantially reduces the predicted torque capacity at high rotational speeds, highlighting the importance of including this effect in accurate CVT models.

14.4 Torque Transmission and Slip Behaviour

The model explicitly distinguishes between no–slip and slipping regimes based on the instantaneous torque capacity of the belt.

Physically, when the demanded torque exceeds the maximum transmissible torque determined by clamping force and friction, the belt must enter a slipping state.

Figure 35: Placeholder: time history of torque demand and maximum transmissible torque during a transient loading event. Regions where torque demand exceeds capacity correspond to slipping behaviour.

Figure 35 illustrates a representative transient in which the torque demand temporarily exceeds the belt's torque capacity. In these intervals the transmitted torque is limited by frictional capacity rather than by the engine torque itself.

The model also records the instantaneous slip state.

Figure 36: Placeholder: slip-state indicator and ratio derivative during the transient event shown in Figure 35.

As shown in Figure 36, the onset of slip coincides with changes in the rate of ratio evolution. This coupling between torque transmission and ratio dynamics is a fundamentally transient phenomenon and cannot be captured by purely steady–state models.

14.5 Vehicle Launch from Rest

The full drivetrain model can be used to simulate a complete vehicle launch from rest.

Physically, such a launch typically exhibits three distinct phases:

1. initial clutch engagement,
2. active ratio upshift as vehicle speed increases,
3. eventual stabilization near the shift–out ratio.

Figure 37: Placeholder: transmission ratio as a function of time during a simulated vehicle launch from rest.

Figure 37 shows the predicted ratio trajectory during the launch. The model captures the gradual engagement of the belt followed by continuous ratio adjustment as engine speed increases.

Figure 38: Placeholder: engine angular velocity and vehicle speed during the launch simulation.

The relationship between engine speed and vehicle speed is shown in Figure 38. A properly tuned CVT maintains the engine near its optimal operating range during acceleration.

Figure 39: Placeholder: trajectory of the engine operating point during launch plotted against the engine torque curve.

Figure 39 visualizes this behaviour by plotting the engine operating point over the torque curve. The ability of the CVT to keep the engine near its optimal torque or power region is one of the primary motivations for using continuously variable transmissions.

14.6 Parameter Sensitivity

Finally, selected parameter variations were performed to illustrate how the model responds to changes in key clutch parameters.

Rather than performing a full factorial sweep, the analysis focuses on parameters known to strongly influence CVT behaviour.

Flyweight mass Increasing flyweight mass increases centrifugal force and therefore lowers engagement RPM.

Figure 40: Placeholder: effect of flyweight mass on predicted ratio trajectory and engagement speed.

Helix angle The helix profile determines how strongly the secondary clutch reacts to transmitted torque.

Figure 41: Placeholder: effect of helix angle variation on ratio dynamics under load.

Friction coefficient The belt friction coefficient determines the torque capacity threshold at which slip occurs.

Figure 42: Placeholder: torque capacity as a function of friction coefficient.

These sensitivity studies demonstrate how the model can be used not only to simulate drivetrain behaviour but also to provide insight into the effects of tuning parameters commonly adjusted during CVT calibration.

15 Conclusion

This work developed a unified first-principles dynamic model for a mechanically actuated dry rubber V-belt CVT and its coupling to the full drivetrain. Starting from geometry and force balance, the derivation produced a closed nonlinear state-space formulation in which rotational dynamics, traction-limited torque transmission, and axial shift dynamics are solved consistently in time.

The completed formulation includes the primary flyweight-ramp actuation model, secondary helix torque feedback, belt geometric closure, centrifugal belt normal-force correction, and a stick–slip switching rule for transmitted torque. Together, these elements provide a physically interpretable framework that bridges the gap between purely quasi-static tuning models and dynamic simulation suitable for transient events such as engagement, upshift, and load-induced slip.

From an engineering perspective, the model is intended to support both understanding and design iteration: it provides traceable equations for mechanism-level reasoning while also enabling numerical studies of parameter sensitivity (e.g., flyweight mass, helix profile, friction level) and their impact on shift behaviour and torque capacity.

The current formulation also has clear limits. It does not yet include thermal effects, wear progression, or detailed tribological contact evolution, and the vehicle side remains a reduced longitudinal model. Efficiency is also treated with simplifying assumptions: torque transfer is modeled with ideal geometric mapping and traction limits, but explicit speed/load-dependent loss partitions (e.g., belt hysteresis, sheave contact dissipation, bearing and seal losses) are not yet parameterized as a standalone efficiency submodel. In addition, experimental calibration and validation data are still limited for the present hardware configuration, so several parameters are currently represented by engineering estimates.

TODO: insert the final quantitative result highlights from Section 14, including the key no-slip/slip transition findings and the measured impact of the centrifugal correction on predicted torque capacity.

TODO: insert final validation-comparison statement once benchmark and/or experimental datasets are finalized.

Future work will prioritize targeted parameter identification, experimental validation across operating regimes, and extension of the model to include higher-fidelity loss and thermal effects so that predictive accuracy and design decision confidence can be improved further.

A Belt Length Small-Angle Approximation

This appendix quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models. Using parameter values representative of the physical system considered in this work, we demonstrate that the approximation introduces substantial error in the computed center-to-center distance, then further in the ratio, which is critical for accurate dynamic modeling.

A.1 Standard Approximation

Many simplified treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this approximation, the belt-length constraint Equation (7.11) reduces to

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{A.1}$$

A.2 Derivation of the Approximate Center Distance

To determine the center-to-center distance at the initial configuration, we substitute $r_p = r_{p,\text{outer,min}}$ and $r_s = r_{s,\text{outer,max}}$ into Equation (A.1) and solve for C.

Step 1: Isolate the square root. Rearranging Equation (A.1):

$$L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) = 2\sqrt{C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}. \tag{A.2}$$

Step 2: Square both sides.

$$\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2 = 4\left[C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2\right]. \tag{A.3}$$

Step 3: Solve for C^2.

$$C^2 = \frac{\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2. \tag{A.4}$$

Step 4: Take the square root. Expanding the squared term in the numerator:

$$\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2 = L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2,$$

the approximate center distance becomes

$$\boxed{C_{\text{approx}} = \sqrt{\frac{L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}.} \tag{A.5}$$

This closed-form expression avoids the need for numerical root finding but neglects the $\sin^{-1}$ term in the exact constraint.

A.3 Error Analysis

We now evaluate the approximation error using parameter values from the physical CVT system considered in this work.

System Parameters

Parameter	Value	SI Value
Belt length L_b	37.538 in	0.9535 m
Belt height h_b	0.613 in	0.015 57 m
Groove half-angle β	11.5°	0.2007 rad
Inner primary radius	0.75 in	0.019 05 m
Primary outer radius $r_{p,\text{outer,min}}$	1.363 in	0.034 62 m
Secondary outer radius $r_{s,\text{outer,max}}$	4.0 in	0.1016 m

The primary outer radius is computed as the inner radius plus belt height: $r_{p,\text{outer,min}} = 0.75 + 0.613 = 1.363$ in.

Computed Center Distances Solving the exact belt-length constraint Equation (7.13) numerically yields

$$C_{\text{exact}} = 0.253\,87\,\text{m} \approx 9.995\,\text{in}.$$

Evaluating the approximate formula Equation (A.5) yields

$$C_{\text{approx}} = 0.271\,16\,\text{m} \approx 10.676\,\text{in}.$$

Error Metrics The absolute and relative errors are:

$$\Delta C = C_{\text{approx}} - C_{\text{exact}} = 0.017\,29\,\text{m} = 0.681\,\text{in}, \tag{A.6}$$

$$\varepsilon_{\text{rel}} = \frac{\Delta C}{C_{\text{exact}}} = \frac{0.01729}{0.25387} = 6.81\%. \tag{A.7}$$

Assessment The small-angle approximation introduces a relative error of nearly 7% in the center-to-center distance for this system. This error arises because the ratio

$$\frac{r_{s,\text{outer,max}} - r_{p,\text{outer,min}}}{C} = \frac{0.1016 - 0.03462}{0.25387} = 0.264$$

is not small. The assumption $\lambda \approx 0$ is therefore not justified for this CVT geometry.

Since the center distance C propagates into all subsequent geometric calculations—wrap angles, tangent span lengths, and derived quantities such as the transmission ratio—a 7% error in C can compound into larger errors in predicted system behavior.

A.4 Propagation to Secondary Radius and Transmission Ratio

To demonstrate how the center distance error propagates into operationally relevant quantities, we now examine the secondary radius and CVT ratio at two shift positions: zero shift and maximum shift.

A.4.1 Derivation of the Approximate Secondary Radius

Given a known primary outer radius $r_{p,\text{outer}}$ and center distance C, the secondary outer radius can be obtained by solving the belt-length constraint. Under the small-angle approximation Equation (A.1), this yields a closed-form expression.

Step 1: Isolate the square root. From Equation (A.1):

$$L_b - \pi(r_{p,\text{outer}} + r_{s,\text{outer}}) = 2\sqrt{C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2}. \tag{A.8}$$

Step 2: Square both sides.

$$\left[L_b - \pi r_{p,\text{outer}} - \pi r_{s,\text{outer}}\right]^2 = 4\left[C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2\right]. \tag{A.9}$$

Step 3: Expand and collect terms. Expanding the left side:

$$(L_b - \pi r_{p,\text{outer}})^2 - 2\pi(L_b - \pi r_{p,\text{outer}})r_{s,\text{outer}} + \pi^2 r_{s,\text{outer}}^2.$$

Expanding the right side:

$$4C^2 - 4r_{s,\text{outer}}^2 + 8r_{p,\text{outer}}r_{s,\text{outer}} - 4r_{p,\text{outer}}^2.$$

Collecting powers of $r_{s,\text{outer}}$ yields a quadratic equation:

$$(\pi^2+4)r_{s,\text{outer}}^2 - 2\left[\pi L_b - \pi^2 r_{p,\text{outer}} + 4r_{p,\text{outer}}\right]r_{s,\text{outer}} + \left[(L_b - \pi r_{p,\text{outer}})^2 + 4r_{p,\text{outer}}^2 - 4C^2\right] = 0. \tag{A.10}$$

Step 4: Apply the quadratic formula. The physically meaningful root (smaller secondary radius for overdrive configuration) is

$$\boxed{r_{s,\text{approx}} = \frac{\pi L_b + (4-\pi^2)r_{p,\text{outer}} - 2\sqrt{(\pi^2+4)C^2 - L_b^2 + 4\pi L_b r_{p,\text{outer}} - 4\pi^2 r_{p,\text{outer}}^2}}{\pi^2+4}.} \tag{A.11}$$

This closed-form expression enables direct computation of the approximate secondary radius without numerical root finding.

A.4.2 Zero Shift Configuration

At zero shift, the CVT is in its lowest ratio (maximum reduction) configuration. This is also the geometry used to compute the approximate center distance C_{approx}, so much of that initial center-distance error cancels when the same configuration is used again to recover the secondary radius.

Primary Radius at Zero Shift At $s = 0$, the primary is within the deadzone, so from Equation (7.3):

$$r_{p,\text{outer}}(0) = r_{p,\text{outer,min}} = 0.034\,62\,\text{m}.$$

Secondary Radius Comparison Solving the exact belt-length constraint with C_{exact} yields

$$r_{s,\text{exact}} = 0.101\,61\,\text{m}.$$

Evaluating the approximate formula Equation (A.11) with $C_{\text{approx}} = 0.271\,16\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.101\,61\,\text{m}.$$

Transmission Ratio Error Computing effective radii with $h_b/2 = 0.007\,79\,\mathrm{m}$:

Quantity	Exact	Approximate
$r_{p,\mathrm{eff}}$	$0.0268\,\mathrm{m}$	$0.0268\,\mathrm{m}$
$r_{s,\mathrm{eff}}$	$0.0938\,\mathrm{m}$	$0.0938\,\mathrm{m}$
$R = r_{s,\mathrm{eff}}/r_{p,\mathrm{eff}}$	3.496	3.497

The ratio error at zero shift is

$$\varepsilon_R = \frac{|3.497 - 3.496|}{3.496} \approx 0.005\%.$$

Thus, although C_{approx} is already 6.81% too large, the induced error has little visible effect at minimum shift because the radius calculation is being evaluated at the same operating point used to construct that approximate center distance in the first place.

Figure 43: Scaled diagram of the CVT at zero shift ($s = 0$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The two solutions nearly overlap at this operating point, indicating negligible ratio error in this case.

A.4.3 Maximum Shift Configuration

At maximum shift, the primary and secondary pulleys are closest in size, representing a near-crossover configuration.

Primary Radius at Maximum Shift Using Equation (7.3) with $s = s_{\max} = 0.75\,\mathrm{in} = 0.019\,05\,\mathrm{m}$ and deadzone $s_{\mathrm{dz}} = 0.002\,489\,2\,\mathrm{m}$:

$$\begin{aligned} r_{p,\mathrm{outer}}(s_{\max}) &= r_{p,\mathrm{outer,min}} + \frac{s_{\max} - s_{\mathrm{dz}}}{2\tan\beta} \\ &= 0.03462 + \frac{0.01905 - 0.0024892}{2 \times \tan(11.5°)} \\ &= 0.03462 + \frac{0.01656}{0.4070} \\ &= 0.0753\,\mathrm{m}. \end{aligned} \tag{A.12}$$

Secondary Radius Comparison Solving the exact belt-length constraint Equation (7.13) numerically with $C_{\mathrm{exact}} = 0.253\,87\,\mathrm{m}$ yields

$$r_{s,\mathrm{exact}} = 0.066\,47\,\mathrm{m}.$$

Evaluating the approximate formula Equation (A.11) with $C_{\mathrm{approx}} = 0.271\,16\,\mathrm{m}$ yields

$$r_{s,\mathrm{approx}} = 0.056\,00\,\mathrm{m}.$$

Transmission Ratio Error The CVT ratio is defined using effective radii, which account for belt seating depth (Equation (7.23)):

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}.$$

With $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	**Exact**	**Approximate**
$r_{p,\text{eff}}$	0.0675 m	0.0675 m
$r_{s,\text{eff}}$	0.0587 m	0.0482 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	0.869	0.714

Both models predict underdrive ($R < 1$), but the approximation substantially underestimates the ratio. The ratio error is

$$\varepsilon_R = \frac{|0.714 - 0.869|}{0.869} = 17.8\%.$$

Unlike the minimum-shift case, this operating point does not benefit from the same cancellation, so the center-distance error propagates into a much larger error in secondary radius and transmission ratio.

Figure 44: Scaled diagram of the CVT at maximum shift ($s = s_{\max}$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The approximation underestimates the secondary radius, leading to a lower predicted ratio than the exact model.

A.5 Summary of Approximation Errors

Configuration	R_{exact}	R_{approx}	**Ratio Error**
Zero shift ($s = 0$)	3.496	3.497	0.005%
Maximum shift ($s = s_{\max}$)	0.869	0.714	17.8%

The small-angle approximation error is strongly operating-point dependent. At zero shift, most of the center-distance error cancels because the approximate center distance was itself constructed from that same geometry. At maximum shift, that cancellation is lost and the ratio error becomes substantial.

These errors arise from the initial 6.81% error in center distance, which propagates nonlinearly through the belt-length geometry in a state-dependent way. Since the transmission ratio directly enters the rotational equations of motion (Section 13), this approximation can still produce materially inaccurate dynamics near maximum shift.

This analysis justifies the use of exact numerical solutions for all geometric computations in this work.

B Common Ramp Profile Designs

This appendix presents common ramp profile parameterizations for CVT primary pulley actuation. From Equation (8.8), the centrifugal axial force is

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds}, \tag{B.1}$$

where m_f is the flyweight mass, $r_{f,0}$ is the initial flyweight radius, and $r_f(s)$ is the radial displacement along the ramp as a function of axial shift s.

The profile $r_f(s)$ must satisfy the admissibility conditions from Section 8: continuously differentiable, non-negative slope, and finite slope throughout $[0, s_{\max}]$.

B.1 Linear Ramp

A linear ramp with slope k_r (radial displacement per unit axial shift):

$$r_f(s) = k_r \cdot s, \qquad \frac{dr_f}{ds} = k_r = \text{const}. \tag{B.2}$$

Substituting into Equation (B.1):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 k_r (r_{f,0} + k_r s) = m_f \omega_p^2 k_r r_{f,0} + m_f \omega_p^2 k_r^2 s. \tag{B.3}$$

At constant ω_p, axial force increases linearly with s because the growing radius is multiplied by a constant slope. Simple to manufacture, but may cause aggressive shifting at high ratio.

B.2 Circular Arc Ramp

A circular arc ramp is designed to reduce force variation across the shift range by achieving *partial* cancellation. The key insight is to parameterize the arc by desired slope angles α_1 and α_2 at the start and end of the segment, where $|dr_f/ds| = \tan\alpha$ specifies the force multiplier. This provides direct control over actuation behavior at engagement (high slope) versus overdrive (low slope).

For a circle of radius R_c centered at (s_c, r_c), the ramp profile satisfies:

$$(s - s_c)^2 + (r_f - r_c)^2 = R_c^2. \tag{B.4}$$

Solving for $r_f(s)$ on the upper arc:

$$r_f(s) = r_c + \sqrt{R_c^2 - (s - s_c)^2}. \tag{B.5}$$

Implicit differentiation yields:

$$\frac{dr_f}{ds} = \frac{-(s - s_c)}{\sqrt{R_c^2 - (s - s_c)^2}} = \frac{-(s - s_c)}{r_f - r_c}. \tag{B.6}$$

Parameterization from slope angles. At any point on the arc where the local slope angle is α (i.e. $dr_f/ds = \tan\alpha$), the circle constraint combined with Equation (B.6) gives the parametric form:

$$s = s_c - R_c \sin\alpha, \qquad r_f = r_c + R_c \cos\alpha. \tag{B.7}$$

The slope angle α decreases monotonically from α_1 (entry) to $\alpha_2 < \alpha_1$ (exit) as s increases along the ramp. Applying Equation (B.7) at both boundary shifts s_1 and s_2 and subtracting:

$$s_2 - s_1 = R_c(\sin\alpha_1 - \sin\alpha_2). \tag{B.8}$$

With $\Delta s = s_2 - s_1$ as the total axial span of the ramp segment, the arc radius and center are determined entirely by the two slope angles:

$$R_c = \frac{\Delta s}{\sin\alpha_1 - \sin\alpha_2}, \qquad s_c = s_1 + R_c \sin\alpha_1. \tag{B.9}$$

The boundary condition $r_f(s_1) = 0$ (flyweights at their initial radius at ramp entry) then fixes the radial center:

$$r_c = -R_c \cos\alpha_1. \tag{B.10}$$

Substituting into Equation (B.1):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f) \cdot \frac{-(s - s_c)}{r_f - r_c}. \tag{B.11}$$

To see the cancellation explicitly, split $r_{f,0} + r_f = (r_{f,0} + r_c) + (r_f - r_c)$:

$$\begin{aligned} F_{p,\text{cent}} &= m_f \omega_p^2 \left[\underbrace{(r_f - r_c)}_{\text{cancels}} \cdot \frac{-(s - s_c)}{r_f - r_c} + (r_{f,0} + r_c) \cdot \frac{-(s - s_c)}{r_f - r_c} \right] \\ &= -m_f \omega_p^2 \, (s - s_c) \left[1 + \frac{r_{f,0} + r_c}{r_f - r_c} \right]. \end{aligned} \tag{B.12}$$

This form reveals the structure clearly. The dominant factor is $-(s - s_c)$, which is linear in shift: the $(r_f - r_c)$ terms cancel exactly in the first part of the split. The bracket $\left[1 + \dfrac{r_{f,0} + r_c}{r_f - r_c}\right]$ is a nonlinear correction multiplier. Using the parametric form Equation (B.7) with Equation (B.10), the bracket evaluates to

$$1 + \frac{r_{f,0} + r_c}{r_f - r_c} = 1 + \frac{r_{f,0} - R_c \cos\alpha_1}{R_c \cos\alpha}, \tag{B.13}$$

where α is the local slope angle at shift s, sweeping from α_1 at entry to α_2 at exit. This makes the effective proportionality between force and $(s - s_c)$ position-dependent.

The numerator $r_{f,0} - R_c \cos\alpha_1$ is the key residual: it is controlled entirely by the choice of α_1, α_2, and Δs through $R_c = \Delta s/(\sin\alpha_1 - \sin\alpha_2)$. Increasing either α_1 or Δs raises R_c, making $R_c \cos\alpha_1$ larger and the residual smaller (or negative). The entry force scale, the exit force scale, and the degree of position-dependence between them are therefore all coupled to the same two angle choices. There is no freedom to set the overall force scale independently of the nonlinearity — changing α_1 and α_2 adjusts both simultaneously.

The condition $r_{f,0} = R_c \cos\alpha_1$ makes the residual zero, so the bracket equals 1 everywhere and force is exactly proportional to shift. This is the angle configuration at which the circular arc best approximates the hyperbolic ideal — it gives a practical design rule for tuning.

The practical consequence is important: at a fixed engine speed ω_p, the centrifugal force is approximately proportional to $(s - s_c)$, growing linearly with shift. This directly matches the linear restoring force of the primary spring (force $\propto k_s \cdot s$). A spring-loaded primary can therefore be tuned to balance the centrifugal flyweights at a target shift, and the linear character of the circular arc ensures the equilibrium is stable across the shift range rather than concentrated at a single point.

Circular arcs are widely used in practice due to ease of manufacture and sufficient compensation. The nonlinear residual is small when the ramp center offset $(r_{f,0} + r_c)$ is small relative to the arc radius R_c, but it cannot be eliminated entirely. The following section shows how to *exactly* achieve the linear target by choosing a different ramp geometry.

B.3 Hyperbolic Ramp (Perfect Cancellation)

From Equation (B.13), the circular arc force-to-shift proportionality constant is:

$$\frac{F_{p,\text{cent}}}{-m_f \omega_p^2 (s - s_c)} = 1 + \frac{r_{f,0} - R_c \cos\alpha_1}{R_c \cos\alpha}. \tag{B.14}$$

This varies with position because α changes along the arc. The natural question is: what ramp profile makes this ratio a *true constant*? That constant is what we call K, and the profile that achieves it satisfies:

$$(r_{f,0} + r_f(s)) \frac{dr_f}{ds} = Ks, \tag{B.15}$$

where $K > 0$. This produces centrifugal force exactly proportional to shift:

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 K s. \tag{B.16}$$

K is not a free parameter any more than R_c is for the circular arc: just as R_c is uniquely determined by α_1, α_2, and Δs (Equation (B.9)), K is uniquely determined by α_1, α_2, s_1, and s_2 (derived in the parameterization below). What differs is not the design inputs, but that the hyperbolic profile makes K exactly constant along the entire ramp, whereas the circular arc's equivalent ratio changes with position.

Deriving the profile. Equation (B.15) is separable. Substituting $u = r_{f,0} + r_f$, so $du/ds = dr_f/ds$:

$$u\,du = Ks\,ds. \tag{B.17}$$

Integrating both sides and writing the constant of integration as $\frac{1}{2}Ks_0^2$:

$$\tfrac{1}{2}u^2 = \tfrac{1}{2}K(s^2 + s_0^2). \tag{B.18}$$

Substituting back $u = r_{f,0} + r_f$:

$$\boxed{(r_{f,0} + r_f(s))^2 = K(s^2 + s_0^2),} \tag{B.19}$$

giving the explicit profile:

$$r_f(s) = \sqrt{K(s^2 + s_0^2)} - r_{f,0}. \tag{B.20}$$

In the (s, r_f) plane, the locus $(r_{f,0} + r_f)^2 = Ks^2$ (taking $s_0 = 0$) is a branch of a *hyperbola* — hence the name. Unlike the circular arc, where the residual multiplier in Equation (B.12) could not be removed, this profile forces the product $(r_{f,0} + r_f) \cdot dr_f/ds = Ks$ exactly for all s.

Local slope and its variation. To see how K relates to actual ramp angles, differentiate Equation (B.19) implicitly:

$$\frac{dr_f}{ds} = \frac{Ks}{r_{f,0} + r_f} = \frac{\sqrt{K}\, s}{\sqrt{s^2 + s_0^2}}. \tag{B.21}$$

Since $dr_f/ds = \tan\alpha(s)$ by definition of the ramp slope angle, the local angle satisfies:

$$\tan\alpha(s) = \frac{\sqrt{K}\, s}{\sqrt{s^2 + s_0^2}}. \tag{B.22}$$

Two observations follow. First, the slope is zero at $s = 0$ and grows monotonically toward the asymptote $\sqrt{K}$ as $s \to \infty$. This is the opposite trend from a circular arc, where the slope decreases from entry to exit. The reason is that the growing lever arm $(r_{f,0}+r_f)$ at large shift would otherwise amplify force super-linearly; a rising slope exactly compensates to keep the product Ks linear. Second, the asymptotic slope $\lim_{s\to\infty} \tan\alpha(s) = \sqrt{K}$ gives a simple interpretation: $K = \tan^2\alpha_\infty$ where α_∞ is the slope the profile approaches at large shift.

Parameterization by entry and exit angles. Following the same convention as Appendix B.2, specify desired slope angles α_1 and α_2 at entry shift s_1 and exit shift s_2. From Equation (B.22):

$$\tan^2\alpha_1 = \frac{Ks_1^2}{s_1^2 + s_0^2}, \qquad \tan^2\alpha_2 = \frac{Ks_2^2}{s_2^2 + s_0^2}. \tag{B.23}$$

Dividing the two equations eliminates K:

$$\frac{\tan^2\alpha_1}{\tan^2\alpha_2} = \frac{s_1^2\,(s_2^2 + s_0^2)}{s_2^2\,(s_1^2 + s_0^2)}. \tag{B.24}$$

Solving for s_0^2:

$$s_0^2 = \frac{s_1^2 s_2^2\,(\tan^2\alpha_2 - \tan^2\alpha_1)}{\tan^2\alpha_1\, s_2^2 - \tan^2\alpha_2\, s_1^2}. \tag{B.25}$$

Then K follows from either condition in Equation (B.23):

$$K = \frac{\tan^2\alpha_1\,(s_1^2 + s_0^2)}{s_1^2}. \tag{B.26}$$

For $s_0^2 > 0$ a consistent solution requires the numerator and denominator of Equation (B.25) to have the same sign. Since the hyperbolic slope increases with s (Eq. (B.21)), the physically consistent case is $\alpha_2 > \alpha_1$ (exit angle larger than entry), with $\tan^2\alpha_1 s_2^2 > \tan^2\alpha_2 s_1^2$ ensuring a positive denominator. No solution exists when the two angle conditions give a degenerate system ($\tan^2\alpha_1 s_2^2 = \tan^2\alpha_2 s_1^2$).

Practical impact. In most cases the difference between circular and hyperbolic profiles is negligible. For the specific ramps implemented in this work, the integrated force deviation across the full shift range is less than 1%. The hyperbolic derivation is nonetheless useful: it establishes the theoretical ideal and shows that the circular arc's partial cancellation is not a fundamental limitation but rather a small correction from the constant offset $(r_{f,0} + r_c)$.

Limitations. Both profiles address only the centrifugal force term. The secondary pulley torque feedback ($F_{s,\mathrm{ax}}(s, \tau_s)$ in Equation (13.12)) introduces load-dependent nonlinear terms that no ramp geometry alone can cancel. Circular arcs therefore remain the practical standard—simpler to manufacture, well-understood, and sufficient given these irreducible secondary-side effects.

B.4 Piecewise Ramps and Smoothing

Any of the profiles above can be combined in a piecewise fashion: multiple linear segments, multiple circular arcs with different angle pairs, or a mix of profile types joined at transition shifts $s_1, s_2, \ldots, s_{n-1}$. This allows region-specific tuning — steeper slopes at engagement, moderate for cruise, gentler near overdrive — without committing to a single global geometry.

Admissibility at junctions. The admissibility conditions from Section 8 require $r_f(s)$ to be continuously differentiable (C^1). A piecewise ramp satisfies C^0 by construction (the profile value is continuous if segments are joined at a common point), but slope continuity may be violated: if the left-segment slope at s_k differs from the right-segment slope, Eq. (B.1) gives a finite force jump at that point. No kinematic singularity occurs, but the force has a step discontinuity.

Numerical stiffness considerations. A sharp slope discontinuity is not the same as a smooth rapid transition: many real implementations use a finite but steep blend region, which the ODE solver sees as an extremely large second derivative $d^2 r_f/ds^2$ near the junction. This can create a locally stiff region and force an adaptive integrator to take very small steps near the knot, even if the discontinuity is physically mild. If simulation performance is a concern, slope transitions should be blended — a short circular arc segment tangent to both adjoining profiles is sufficient, as it restores C^1 continuity and bounds $d^2 r_f/ds^2$ throughout the ramp.

C Numerical Methods Summary

This section documents the numerical algorithms used for belt-geometry root finding and dynamic simulation, including the specific tolerances, bracketing strategies, and solver configurations employed in the implementation.

C.1 Root Finding for Belt-Length Constraint

Given a known primary outer radius r_1 and fixed center-to-center distance C, the secondary outer radius r_2 is the unique root of the belt-length constraint. This scalar root problem is obtained by rearranging the exact belt length closure from Equation (7.13) into the residual form:

$$g(r_2;\, r_1) = \pi(r_1 + r_2) + 2(r_2 - r_1)\arcsin\left(\frac{r_2 - r_1}{C}\right) + 2\sqrt{C^2 - (r_2 - r_1)^2} - L_b = 0, \tag{C.1}$$

where L_b is the fixed belt length. A separate invocation solves for the center-to-center distance C itself at initialization, using the same equation evaluated at the boundary radii $(r_{1,\min},\, r_{2,\max})$ with

C as the unknown; this is exactly the initialization step implied by Equation (7.13) before the ratio law Equation (13.2) can be evaluated over the shift range.

Bracketing strategy. The arcsin term in Equation (C.1) is real only when $|r_2 - r_1| \le C$, so the natural domain for r_2 is the open interval $(r_1 - C,\ r_1 + C)$. Within this interval a sign change is guaranteed: at the lower boundary $r_2 \to r_1 - C$, the arcsin saturates to $-\pi/2$ and the square root vanishes, giving $g \to 2\pi r_1 - 2\pi C - L_b < 0$ for all physically valid parameters. At the upper boundary $r_2 \to r_1 + C$, by symmetry $g \to 2\pi r_1 + 2\pi C - L_b > 0$, satisfied whenever the belt length is less than the circumference subtended by the wider pulley pair — a requirement any real CVT must satisfy. A small safety margin $\varepsilon = 10^{-9}$ m is applied at both ends to keep arcsin away from its domain boundary where the function becomes numerically steep.

For center-to-center distance solving, the bracket is instead the interval $(\Delta r + \varepsilon,\ L_b/2)$, where $\Delta r = |r_{2,\text{eff}} - r_{1,\text{eff}}|$ is the effective-radius difference at the boundary configuration. The lower bound ensures arcsin is defined; the upper bound is a conservative geometric limit (a straight two-span belt cannot exceed L_b across a gap of $L_b/2$). Both brackets are validated by an explicit sign-change check before invoking the solver, so any failure to bracket immediately identifies an infeasible geometry rather than producing a silent wrong answer.

Brent's method. Both roots are found using `scipy.optimize.brentq`, an implementation of Brent's algorithm Brent (1973). The method combines three strategies: bisection (guaranteed bracket-halving), secant interpolation (superlinear convergence near smooth roots), and inverse quadratic interpolation (order-≈ 1.84 convergence when three previous iterates are well-conditioned). Whenever the faster interpolation steps would place the next iterate outside the current bracket, the algorithm falls back to bisection, preserving the worst-case $O(\log_2(1/\varepsilon))$ guarantee of pure bisection. Because the sign-change brackets described above are guaranteed for all valid CVT geometries, `brentq` is certain to converge. Both root solves use an absolute tolerance of 10^{-9} m, well below any physically meaningful length precision.

C.2 ODE Integration for Dynamic Simulation

The full CVT state vector $\mathbf{x} = (\omega_p, \omega_s, s, \dot{s})$ is integrated using `scipy.integrate.solve_ivp`. The governing system is the assembled first-order model in Equation (13.13), with rotational sub-equations Equation (13.9) and Equation (13.10), axial dynamics Equation (13.12), and algebraic geometry closures Equation (13.2) and Equation (13.3). The time integrator is used with the `RK45` method — the Dormand–Prince explicit embedded Runge–Kutta pair Dormand and Prince (1980). The method propagates the state with a 5th-order formula and simultaneously computes a 4th-order companion estimate for local error control, using the same six function evaluations. The tableau is FSAL (first-same-as-last): the final stage of each accepted step is reused as the first stage of the next, so only five new evaluations are needed per step in steady operation.

Adaptive step size control. At each candidate step the per-component local error estimate $\hat{e}_i$ is compared to a mixed tolerance threshold. A step is accepted when

$$\max_i \frac{|\hat{e}_i|}{\text{atol} + \text{rtol} \cdot |y_i|} \le 1, \tag{C.2}$$

and the next step size is scaled proportionally to $(\text{tol}/\hat{e})^{1/5}$. The tolerances used in this work are $\text{atol} = 10^{-6}$ and $\text{rtol} = 10^{-4}$ (Table 7). The absolute floor prevents spurious step rejection when a state variable passes through zero (e.g. $\dot{s} \approx 0$ at shift equilibrium or $\omega_p \approx 0$ at engine start), where a purely relative criterion would demand unreachable accuracy.

Parameter	Value	Interpretation
`atol`	10^{-6}	absolute floor: 1 µm on s, 1 µrad/s on ω
`rtol`	10^{-4}	relative tolerance: 0.01% of current state magnitude
`t_eval`	10 000 pts	uniform output grid over 30 s

Table 7: ODE solver tolerance parameters.

Event detection. Phase transitions (shift engagement, boundary contact, back-shift onset) are located using `solve_ivp`'s event mechanism. These events correspond to qualitative changes in the dynamics around the bounded shift coordinate used in Equation (13.12) and the full state evolution in Equation (13.13). After each accepted step the solver evaluates the event functions on a degree-4 dense-output interpolant and applies a root-finding pass to detect any sign change within the step. When a crossing is found it is bracketed at sub-tolerance precision before being reported, so transitions are located with the same accuracy as the integration tolerance rather than being snapped to the coarse output grid.

Phase splitting at hard constraints. The shift bounds $s \in [0, s_{\max}]$ with $\dot{s} = 0$ at both endpoints are handled by partitioning the trajectory into separate `solve_ivp` calls rather than imposing inequality constraints inside the ODE. Each phase — normal shifting, clamped-at-maximum, and back-shifting — presents a smooth right-hand side to the integrator, allowing RK45 to operate at full 5th-order accuracy throughout. The terminal event of each phase supplies the initial condition of the next. This is numerically cleaner than directly clamping Equation (13.12) inside a single solve because it preserves a smooth right-hand side within each phase. Up to ten alternating full-shift / back-shift cycles are supported before the simulation terminates, preventing unbounded phase loops for unusual parameter regimes.

C.3 Slip-Law Regularization and Blend Parameter Selection

The traction law in ?? is physically clear but discontinuous at $v_{\text{rel}} = 0$, which can induce step-size chatter near stick–slip transitions. A practical regularized formulation uses three ideas: a smooth saturation law, a clamped no-slip torque request, and a blending rule between those two limits.

First define a smooth saturation torque

$$\tau_{\text{sat}} = \tau_{\max} \tanh\left(\frac{v_{\text{rel}}}{v_{\text{smooth}}}\right), \tag{C.3}$$

where $v_{\text{smooth}} > 0$ sets the transition width in slip-speed units. For $|v_{\text{rel}}| \gg v_{\text{smooth}}$, this approaches the usual friction-limited value $\pm\tau_{\max}$, while near zero it removes the sharp sign discontinuity.

Next, clamp the no-slip torque request:

$$\tau_{\text{req}} = \text{clip}(\tau_{\text{ns}},\ -\tau_{\max},\ \tau_{\max})\,. \tag{C.4}$$

This is the largest admissible torque consistent with the no-slip request, without allowing the requested value to exceed traction capacity.

Then define the blend factor

$$\alpha = \mathrm{clip}\left(\frac{|v_{\mathrm{rel}}|}{v_{\mathrm{blend}}},\, 0,\, 1\right), \tag{C.5}$$

with $v_{\mathrm{blend}} > 0$ defining the width of the blending region, and use the transmitted torque

$$\tau_c = (1-\alpha)\,\tau_{\mathrm{req}} + \alpha\,\tau_{\mathrm{sat}}. \tag{C.6}$$

This has a useful interpretation. For very small mismatch, $\alpha \approx 0$, so the model stays close to the clamped no-slip demand. For larger mismatch, $\alpha \to 1$, so the law transitions to the smooth Coulomb limit. In other words, the regularized model behaves like a requested no-slip torque near stick and like a friction-limited sliding law once the mismatch becomes appreciable.

How to choose v_{smooth}. Choose the smoothing scale small enough that $\tanh(v_{\mathrm{rel}}/v_{\mathrm{smooth}})$ is close to a sign law through most of the slip regime, but not so small that the solver again sees an effectively discontinuous transition. A practical workflow is:

1. Start with v_{smooth} around 0.5–2% of a representative relative-speed magnitude seen during actual slip.
2. Decrease it until predicted macro outputs (ratio trajectory, launch time, peak speeds) stop changing materially.
3. Increase it slightly if event localization or adaptive step-size behavior becomes noisy near $v_{\mathrm{rel}} \approx 0$.

The blend scale v_{blend} should generally be chosen greater than or equal to v_{smooth}, so that the handoff from clamped demand to the smooth Coulomb limit is not sharper than the smoothing law itself.

D Secondary Capacity Closure

This remains a placeholder for now, as it relies on outdated equations, but one can see how the circular dependency can be resolved.

$$\tau_{max} = 2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\left(\frac{2F_{clamp}\tan(\beta)}{\phi} + w^2r^2\rho A\right)$$

$$F_{clamp} = \frac{K_T\theta_s + \tau_c i}{2r_h\tan(\theta_r)} + K_C d$$

$$\tau_c \le \tau_{max} = 2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\left(\frac{2\left(\frac{K_T\theta_s + \tau_c i}{2r_h\tan(\theta_r)} + K_C d\right)\tan(\beta)}{\phi} + w^2r^2\rho A\right)$$

$A = w^2r^2\rho A$ $\quad B = 2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}$ $\quad C = K_T\theta_s$ $\quad D = 2r_h\tan(\theta_r)$ $\quad E = K_C d$

$$\tau_c \le \tau_{max} = B\left(\frac{2\left(\frac{C+\tau_c i}{D} + E\right)\tan(\beta)}{\phi} + A\right)$$

$$\tau_c \le \frac{2B\tan(\beta)}{\phi}\left(\frac{C+\tau_c i}{D} + E\right) + BA$$

$$F = \frac{2B\tan(\beta)}{\phi}$$

$$\tau_c \le F\left(\frac{C+\tau_c i}{D} + E\right) + BA$$

$$\tau_c \le \frac{FC}{D} + \frac{F\tau_c i}{D} + FE + BA$$

$$\tau_c - \frac{F\tau_c i}{D} \le \frac{FC}{D} + FE + BA$$

$$\tau_c \le \frac{\frac{FC}{D} + FE + BA}{1 - \frac{Fi}{D}}$$

$A = w^2r^2\rho A$ $\quad B = 2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}$ $\quad C = K_T\theta_s$ $\quad D = 2r_h\tan(\theta_r)$ $\quad E = K_C d$ $\quad F = \frac{2B\tan(\beta)}{\phi}$

$$\tau_c \le \frac{\frac{FC}{D} + E + BA}{1 - \frac{Fi}{D}} \qquad \tau_c \le \frac{\frac{\frac{2B\tan(\beta)}{\phi}K_T\theta_s}{2r_h\tan(\theta_r)} + \frac{2B\tan(\beta)}{\phi}K_C d + \left(2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)w^2r^2\rho A}{1 - \frac{\frac{2B\tan(\beta)}{\phi}}{2r_h\tan(\theta_r)}i}$$

$$\tau_c \le \frac{\frac{\frac{2\cdot 2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\tan(\beta)}{\phi}K_T\theta_s}{2r_h\tan(\theta_r)} + \frac{2\cdot 2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\tan(\beta)}{\phi}K_C d + \left(2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)w^2r^2\rho A}{1 - \frac{\frac{2\cdot 2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\tan(\beta)i}{\phi}}{2r_h\tan(\theta_r)}}$$

$$\left(2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)w^2r^2\rho A = \frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\left(2w^2r^3\rho A\right)$$

$$1 - \frac{\frac{2\cdot 2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\tan(\beta)i}{\phi}}{2r_h\tan(\theta_r)} = 1 - \frac{8r\tan(\beta)i\,r_h\tan(\theta_r)}{\phi}\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)$$

$$\frac{\frac{2\cdot 2r\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)\tan(\beta)}{\phi}K_T\theta_s}{2r_h\tan(\theta_r)} = \left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)\frac{K_T\theta_s 2r\tan(\beta)}{r_h\tan(\theta_r)\phi}$$

$$1 - \frac{2\cdot 2r\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)\tan(\beta)i}{\left(\frac{\phi}{2r_h\tan(\theta_r)}\right)}$$

$$1 - \left(\frac{4r\tan(\beta)i}{1}\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)\frac{1}{\left(\frac{\phi}{2r_h\tan(\theta_r)}\right)}\right)$$

$$1 - \left(\frac{4r\tan(\beta)i}{1}\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)\frac{2r_h\tan(\theta_r)}{\phi}\right)$$

$$1 - \frac{8r\tan(\beta)i\,r_h\tan(\theta_r)}{\phi}\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)$$

$$\frac{2\cdot 2r\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\tan(\beta)}{\phi}K_C d = \frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\,\frac{4r\tan(\beta)K_C d}{\phi}$$

$$\tau_c \le \frac{\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)\frac{K_T\theta_s 2r\tan(\beta)}{r_h\tan(\theta_r)\phi} + \frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\,\frac{4r\tan(\beta)K_C d}{\phi} + \frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\left(2w^2r^3\rho A\right)}{1 - \frac{8r\tan(\beta)i\,r_h\tan(\theta_r)}{\phi}\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)}$$

$$\tau_c \le \frac{\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)\frac{K_T\theta_s 2r\tan(\beta)}{r_h\tan(\theta_r)\phi} + \frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\,\frac{4r\tan(\beta)K_C d}{\phi} + \frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\left(2w^2r^3\rho A\right)}{1 - \frac{8r\tan(\beta)i\,r_h\tan(\theta_r)}{\phi}\left(\frac{e^{\mu\phi}-1}{e^{\mu\phi}+1}\right)} \cdot \frac{\frac{e^{\mu\phi}+1}{e^{\mu\phi}-1}}{\frac{e^{\mu\phi}+1}{e^{\mu\phi}-1}}$$

$$\tau_c \le \frac{\frac{K_T\theta_s 2r\tan(\beta)}{r_h\tan(\theta_r)\phi} + \frac{4r\tan(\beta)K_C d}{\phi} + \left(2w^2r^3\rho A\right)}{\frac{e^{\mu\phi}+1}{e^{\mu\phi}-1} - \frac{8r\tan(\beta)i\,r_h\tan(\theta_r)}{\phi}} \cdot \frac{\frac{1}{2r}}{\frac{1}{2r}}$$

$$\tau_c \le \frac{\frac{K_T\theta_s\tan(\beta)}{r_h\tan(\theta_r)\phi} + \frac{2\tan(\beta)K_C d}{\phi} + \left(w^2r^2\rho A\right)}{\frac{e^{\mu\phi}+1}{2r(e^{\mu\phi}-1)} - \frac{4\tan(\beta)i\,r_h\tan(\theta_r)}{\phi}} \cdot \frac{\phi}{\phi}$$

$$\tau_c \le \frac{\frac{K_T\theta_s}{r_h\tan(\theta_r)}\tan(\beta) + 2K_C d\tan(\beta) + \left(\phi w^2r^2\rho A\right)}{\frac{\phi(e^{\mu\phi}+1)}{2r(e^{\mu\phi}-1)} - 4i\,r_h\tan(\theta_r)\tan(\beta)} \cdot \frac{\frac{1}{2}}{\frac{1}{2}}$$

$$\tau_c \le \frac{\left(\frac{K_T\theta_s}{2r_h\tan(\theta_r)} + K_C d\right)\tan(\beta) + \frac{1}{2}\left(\phi w^2r^2\rho A\right)}{\frac{\phi(e^{\mu\phi}+1)}{4r(e^{\mu\phi}-1)} - 2i\,r_h\tan(\theta_r)\tan(\beta)}$$

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